

ROBOT PROGRAMMING FRAMEWORKS AND IOT PLATFORMS

Scuola Superiore Sant'Anna

Academic Year 2025/2026

Gastone Ciuti



Organization

11/11/2025	Introduction and Communication Protocols (1/2) (3 hours – SLOT 1)
17/11/2025	Communication Protocols (2/2) (3 hours – SLOT 1)
18/11/2025	IoT Platforms and Main Features (3 hours – SLOT 2)
24/11/2025	Set-up of Particle Development Boards & Basic Functions (2 hours) • Web-based programming, cloud functions and examples
25/11/2025	How to Program an IoT System: Functions and Coding (1/3) (3 hours – SLOT 3) • Other Programming Environments, <i>i.e.</i> Particle Workbench based on Visual Studio Code
01/12/2025	How to Program an IoT System: Functions and Coding (2/3) (2 hours – SLOT 3) • Advanced Programming Functionalities for IoT Platforms and Use Cases
02/12/2025	How to Program an IoT System: Functions and Coding (3/3) (3 hours – SLOT 3) • Integration with Web Services, <i>i.e.</i> IFTTT (if this than that)
09/12/2025	Examples with IoT Systems and Exercises (3 hours – SLOT 4)



Introduction (11.11.2025) – SLOT 1

Gastone Ciuti





2008

Master degree in Biomedical Engineering, curriculum **Biomedical Technology** (110/110 magna cum laude), Biomedical Engineering, University of Pisa (Pisa, Italy)



2012

PhD degree in Innovative Technologies of Info. & Com. Eng. and Robotics, curriculum **BioRobotics** (100/100 magna cum laude), The BioRobotics Institute, Scuola Superiore Sant'Anna (Pisa, Italy)



2024

Full Professor of Bioengineering, The BioRobotics Institute, Scuola Superiore Sant'Anna (Pisa, Italy)



2017

Founder and Scientific Advisor of Mediate s.r.l. (**Spin-Off company / Scuola Superiore Sant'Anna**), Pisa, Italy



IEEE EMBS Technical Community on BioRobotics

IEEE Senior Member (2021) and "Roadmapping Sub-Committee Member" (2016)

Academic role: Full Professor in Bioengineering | Deputy Director of the BioRobotics Institute of Sant'Anna School of Advanced Studies



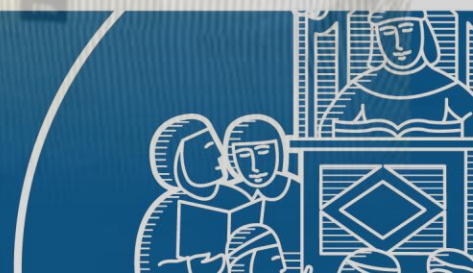
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MEDIATE
PROXIMAL COLLABORATION



Gastone Ciuti, PhD





Sant'Anna
School of Advanced Studies – Pisa

THE BIOROBOTICS
INSTITUTE



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INTERDISCIPLINARY
CENTER
Sant'Anna
School of Advanced Studies – Pisa

Healthcare MechatronicsLab

GASTONE CIUTI

The BioRobotics Institute, Scuola Superiore Sant'Anna, 56025 Pisa, Italy

Department of Excellence in Robotics & AI, Scuola Superiore Sant'Anna, 56127 Pisa, Italy

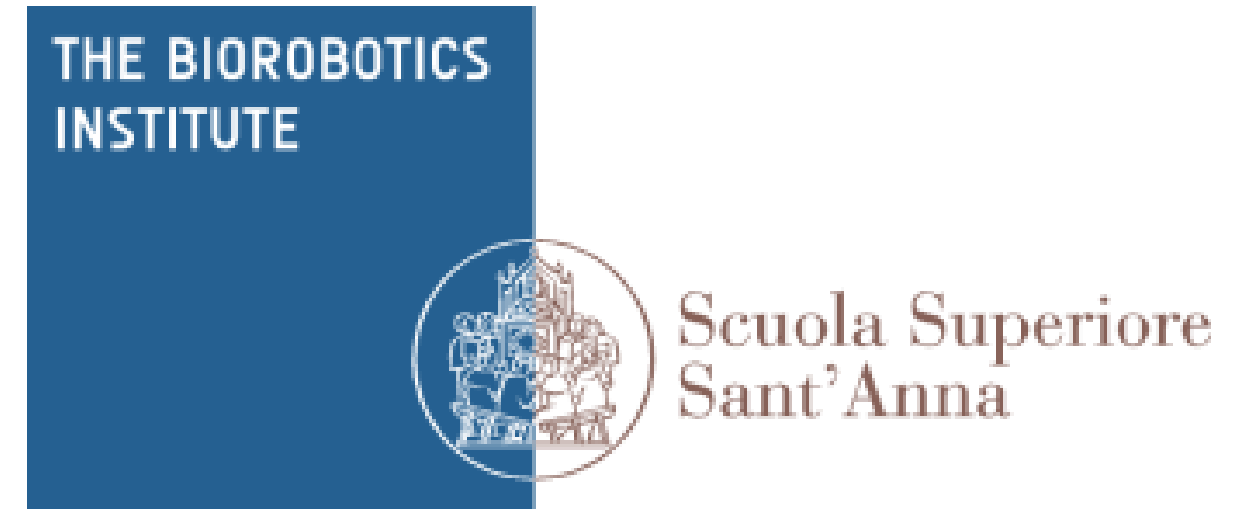
Health Science Interdisciplinary Center, Scuola Superiore Sant'Anna, 56127 Pisa, Italy



Gastone Ciuti

Mobile: +39 348 0015616

E-mail: gastone.ciuti@santannapisa.it



Class, academic year 2025/2026

#	Name	B.Sc. Background	Curricula
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			



Thanks to my research team!



What kind of research do we conduct at the Healthcare Mechatronics lab?



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School of Advanced Studies – Pisa

DIPARTIMENTO
DI ECCELLENZA
2023-2027

ROBOTICS & AI



Sant'Anna
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**INTERDISCIPLINARY
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Healthcare Mechatronics Lab

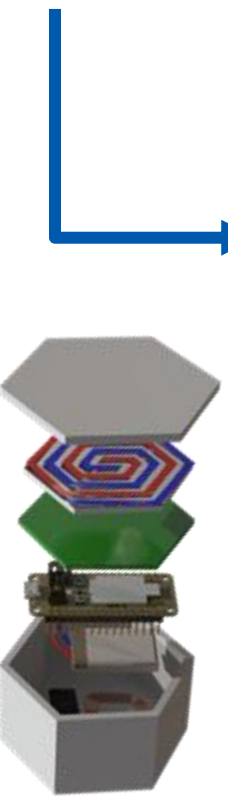
Intelligent and Integrated **Macro-to-Meso Scale Medical Robots** and **eHealth Technologies** for Human-Centred Longitudinal Healthcare

Assistive Robotics

Collaborative Technologies and Context Awareness Sensors for Human Care

Endoluminal Robotics

Macro-to-Meso Scale Robots for Endoluminal Diagnosis and Intervention



Wearable and collaborative Technologies



Artificial organs and prostheses



Medical Robotics and regenerative medicine



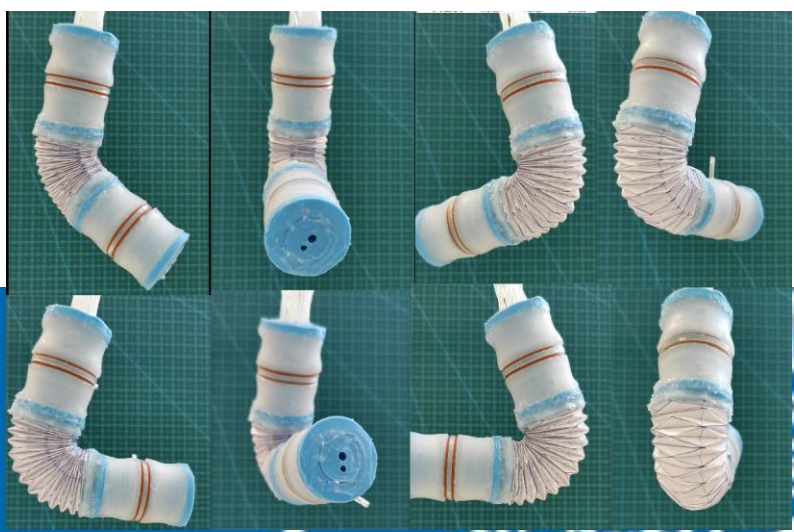
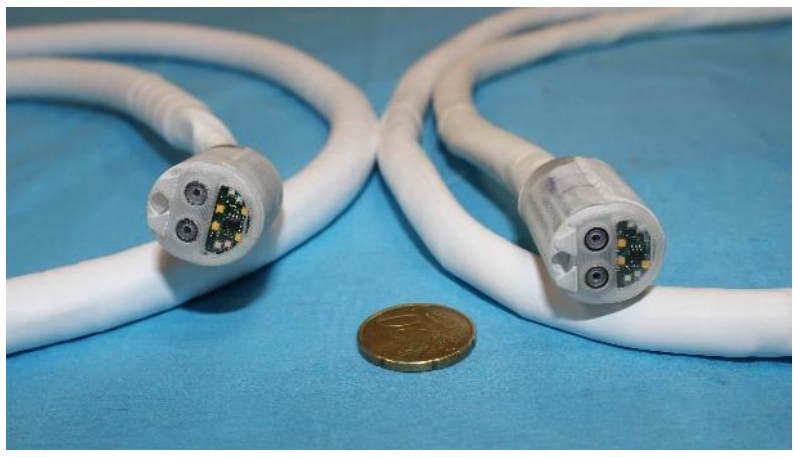
Mathematical and computational modelling



Bioinspired and soft Robotics



Neuroscience Robotics



Research Themes at The BioRobotics Institute



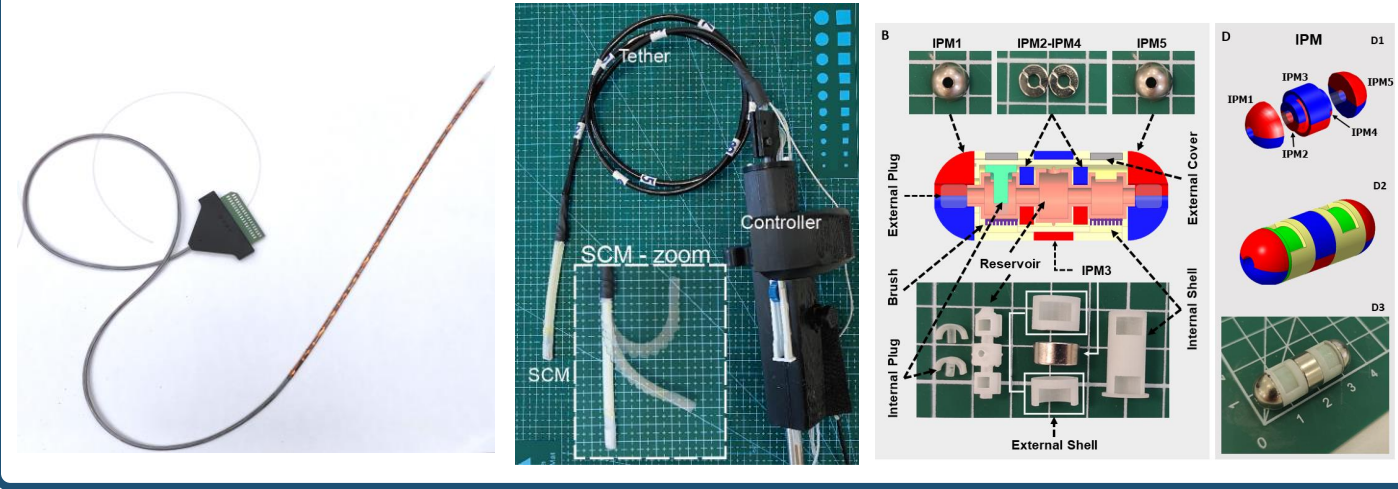
Healthcare Mechatronics Lab

Endoluminal Robotics and Healthcare Technologies

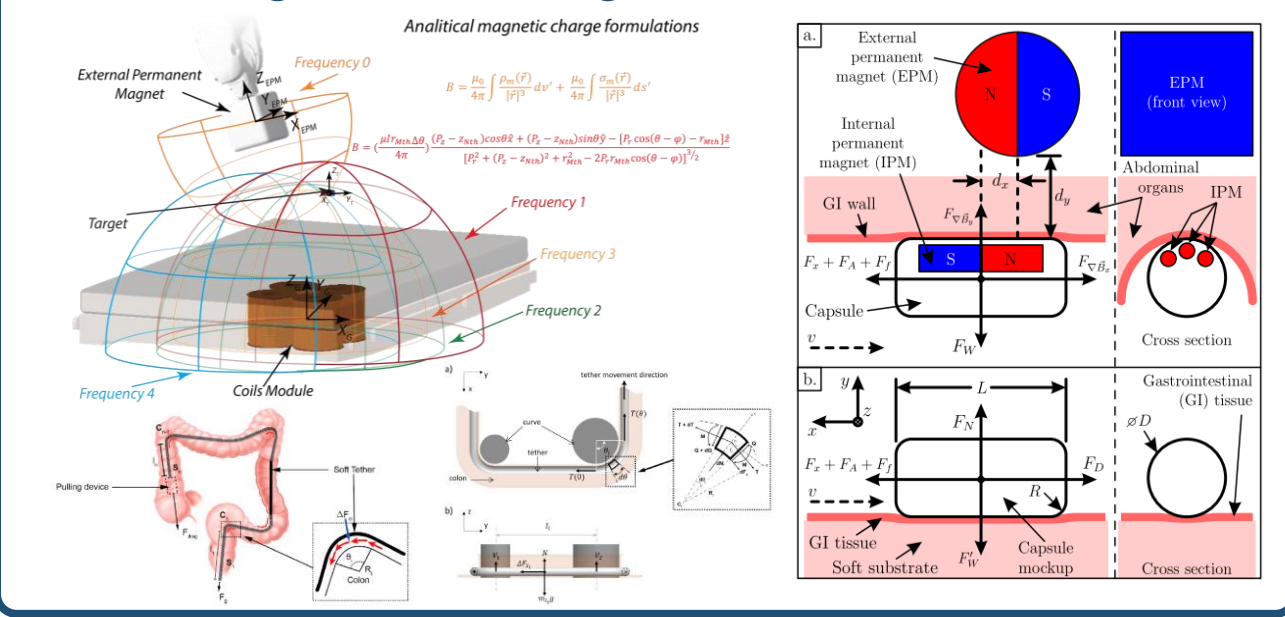
Intelligent and Integrated **Macro-to-Meso Scale Endoluminal Robots** and **Biomedical Technologies** for Human-Centred Longitudinal Healthcare

Endoluminal Robotics

Oesophagus, Stomach and Small Bowel

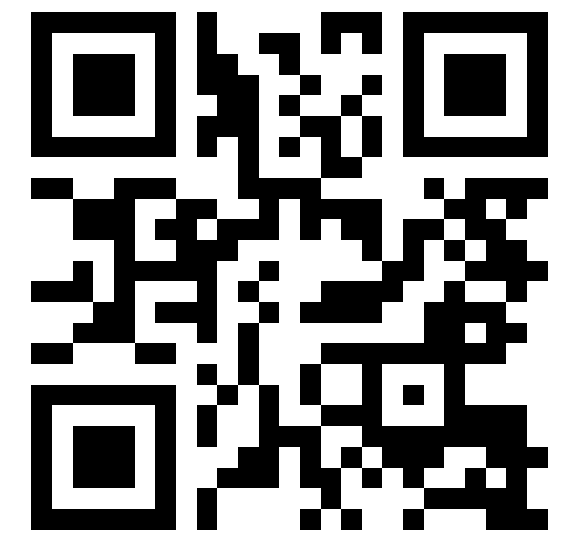


Modelling & Control Algorithms

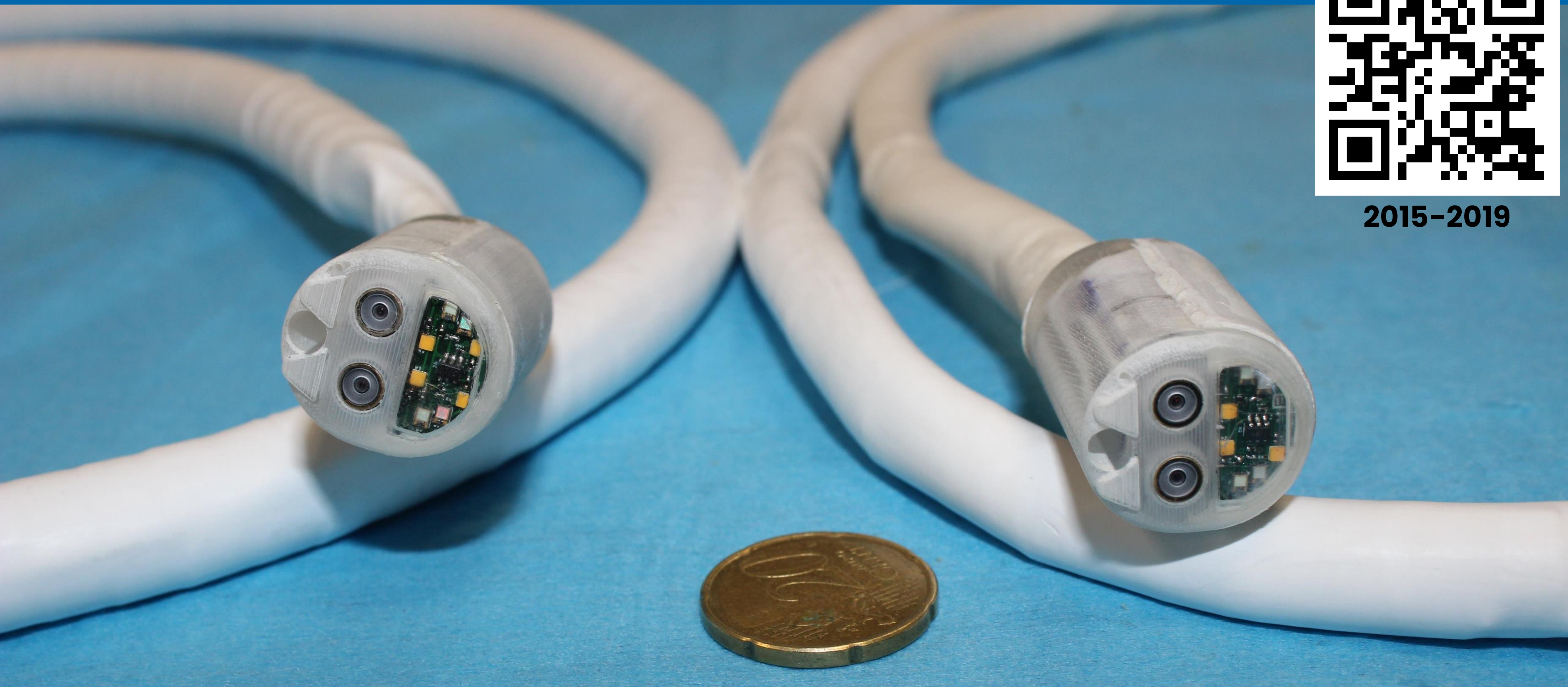


Prof. Gastone Ciuti (gastone.ciuti@santannapisa.it)

Soft-tethered front-wheel magnetic colonoscope



2015–2019



Endoo-project.eu
Endoscopic versatile robotic guidance, diagnosis and therapy
of magnetic-driven soft-tethered endoluminal robots



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o o o o v e s c o





From an idea to the real implementation

A new inchworm-like robot with a double-balloon chamber



SUBMITTED



A

Soft tether

Central Actuator (CA)

Suction Cavity (SC)

Rear Anchoring Actuator (RAA)

Front Anchoring Actuator (FAA)



B

P_N0

P_N1

P_N2

P_N3

P_N4

P_{N5}

P_N6

R-AA
CA
F-AA

R-AA
CA ←
F-AA

R-AA
CA
F-AA

R-AA
CA
F-AA

R-AA
CA
F-AA

R-AA
CA
F-AA

K-AA
CA —
F-AA

Iteration

Soft robotic origami-inspired colonoscope

A new inchworm-like robot: soon on the screen

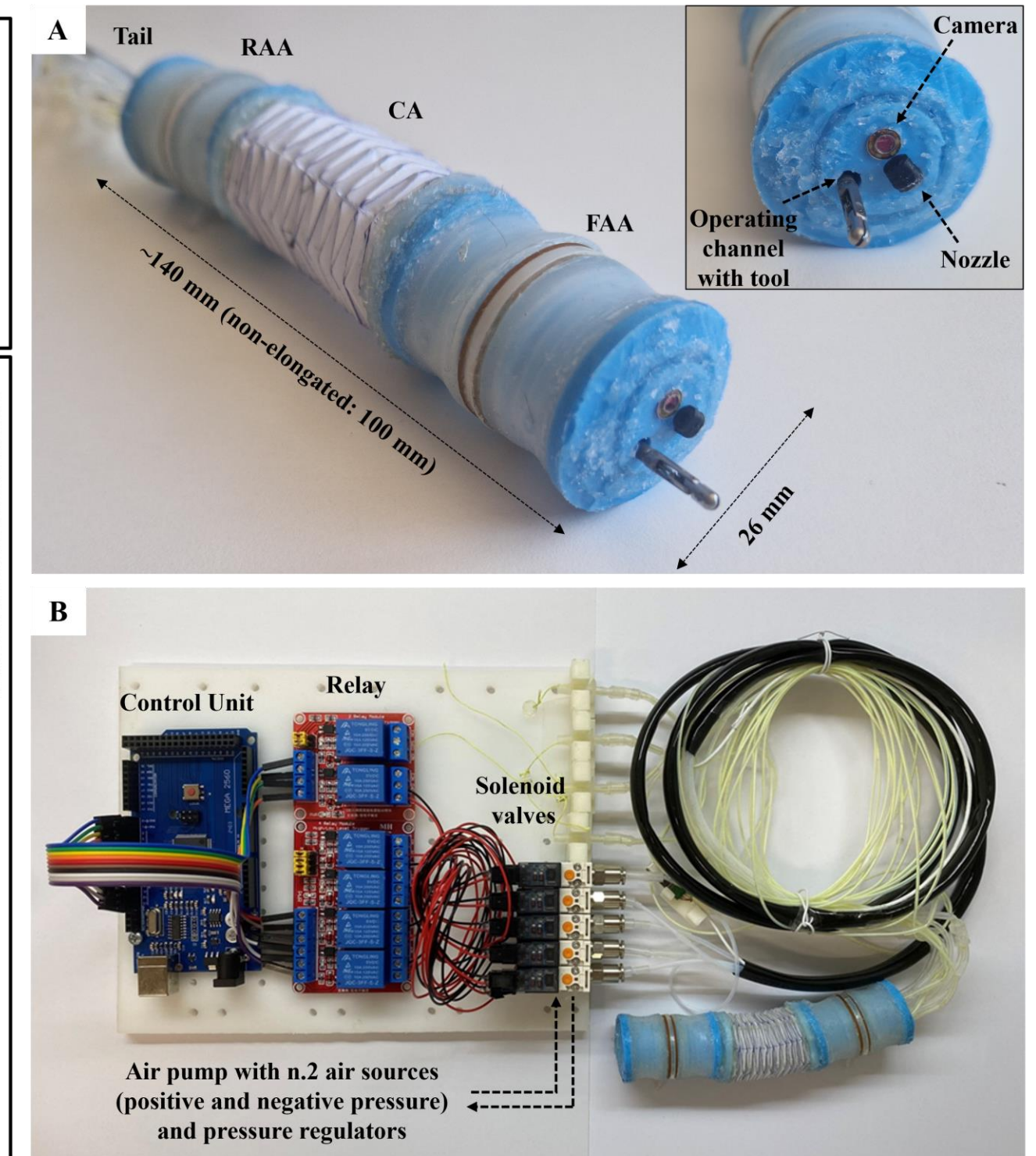
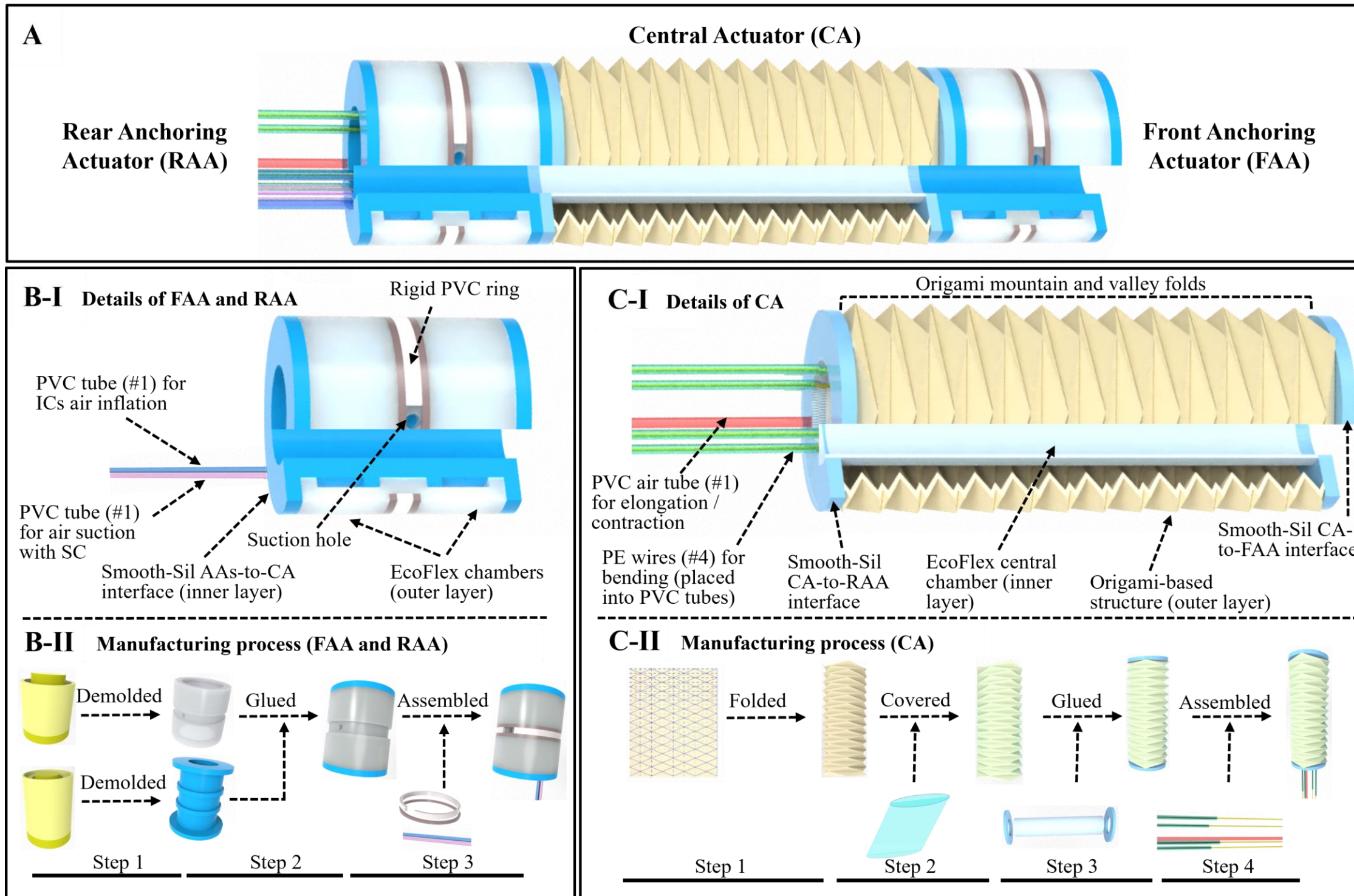
PAPER

COMING
SOON



PATENTED

SUBMITTED



[A]: ROSE navigation in a rigid (*i.e.*, non-deformable) straight tube with variable diameters

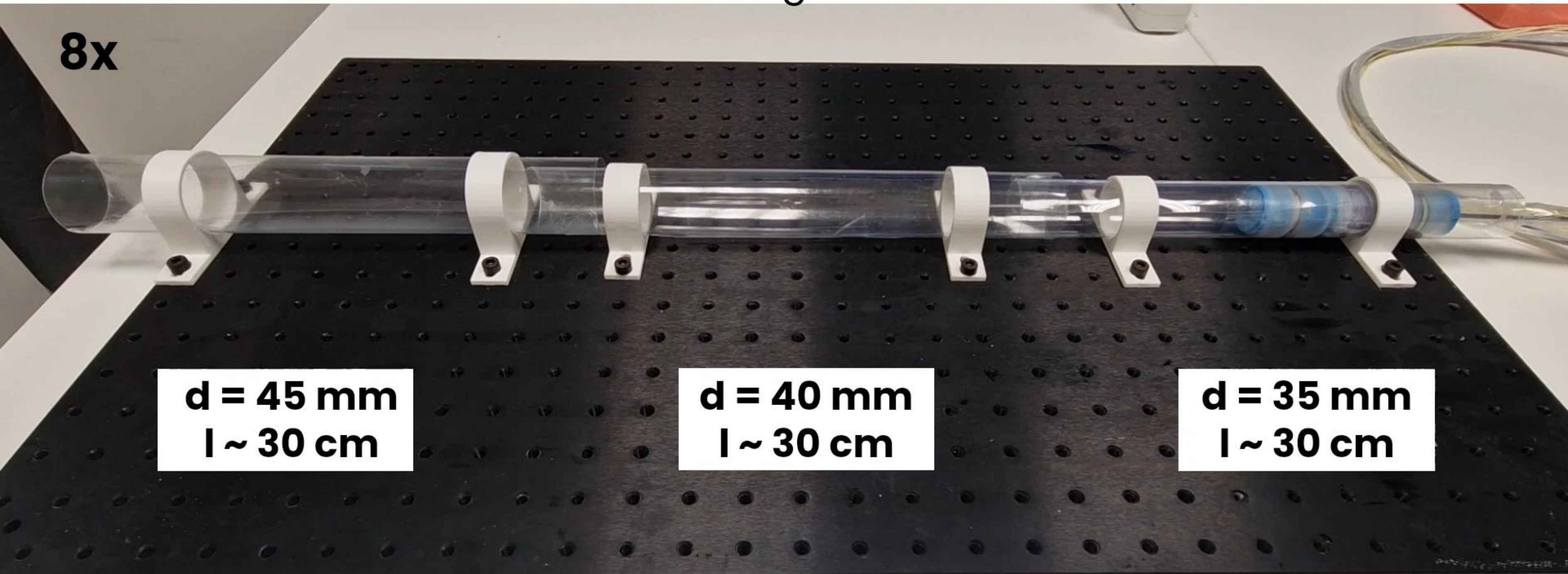
ROSE's adaptability to different diameters (d) was tested using transparent rigid tubes, each segment measuring approximately 30 cm in length, for a total tube length of ~90 cm

8x

$d = 45 \text{ mm}$
 $l \sim 30 \text{ cm}$

$d = 40 \text{ mm}$
 $l \sim 30 \text{ cm}$

$d = 35 \text{ mm}$
 $l \sim 30 \text{ cm}$



L'IMPORTANZA DEL MICROBIOTA ATTRAVERSO I NUMERI

90%

Il 90% di tutte le malattie può coinvolgere in qualche modo l'intestino e lo stato di salute del microbiota.

>10.000

Differenti specie microbiche identificate nel corpo umano.

100 a 1

Rapporti tra i geni del microbiota e quelli umani.

3.3 milioni

Il numero di geni non ridondanti codificati dal microbiota intestinale.

99,9%

Percentuale di individui con il genoma dell'ospite identico.

100 miliardi di miliardi

Il numero di microrganismi simbiotici ospitati da ciascun individuo, fondamentalmente sono batteri intestinali, che predominano il microbiota umano.

10x

Il numero degli microrganismi ospiti del corpo umano è 10 volte le cellule umane.

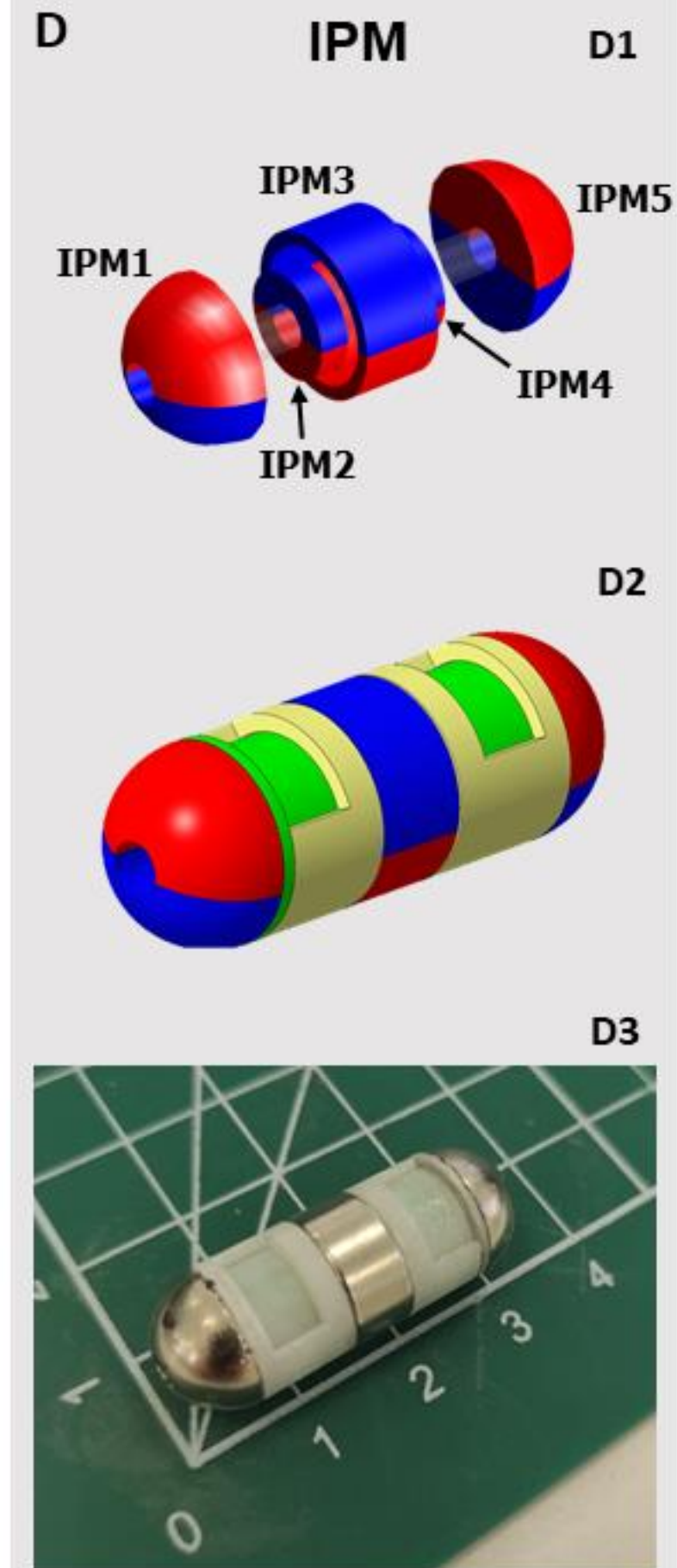
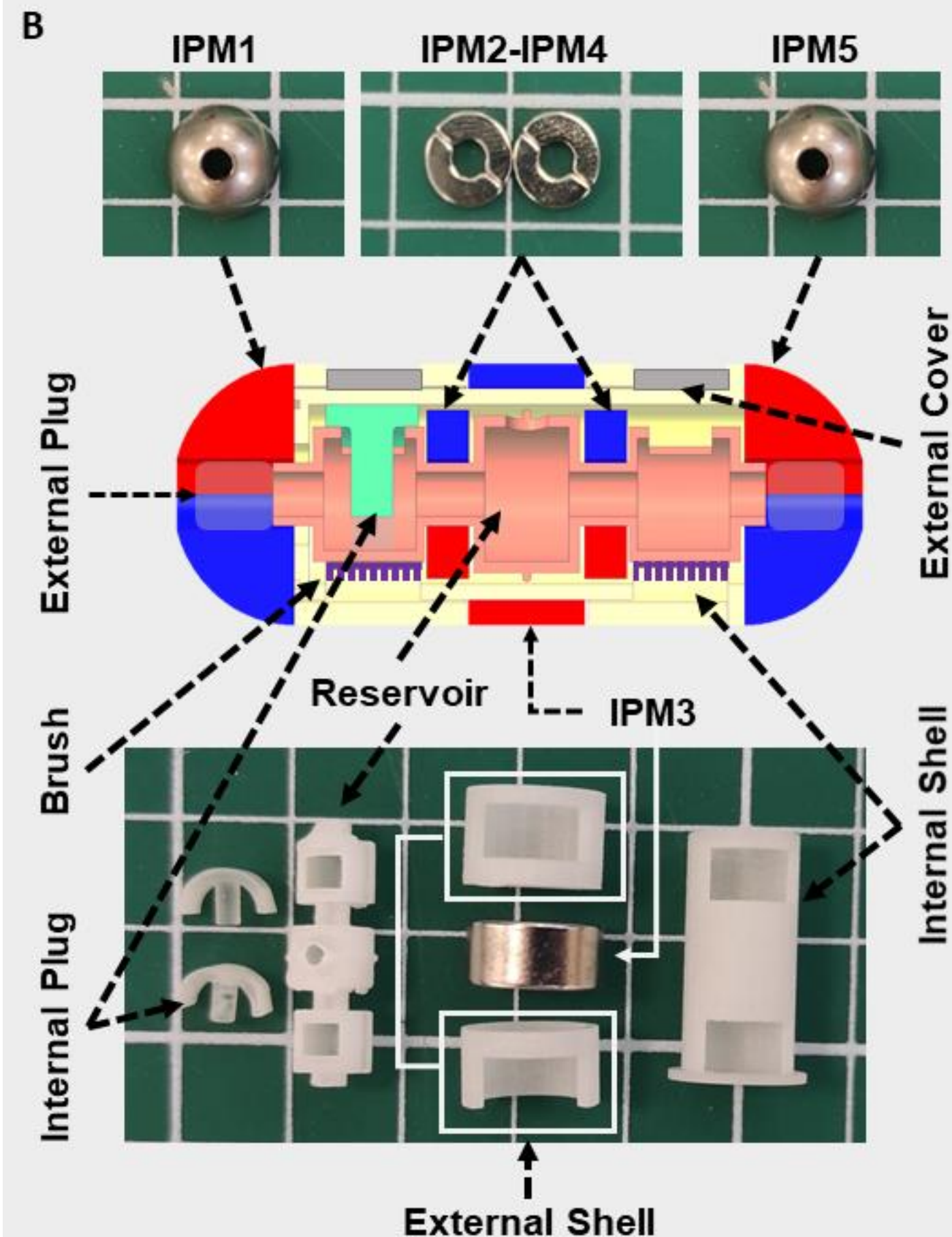
22,000

Numero di geni presenti nel genoma umano.

80% - 90%

Percentuale di individui con differente espressione genica del microbiotica.

MECCA: Microbiota Endoscopic Capsule with Compliant Activation





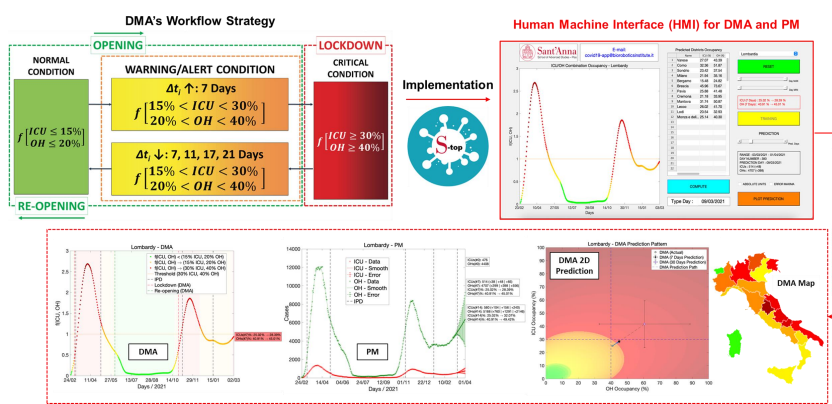


Healthcare Mechatronics Lab

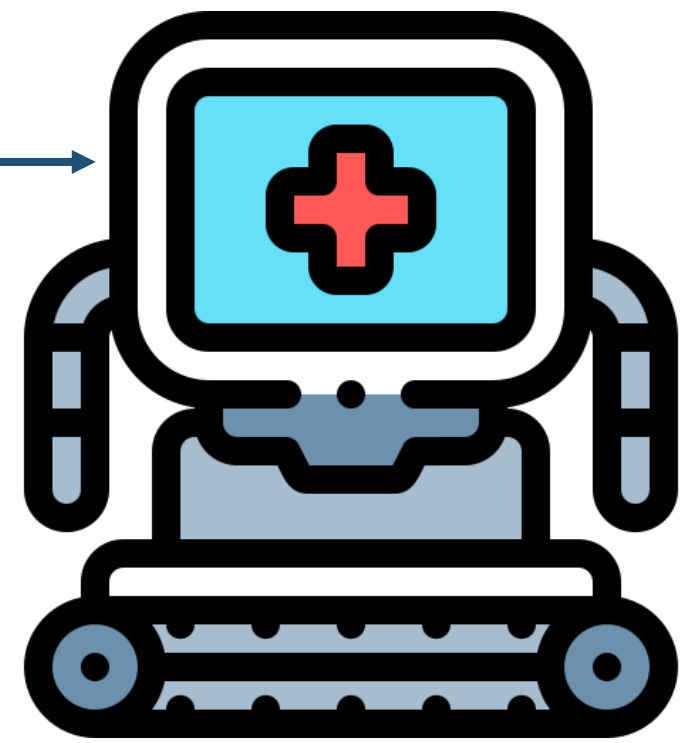
Endoluminal Robotics and Healthcare Technologies

Intelligent and Integrated **Macro-to-Meso Scale Endoluminal Robots** and **Biomedical Technologies** for Human-Centred Longitudinal Healthcare

Healthcare Decision Support Algorithms



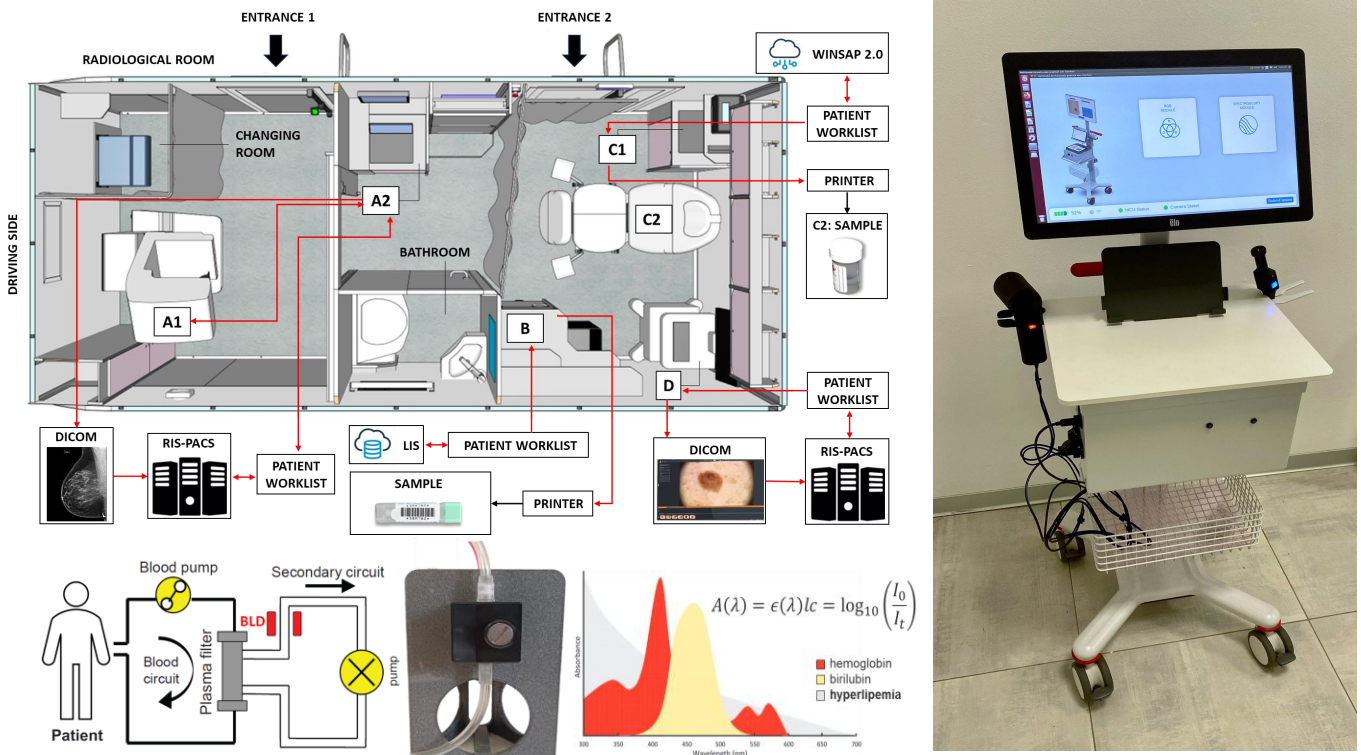
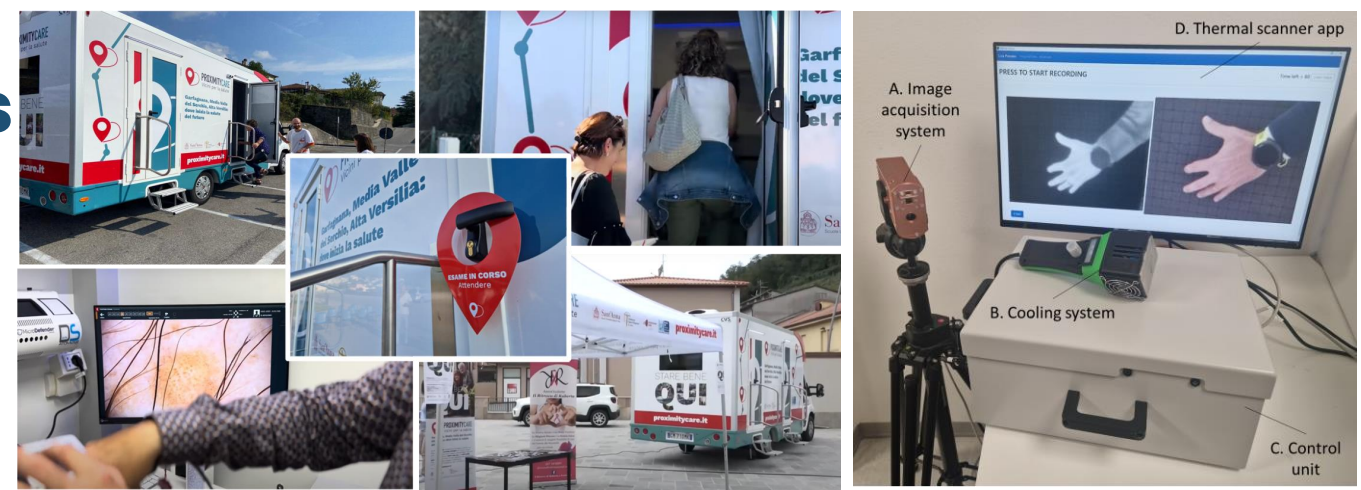
Healthcare Technologies



Hospital Logistics and Monitoring



Screening and Diagnosis

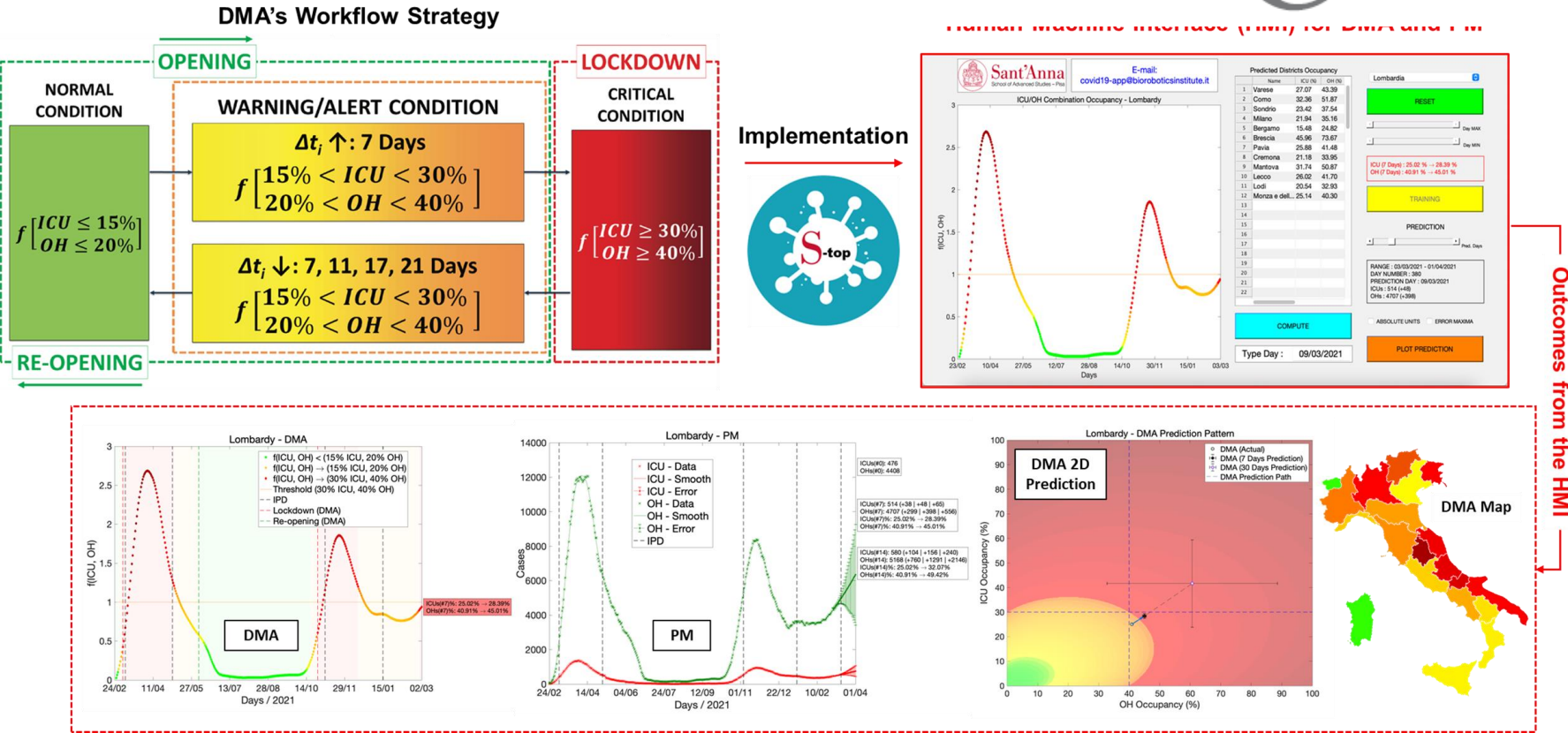
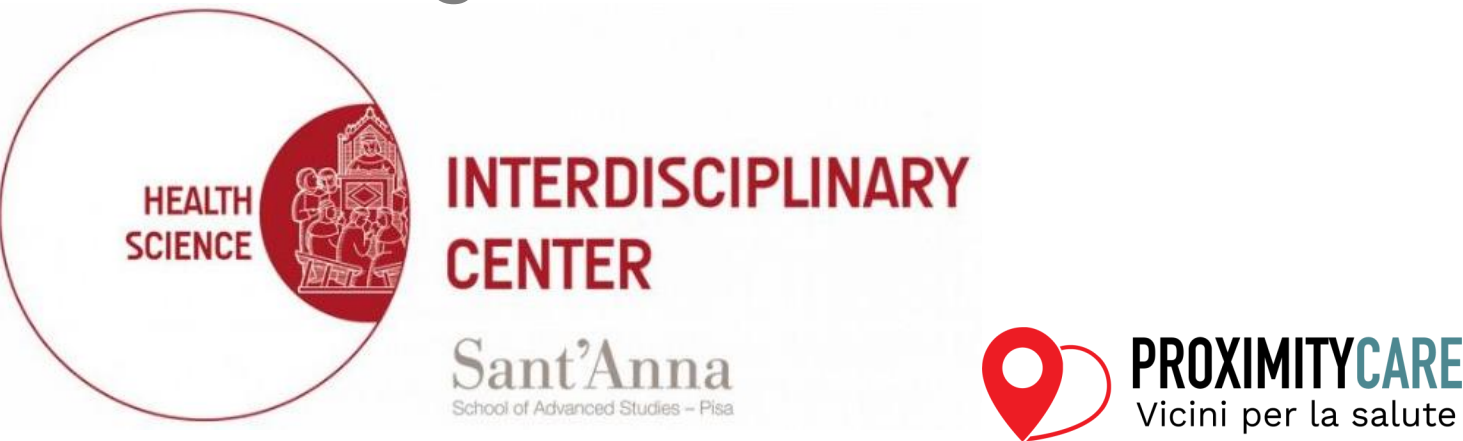


Prof. Gastone Ciuti (gastone.ciuti@santannapisa.it)

Collaborations with other Institutes and Interdisciplinary Centers (recent examples)

eHealth Technologies

Digital Platforms and Machine Learning-based Models for Healthcare



Mainardi et al. "A Proximity Care Paradigm for Cancer Screening with a Mobile Multi-Screening Unit in Inner Areas." IEEE PerCom Conference, France 2024



A. Damone et al., "Decision-Making Algorithm and Predictive Model to Assess the Impact of Infectious Disease Epidemics on the Healthcare System:...", in IEEE JBHI 2022





PROXIMITYCARE
Vicini per la salute

**Garfagnana, Media Valle
del Serchio, Alta Versilia:
dove inizia la salute
del futuro**

Sant'Anna
Servizio di Diagnostica e Cura

Azienda USL Toscana Nord Ovest
Servizio Sanitario della Toscana

proximitycare.it

CVS

PROXIMITYCARE
Vicini per la salute

PROXY SCREENING
effettuato
presso la
Villa Colonna
di Massa Marittima
in 11 giorni di permanenza

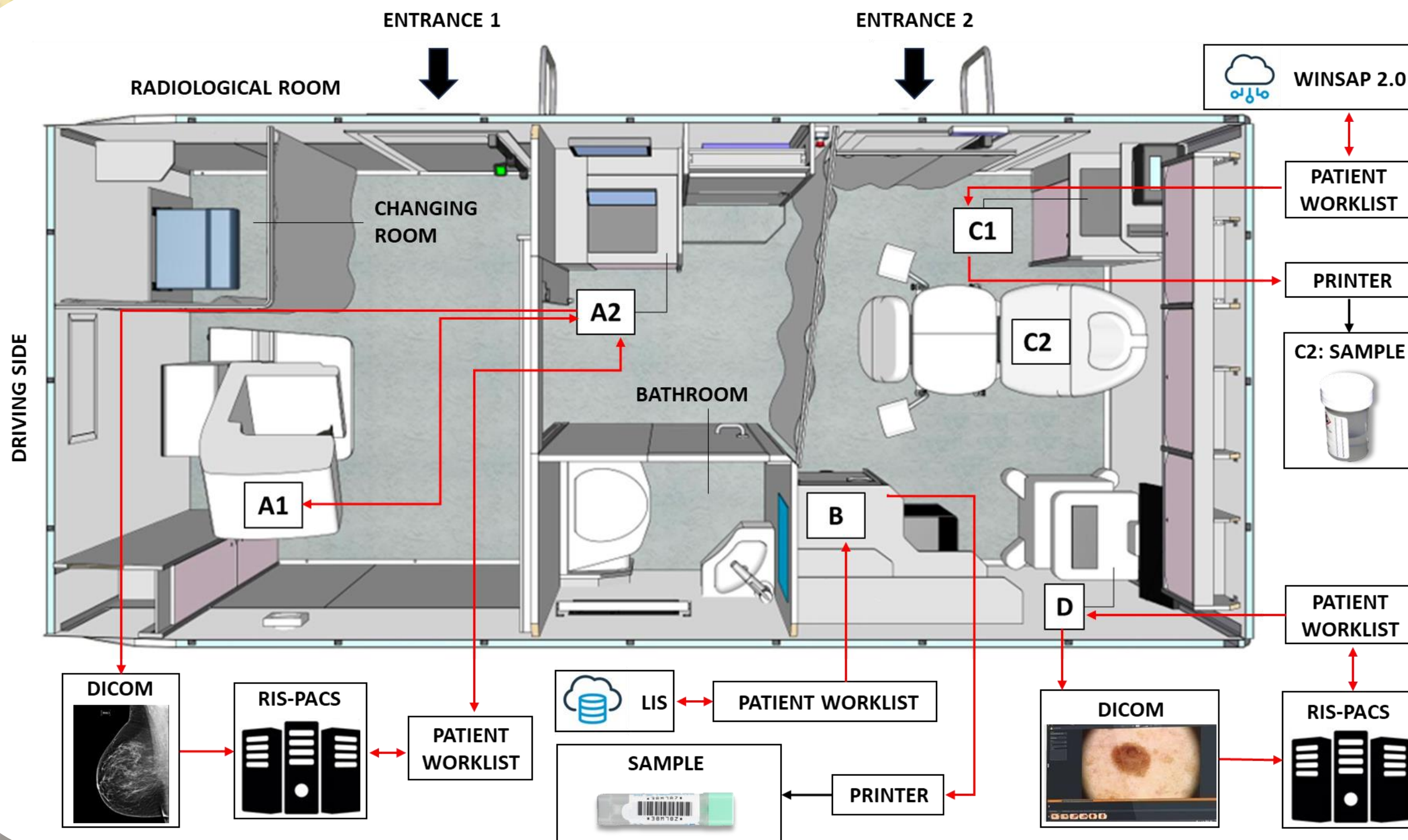


PROXIMITYCARE
Vicini per la salute

FIAT

GM-710MK

Proxy Screening



CVS
Design



HEALTH SCIENCE
INTERDISCIPLINARY CENTER
Sant'Anna
Scuola Universitaria Superiore Pisa

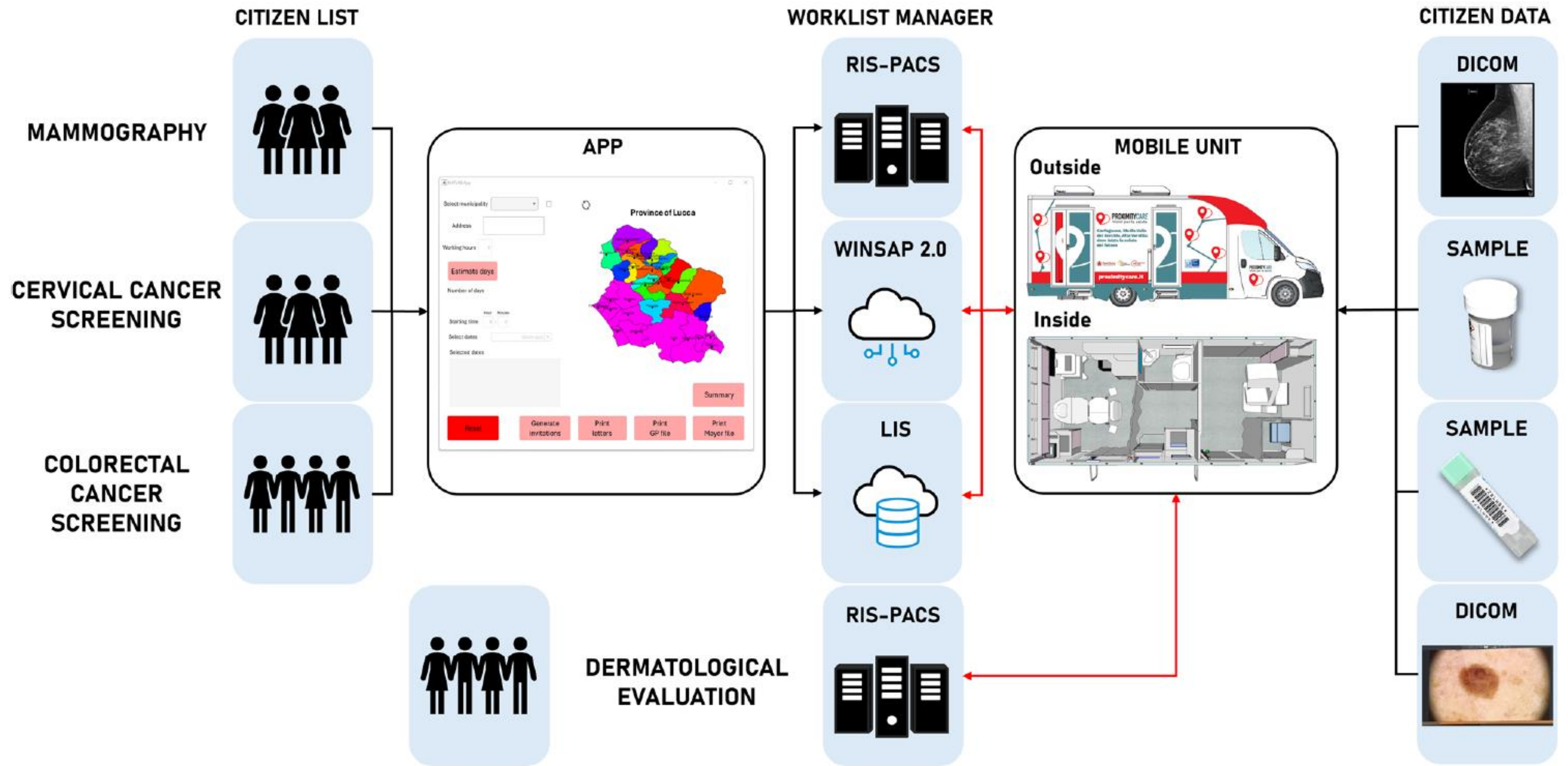


PROXIMITYCARE
Vicini per la salute

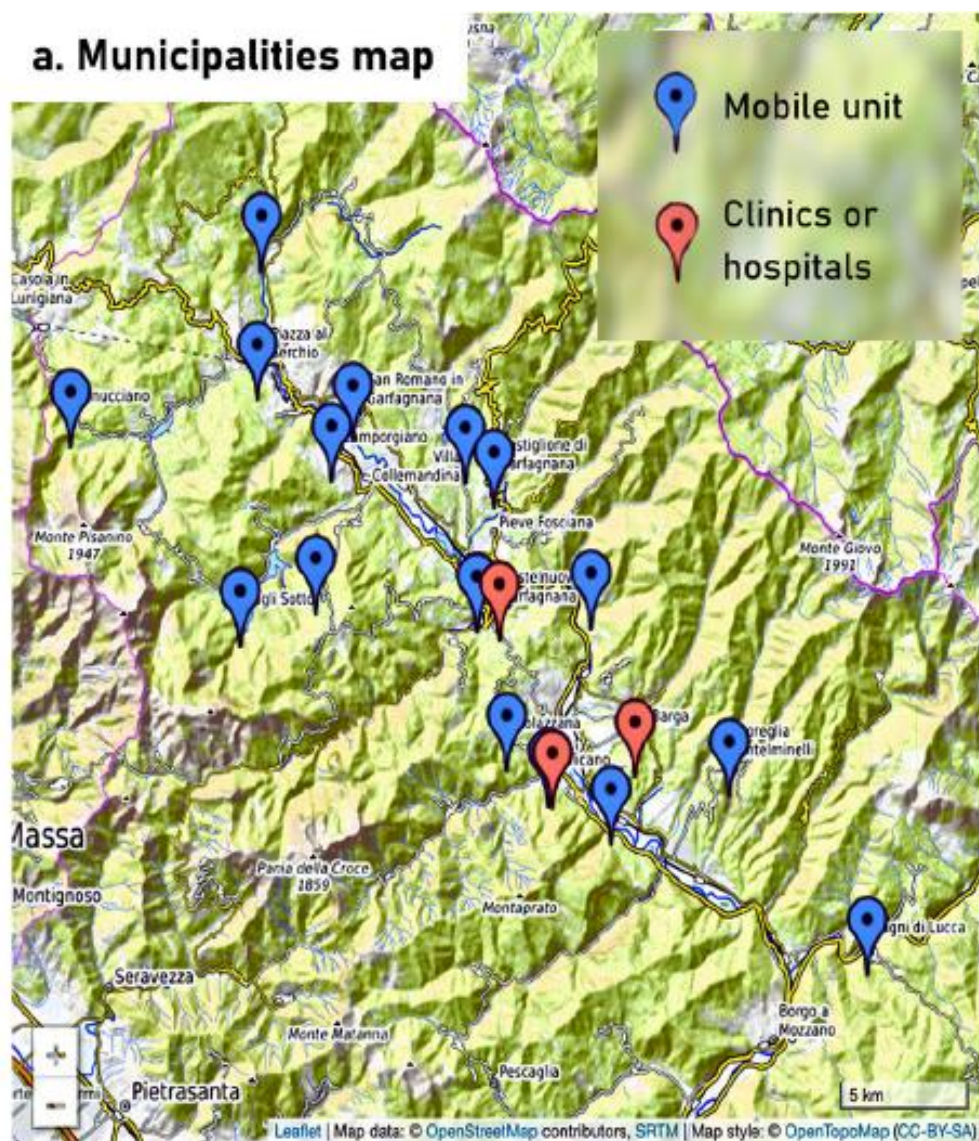
Project realised with the contribution of
Fondazione Cassa di Risparmio di Lucca



Fondazione
Cassa di Risparmio
di Lucca



"Orchestratore" (multi-screening invitations)



a - Select municipality from the map

Select municipality:

Address:

Working hours:

Estimate days:

Number of days:

Starting time:

Select dates:

Selected dates:

Province of Lucca

Municipality

Summary

Reset Generate invitations Print letters Print GP file Print Mayor file

b - Check address and estimate days

Select municipality:

Address:

Working hours:

Estimate days:

Number of days:

Starting time:

Select dates:

Selected dates:

Province of Lucca

Summary

Reset Generate invitations Print letters Print GP file Print Mayor file

c - Select dates for screening

Select municipality:

Address:

Working hours:

Estimate days:

Number of days:

Starting time:

Select dates:

Selected dates:

Province of Lucca

Summary

Reset Generate invitations Print letters Print GP file Print Mayor file

d - Generate invitations and related files

Select municipality:

Address:

Working hours:

Estimate days:

Number of days:

Starting time:

Select dates:

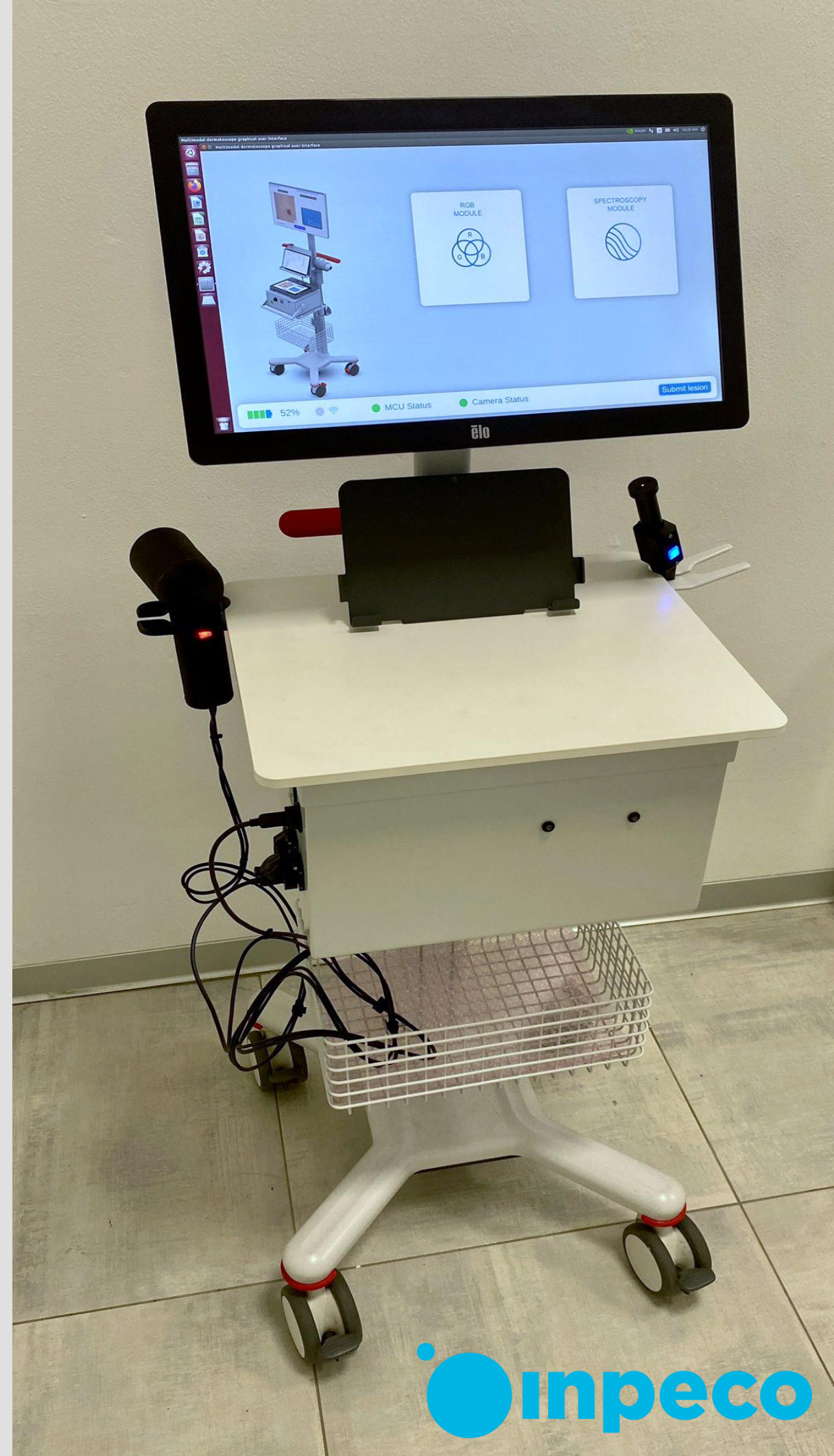
Selected dates:

Province of Lucca

Summary

Reset Generate invitations Print letters Print GP file Print Mayor file

MULTI-MODAL DERMATOLOGY SYSTEM

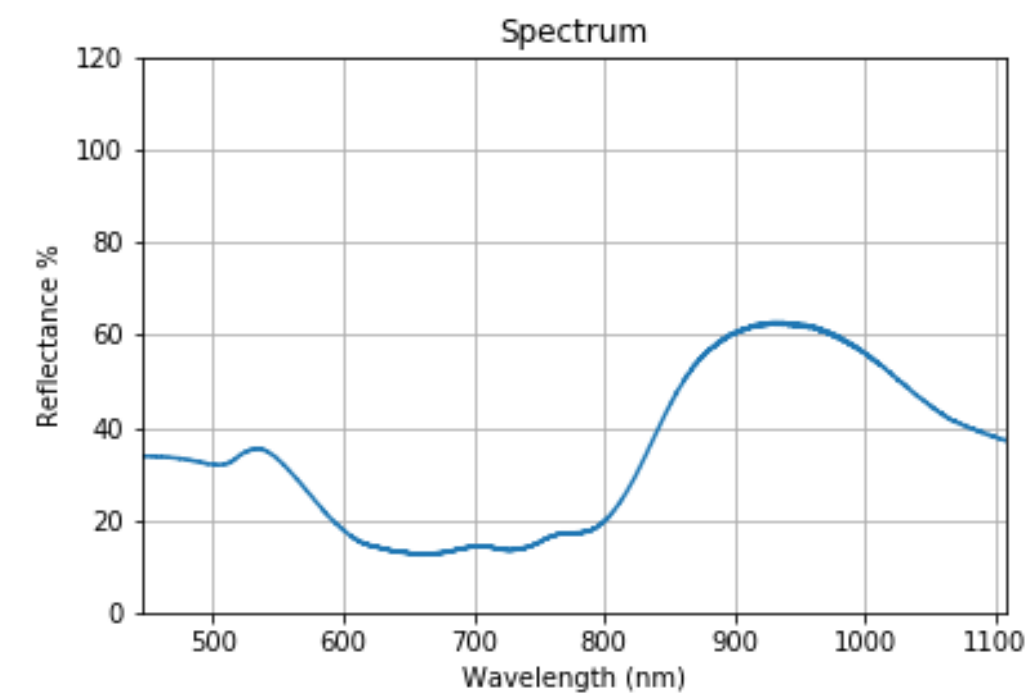


1



Dermatoscopic image

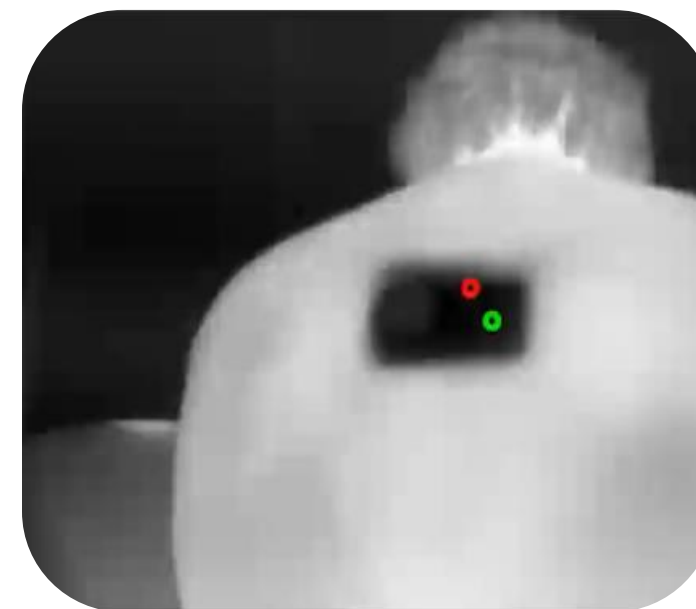
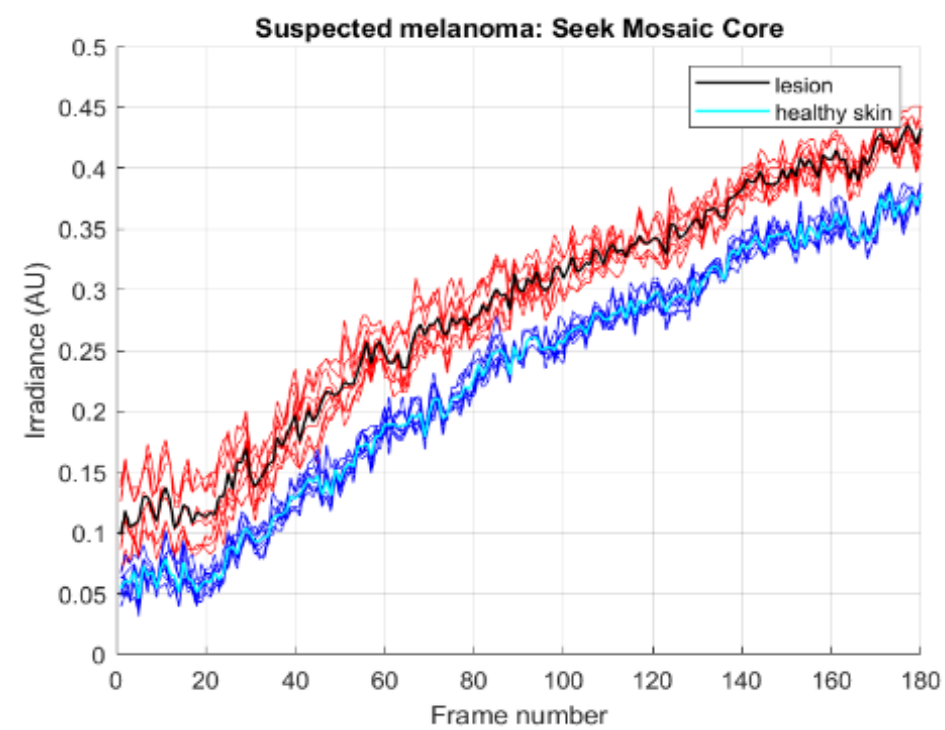
2



Reflectance spectrum



3



Skin Thermal recovery



OUTLINE

- GENERAL OVERVIEW ON INTEGRATED/INTELLIGENT ROBOTICS IN HEALTHCARE
- ROBOTIC TECHNOLOGIES: KEY-FACTORS TO BUILD A HEALTHCARE ROBOT, MACRO-TRENDS AND OPEN ISSUES
- EXAMPLES OF HEALTHCARE TECHNOLOGIES AND APPLICATIONS WITH IOT FUNCTIONALITIES



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June 13, 1961

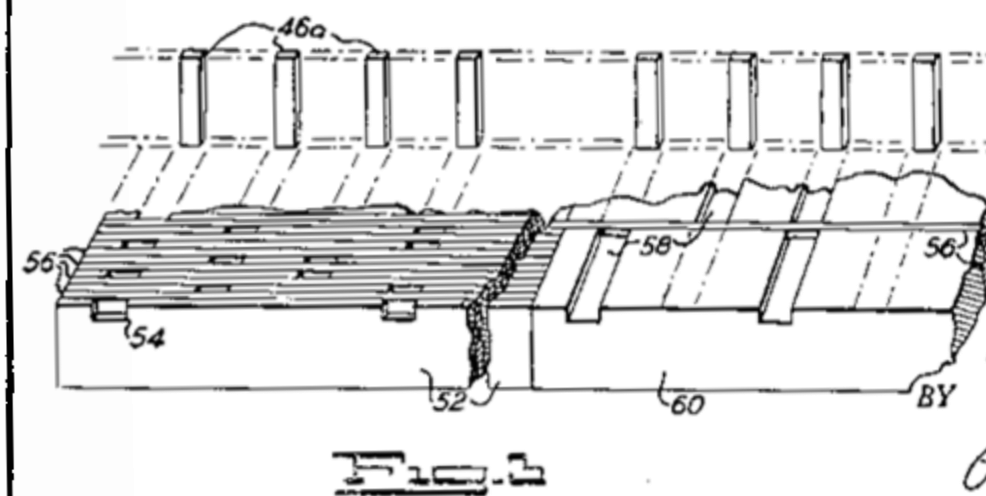
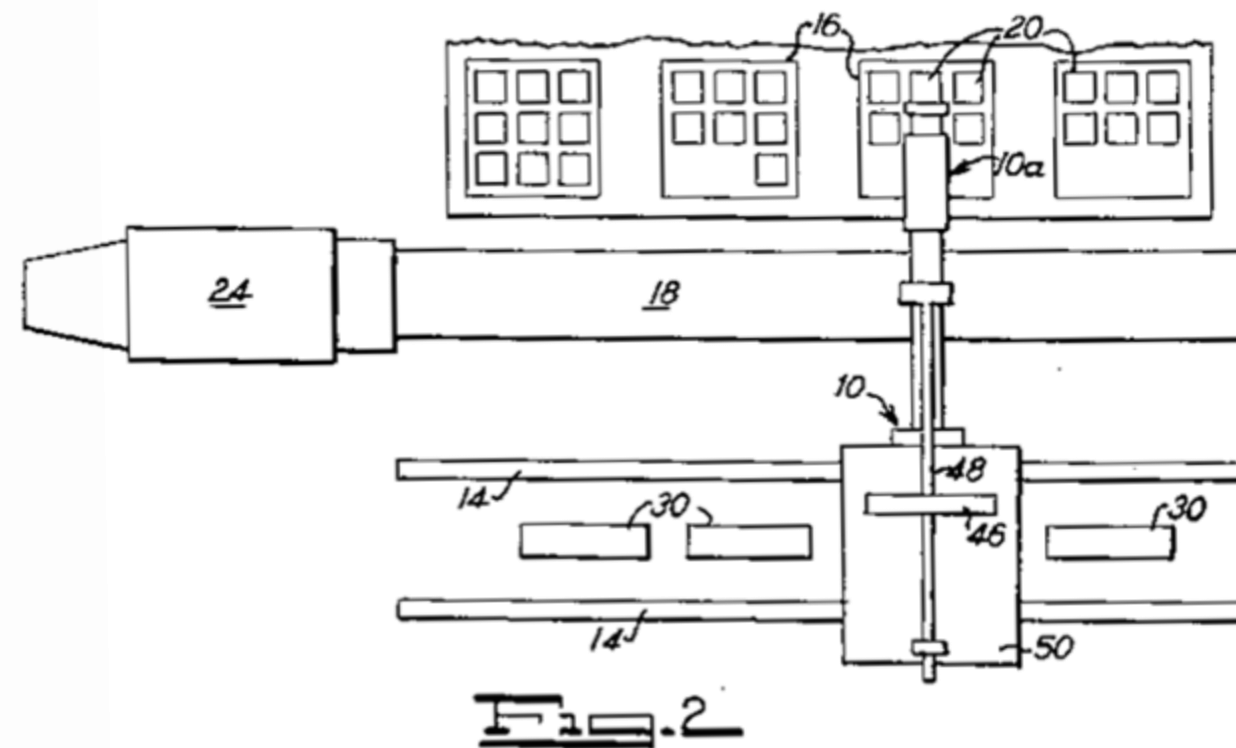
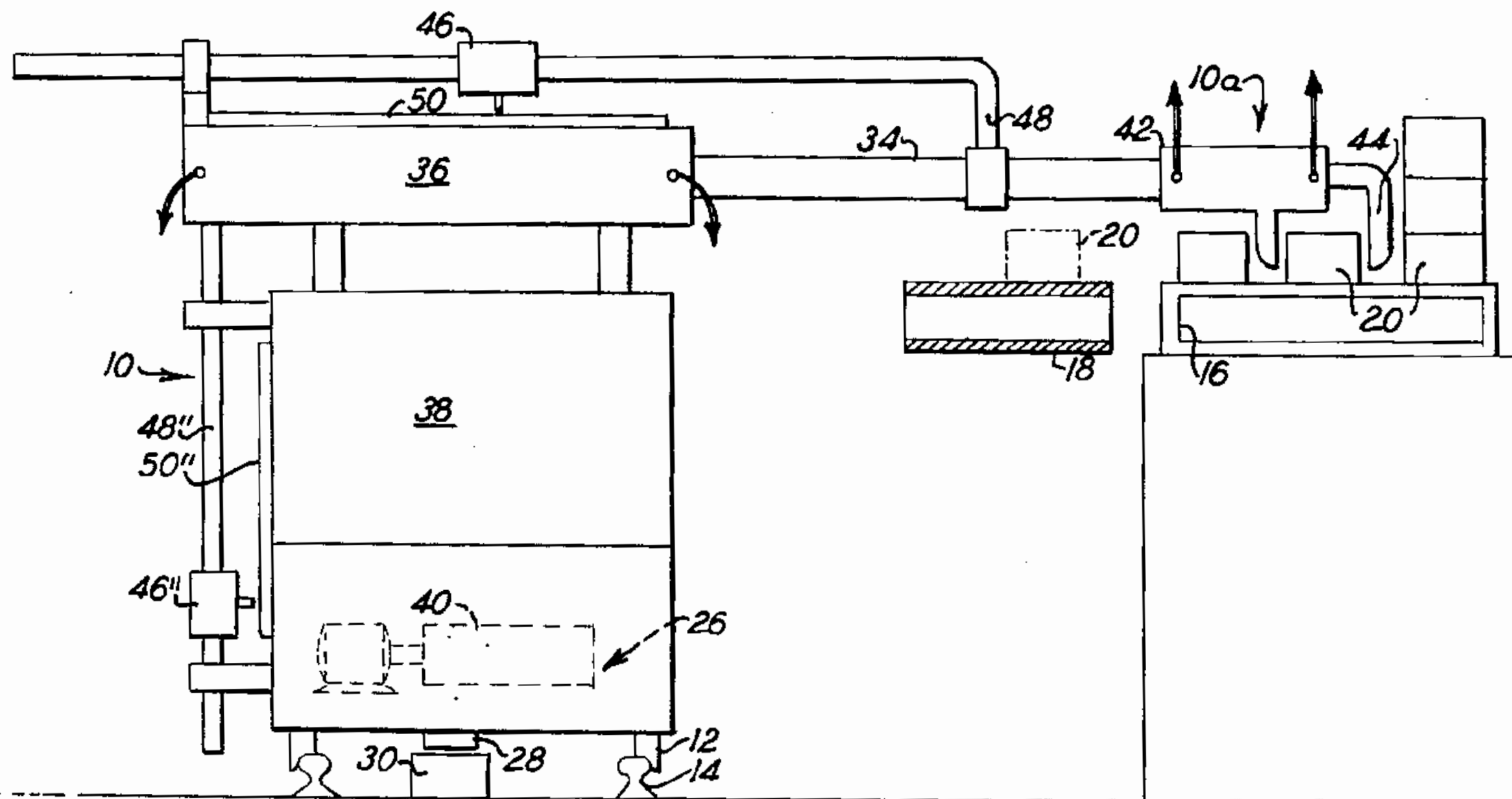
G. C. DEVOL, JR

2,988,237

PROGRAMMED ARTICLE TRANSFER

Filed Dec. 10, 1954

3 Sheets-Sheet 1



INVENTOR.
GEORGE C. DEVOL, JR.

BY
Paul S. Martin
ATTORNEY

1962 – Unimation Inc. (Danbury, USA)



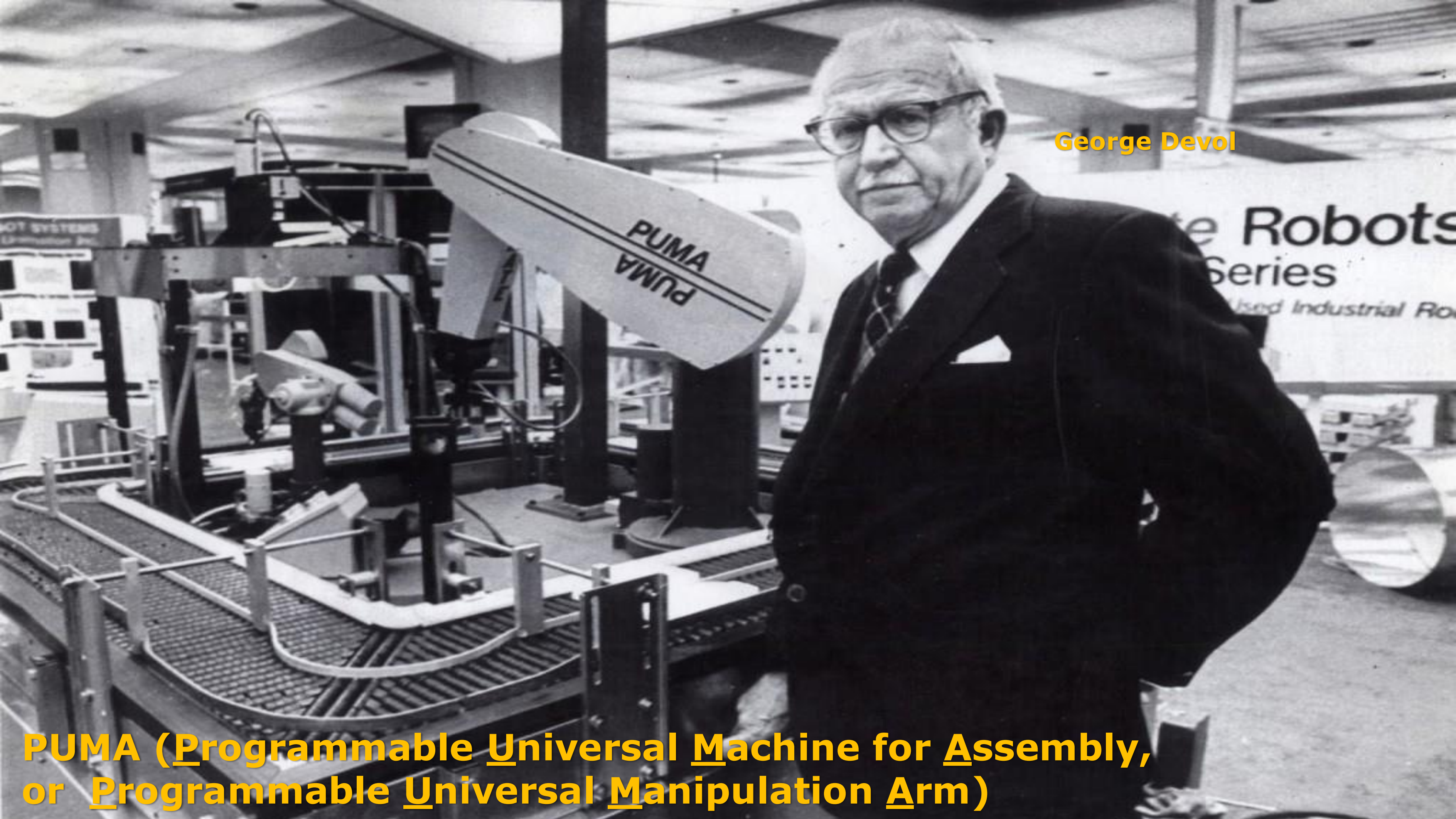
Joseph Engelberger

George Devol



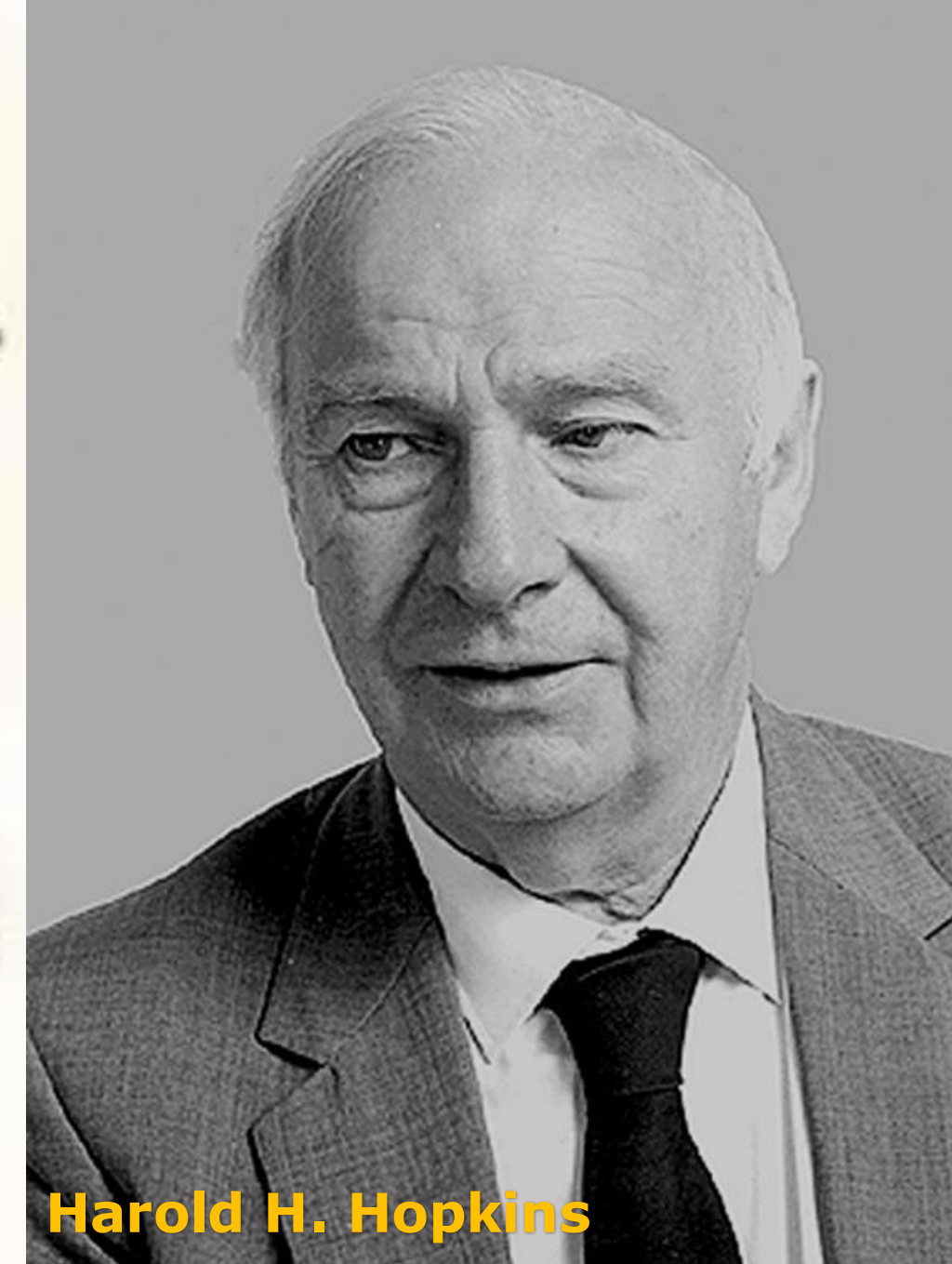
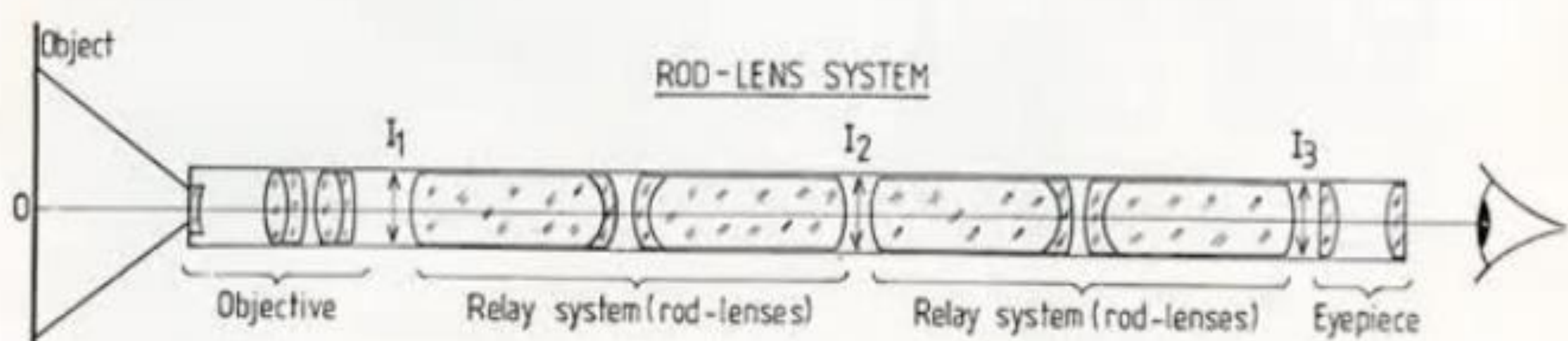
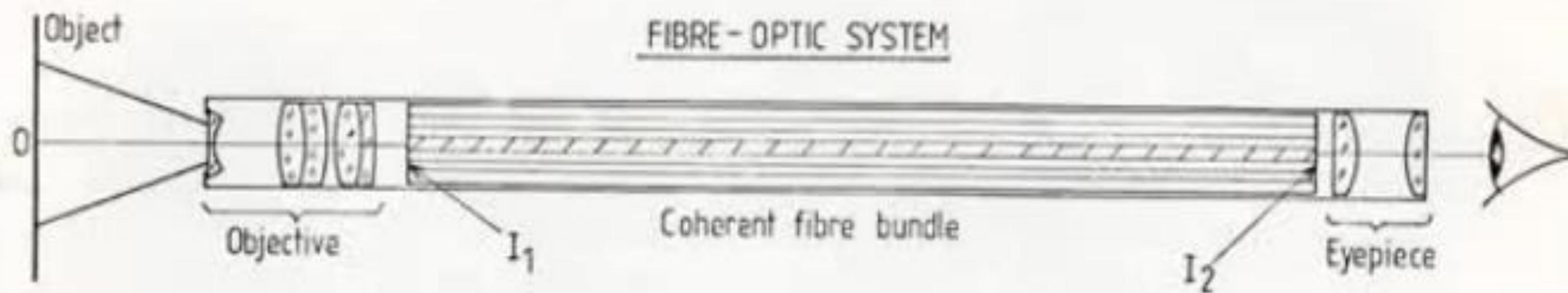
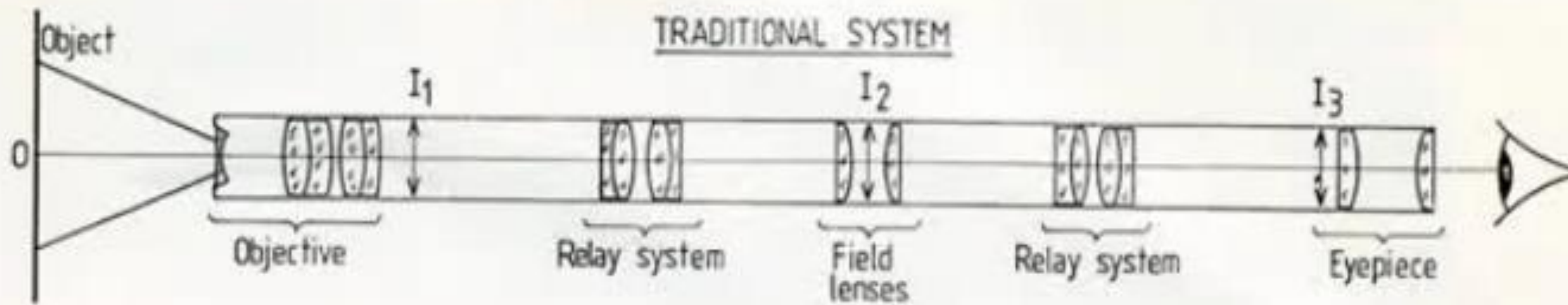
Unimate robot (General Motors in Trenton, New Jersey)





George Devol

**PUMA (Programmable Universal Machine for Assembly,
or Programmable Universal Manipulation Arm)**



Harold H. Hopkins

1950-ish: Harold H. Hopkins invented optical fibres and the cylindrical lens

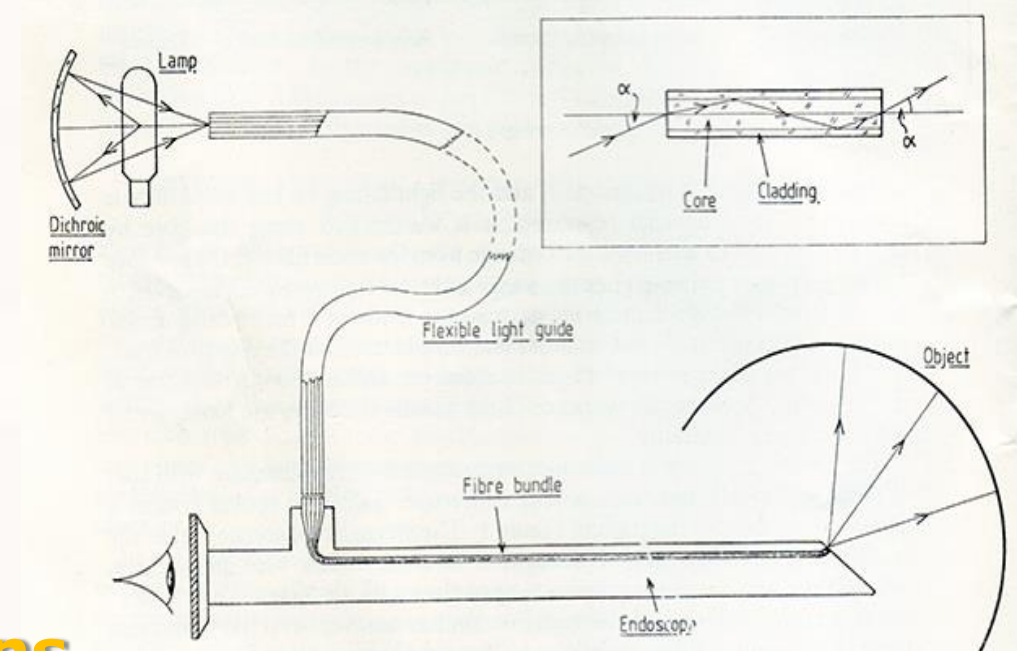
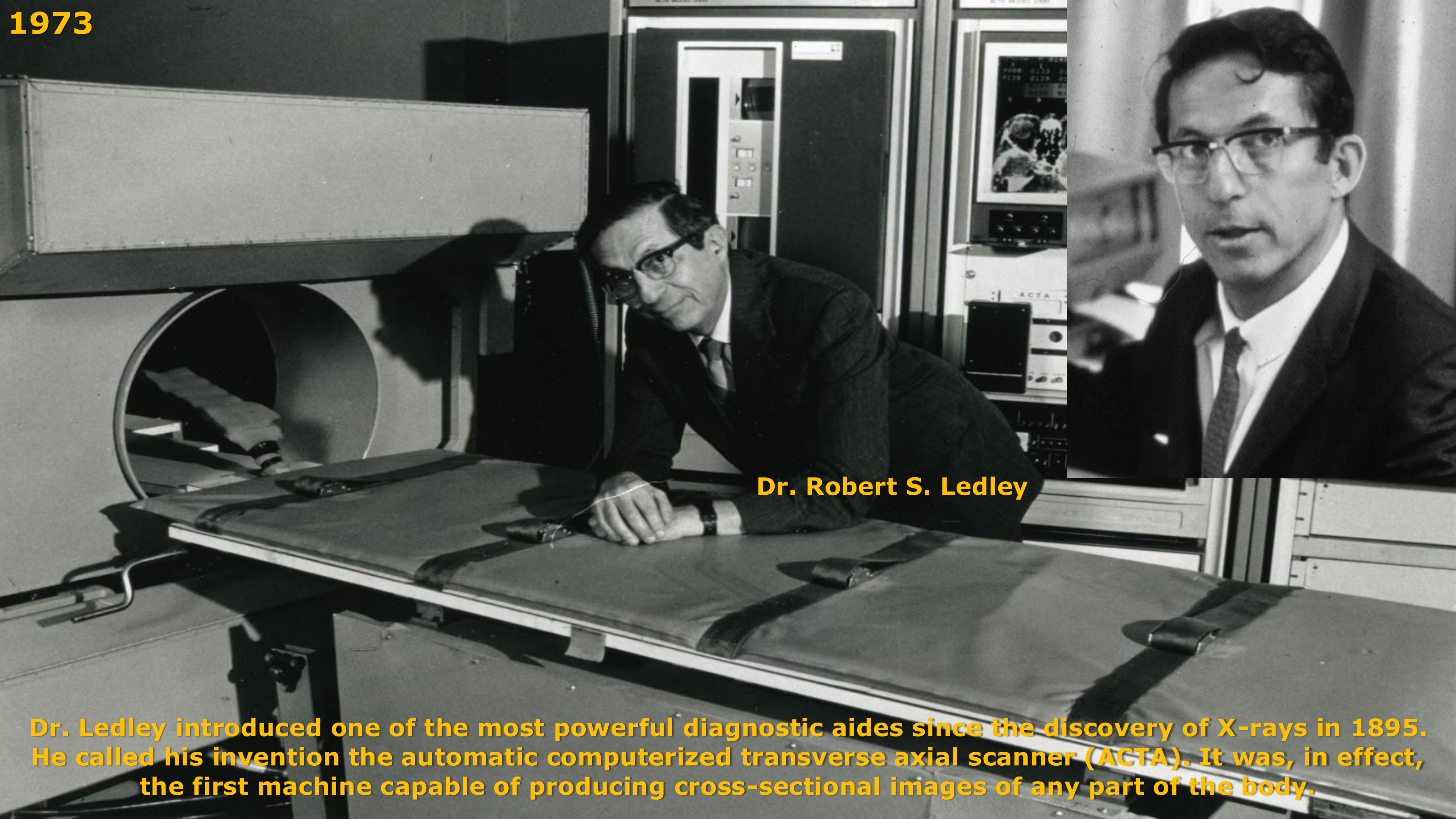


Fig. 2. Fibre-optic illumination system.

1973



Dr. Robert S. Ledley

Dr. Ledley introduced one of the most powerful diagnostic aides since the discovery of X-rays in 1895. He called his invention the automatic computerized transverse axial scanner (ACTA). It was, in effect, the first machine capable of producing cross-sectional images of any part of the body.

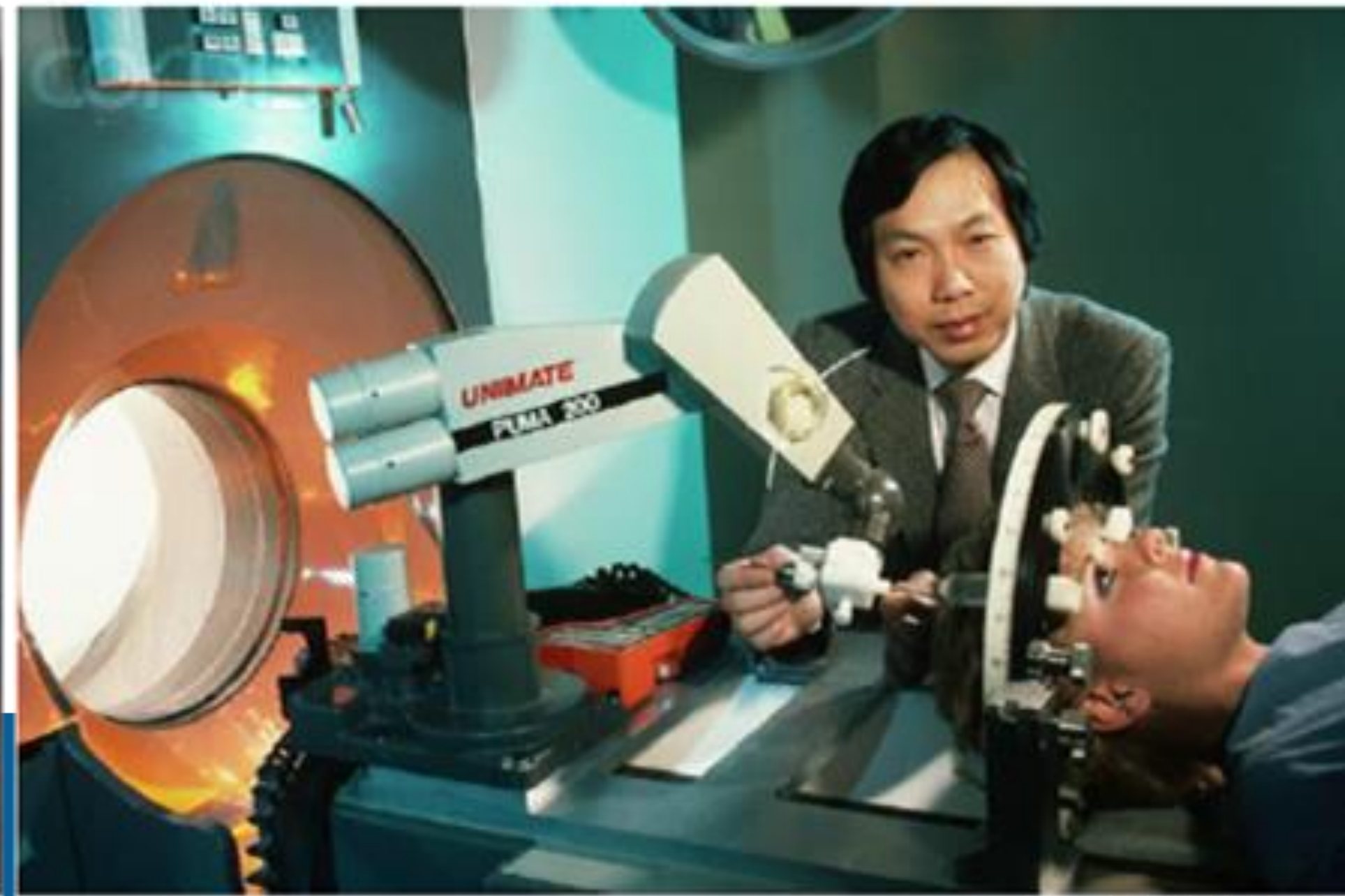
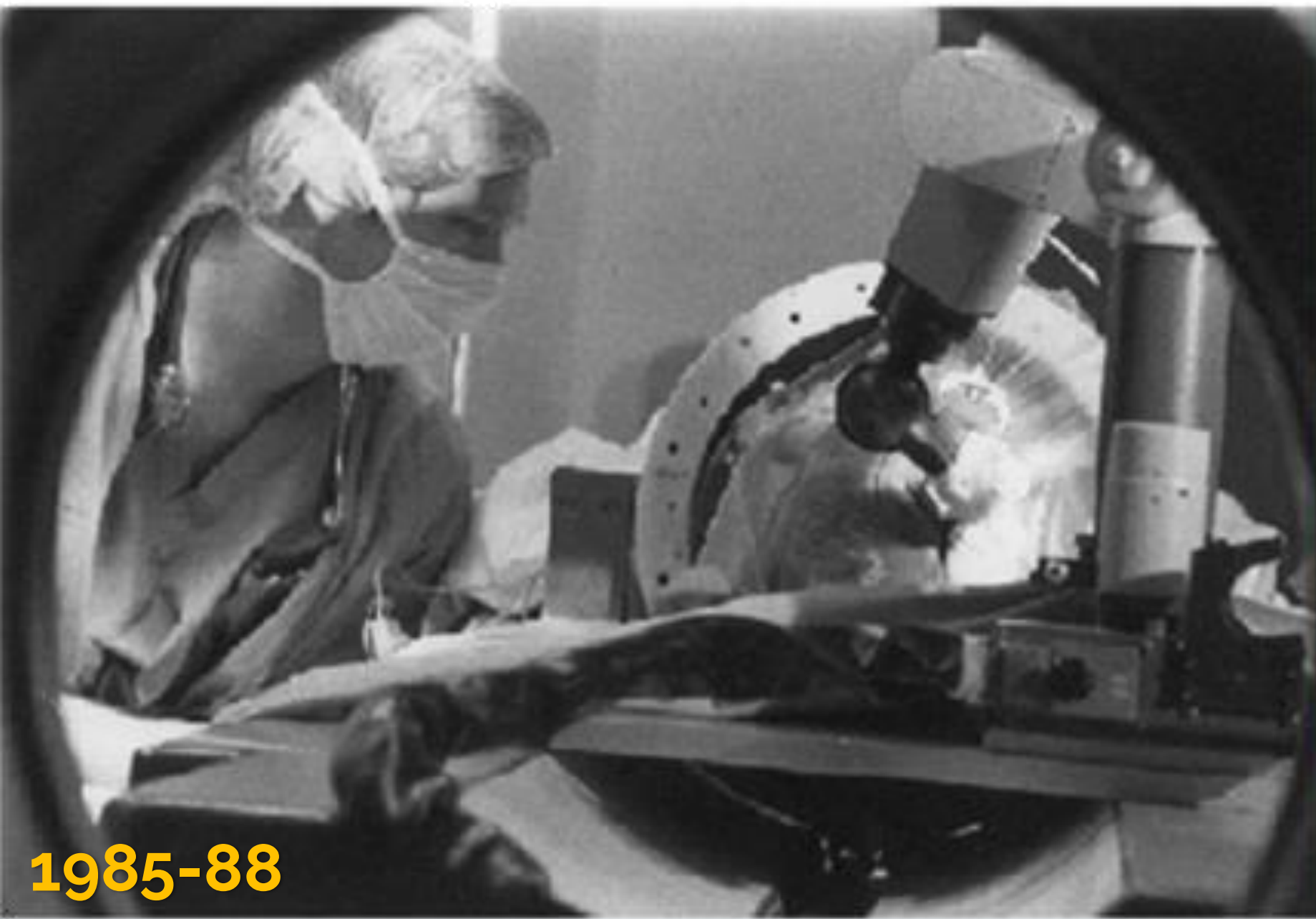
A Robot with Improved Absolute Positioning Accuracy for CT Guided Stereotactic Brain Surgery

YIK SAN KWOH, MEMBER, IEEE, JOAHIN HOU, EDMOND A. JONCKHEERE, SENIOR MEMBER, IEEE, AND SAMAD HAYATI

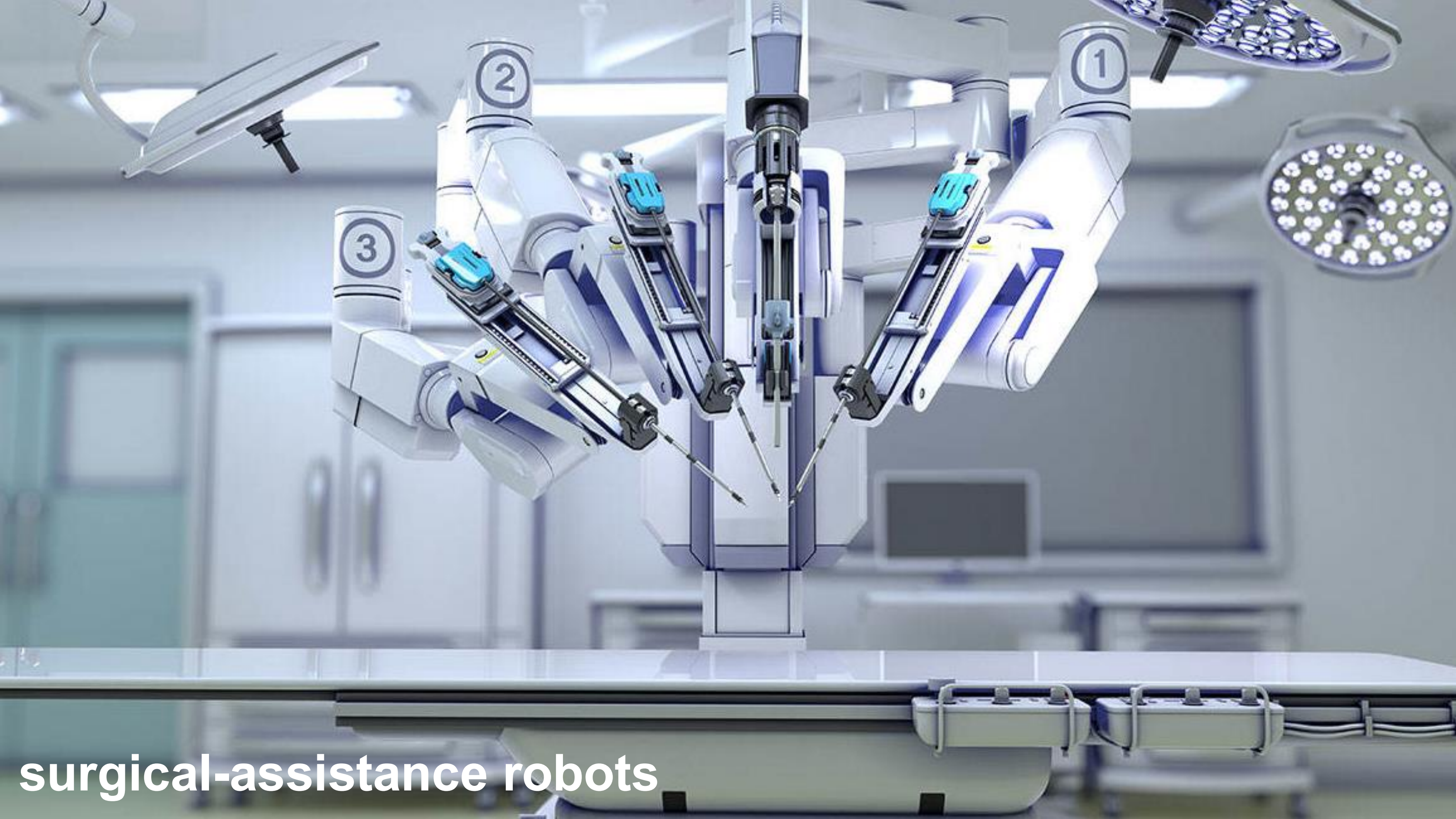
Abstract—In this paper, we describe how a Unimation Puma 200 robot, properly interfaced with a CT scanner and with a probe guide mounted at its end effector, can be used for CT-guided brain tumor biopsies. Once the target is identified on the CT picture, a simple command allows the robot to move to a position such that the end effector probe guide points towards the target. This results in a procedure faster than one with a manually adjustable frame. But probably the most important advantage, as we show in this paper, is the improved accuracy that can be reached by proper calibration of the robot.

The basic process of stereotactic neurosurgery has been used both clinically and as a research tool for nearly 80 years, following the pioneering work of Sir Victor Horsley and Robert Clarke [1] in Queen's Square Hospital, London, England, in the first decade of this century. Horsley and Clarke developed the first 3-D stereotactic apparatus. The Clarke apparatus [2] was an adjustable mechanical structure to be affixed to the cranium of an

Abstract—In this paper, we describe how a Unimation Puma 200 robot, properly interfaced with a CT scanner and with a probe guide mounted at its end effector, can be used for CT-guided brain tumor biopsies. Once the target is identified on the CT picture, a simple command allows the robot to move to a position such that the end effector probe guide points towards the target. This results in a procedure faster than one with a manually adjustable frame. But probably the most important advantage, as we show in this paper, is the improved accuracy that can be reached by proper calibration of the robot.



1985-88

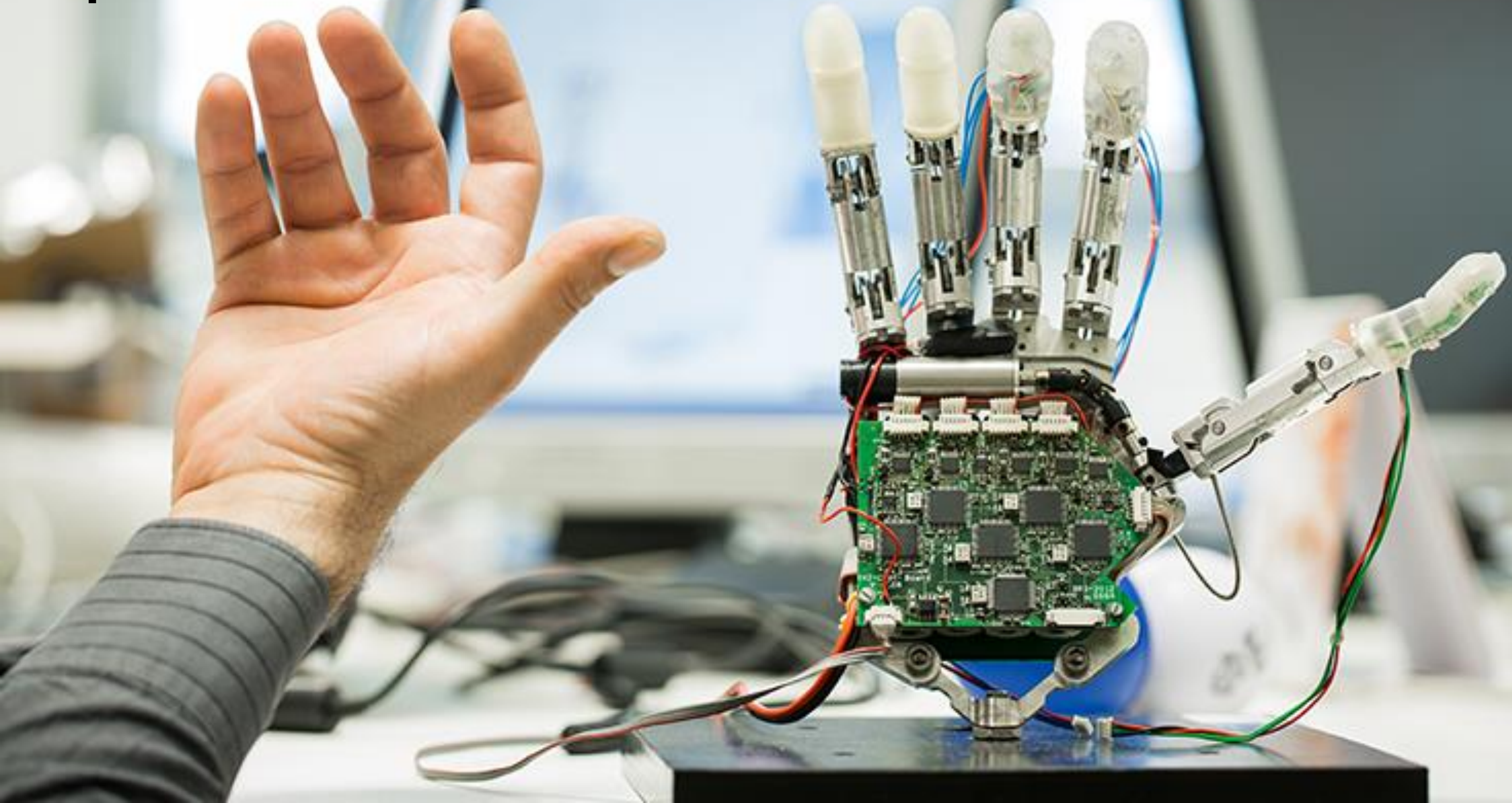


surgical-assistance robots



wearable robots

implantable robots





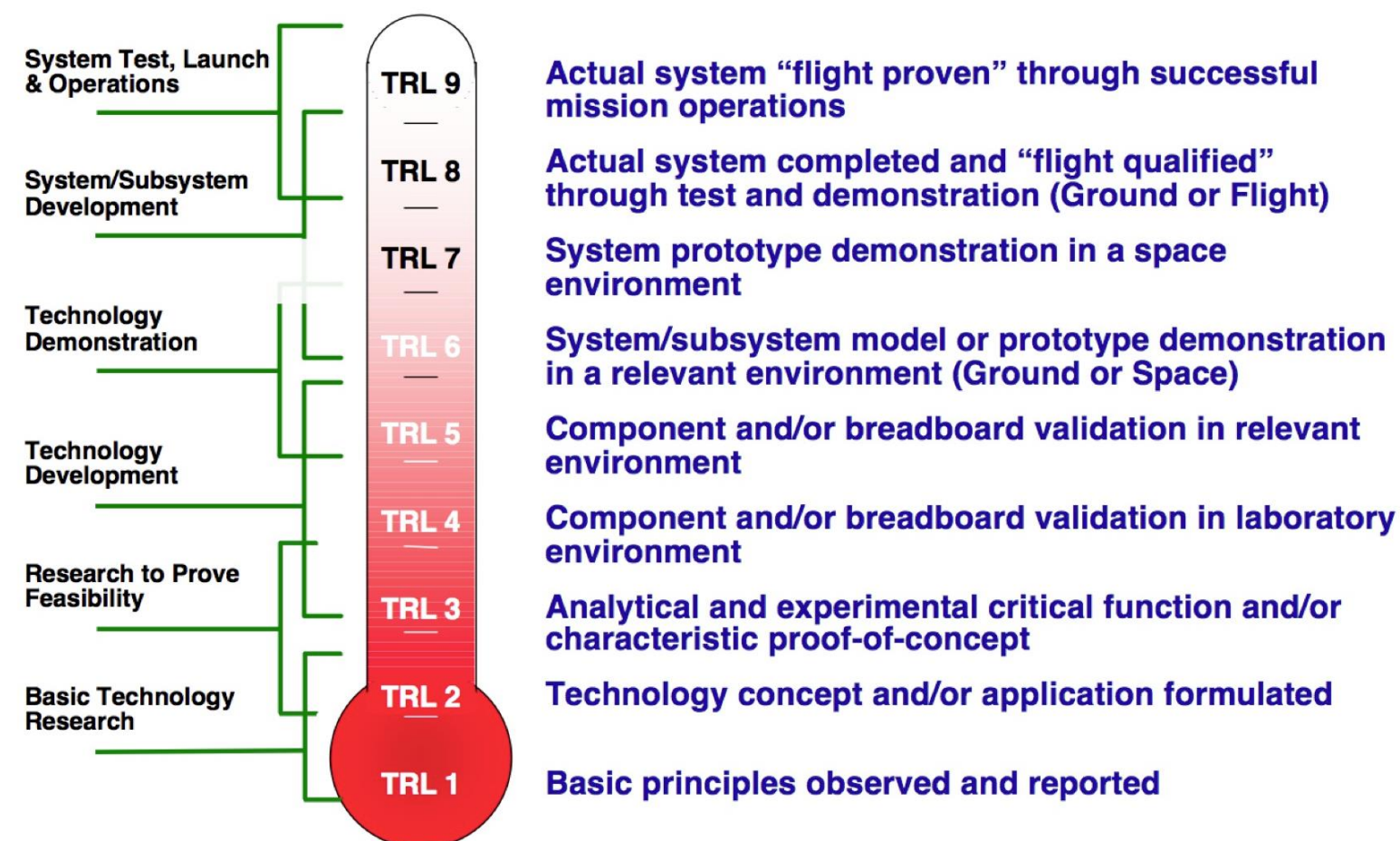
social/service robots



robots for automation and logistics



NASA/DOD Technology Readiness Level



Technology readiness levels (TRL)



Horizon 2020
Work Programme 2016-2017

General annex G of the Work programme 2016-2017,

Where a topic description refers to a TRL, the following definitions apply:

- TRL 1 – basic principles observed
- TRL 2 – technology concept formulated
- TRL 3 – experimental proof of concept
- TRL 4 – technology validated in lab
- TRL 5 – technology validated in relevant environment (industrial environment in the case of key enabling technologies)
- TRL 6 – technology demonstrated in relevant environment (industrial environment in the case of key enabling technologies)
- TRL 7 – system prototype demonstration in operational environment
- TRL 8 – system complete and qualified
- TRL 9 – actual system proven in operational environment (competitive manufacturing in the case of key enabling technologies; or in space)

1 2 3 4 5 6 7 8 9

Basic Principles Observed	Technology Concept Formulated	First Assessment feasibility concept & technologies	Validated Integrated Prototype In Lab	Tested Prototype In User Environment	Pre-Production Product	Low Scale Pilot Production Demonstrated	Manufacturing fully tested, validated, qualified	Production & Product Fully Operational
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Fundamental research/invention

Concept validation

Prototyping & incubation

Pilot production & demonstration

Initial market introduction

Market expansion

Translation of basic research to possible applications. Ideation, technology foresight.

Early technological and applied product and process research. Preparation system integration.

Prototype system integration. Service development.

Prototype testing in real environment, product/process enhancements.

Deployment and performance tests.

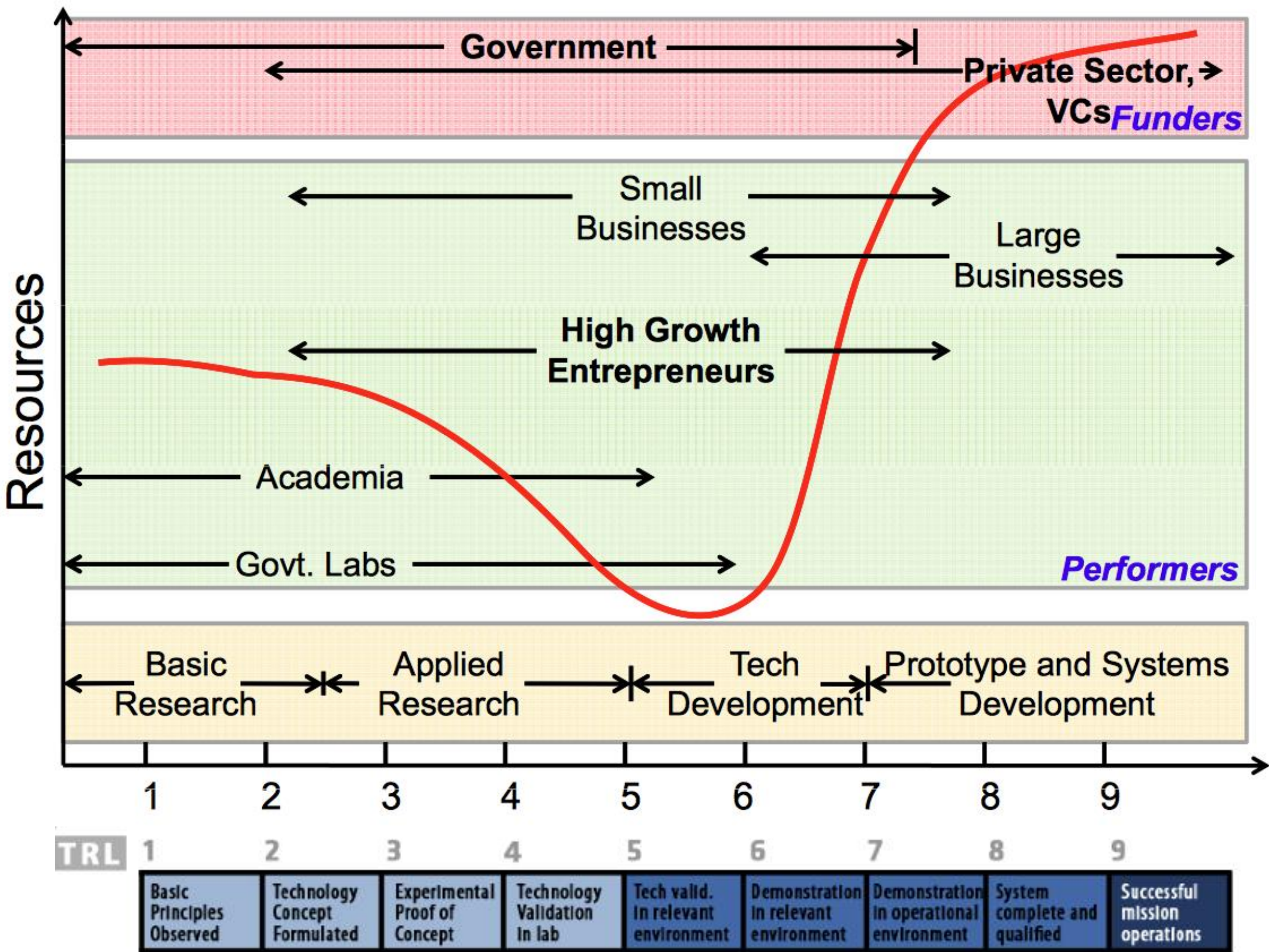
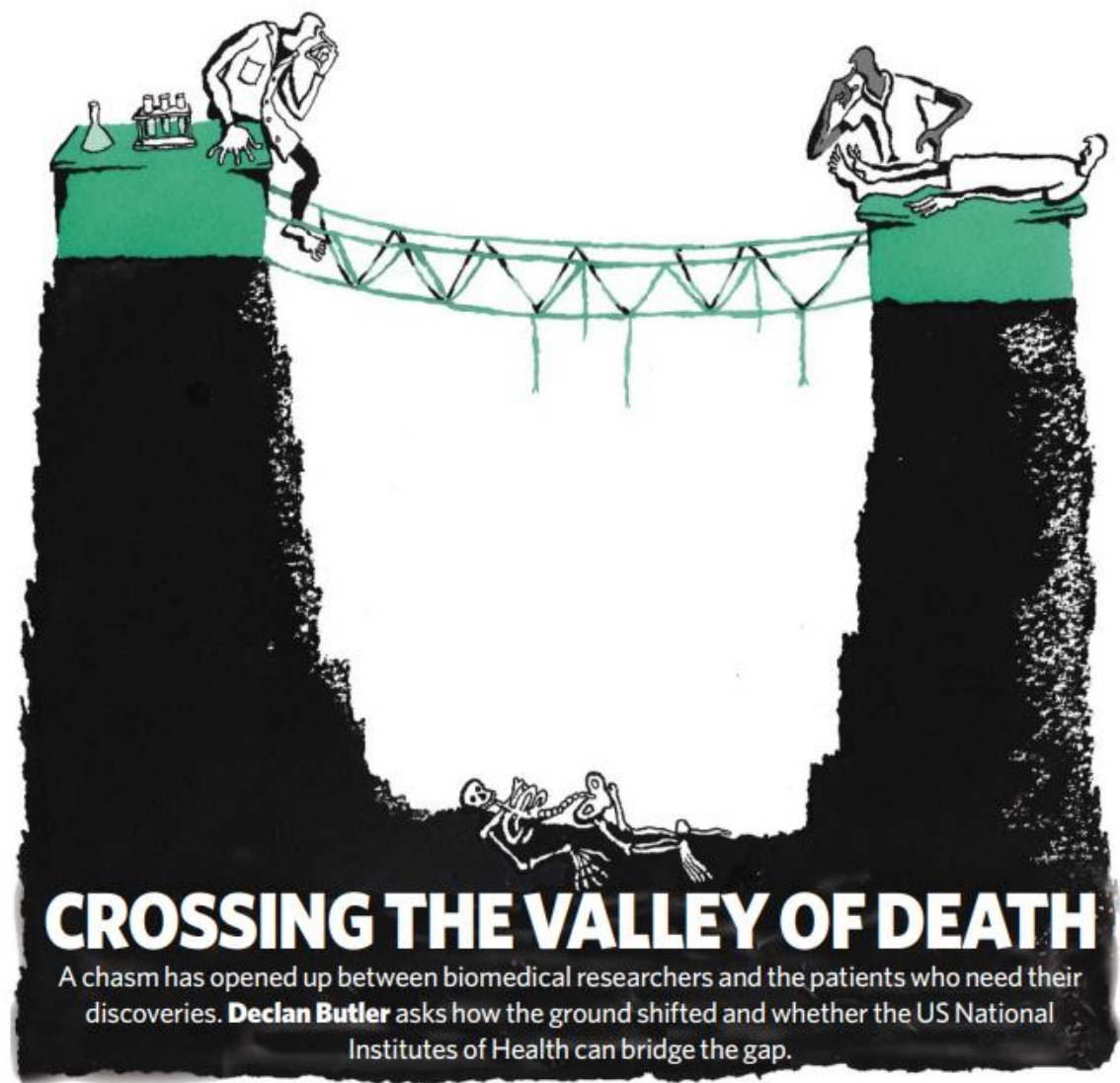
Scale in markets

News Feature

Translational research: Crossing the valley of death

A chasm has opened up between biomedical researchers and the patients who need their discoveries. Declan Butler asks how the ground shifted and whether the US National Institutes of Health can bridge the gap.

NEWS FEATURE TRANSLATIONAL RESEARCH NATURE Vol 453 12 June 2008



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- ROBOTIC TECHNOLOGIES: KEY-FACTORS TO BUILD A HEALTHCARE ROBOT, MACRO-TRENDS AND OPEN ISSUES
- EXAMPLES OF HEALTHCARE TECHNOLOGIES AND APPLICATIONS WITH IOT FUNCTIONALITIES



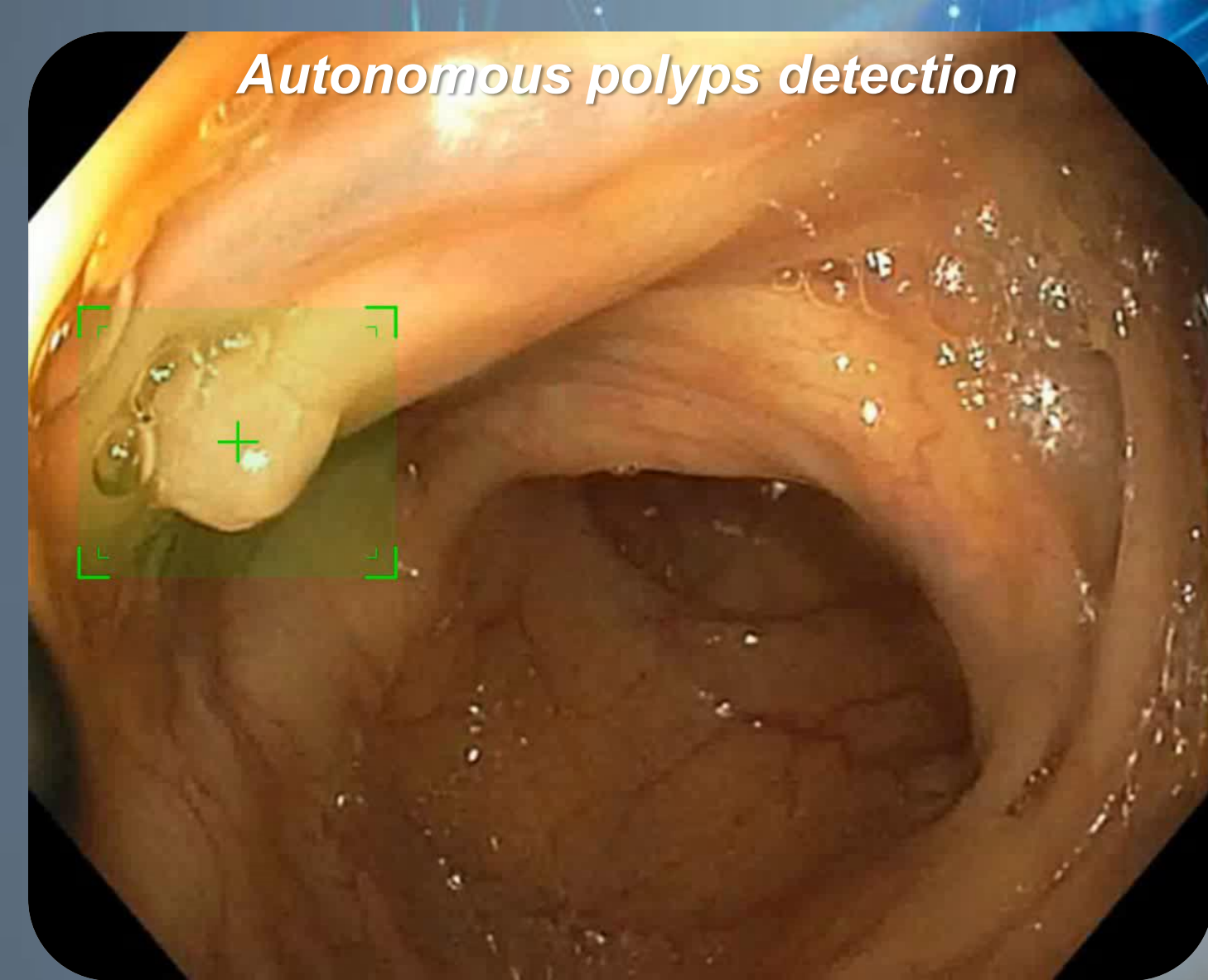


soft robotics



collaborative robotics

Autonomous polyps detection



Brandao, Patrick, et al. "Towards a computed-aided diagnosis system in colonoscopy: automatic polyp segmentation using convolution neural networks." *Journal of Medical Robotics Research* 3.02 (2018): 1840002.

artificial intelligence





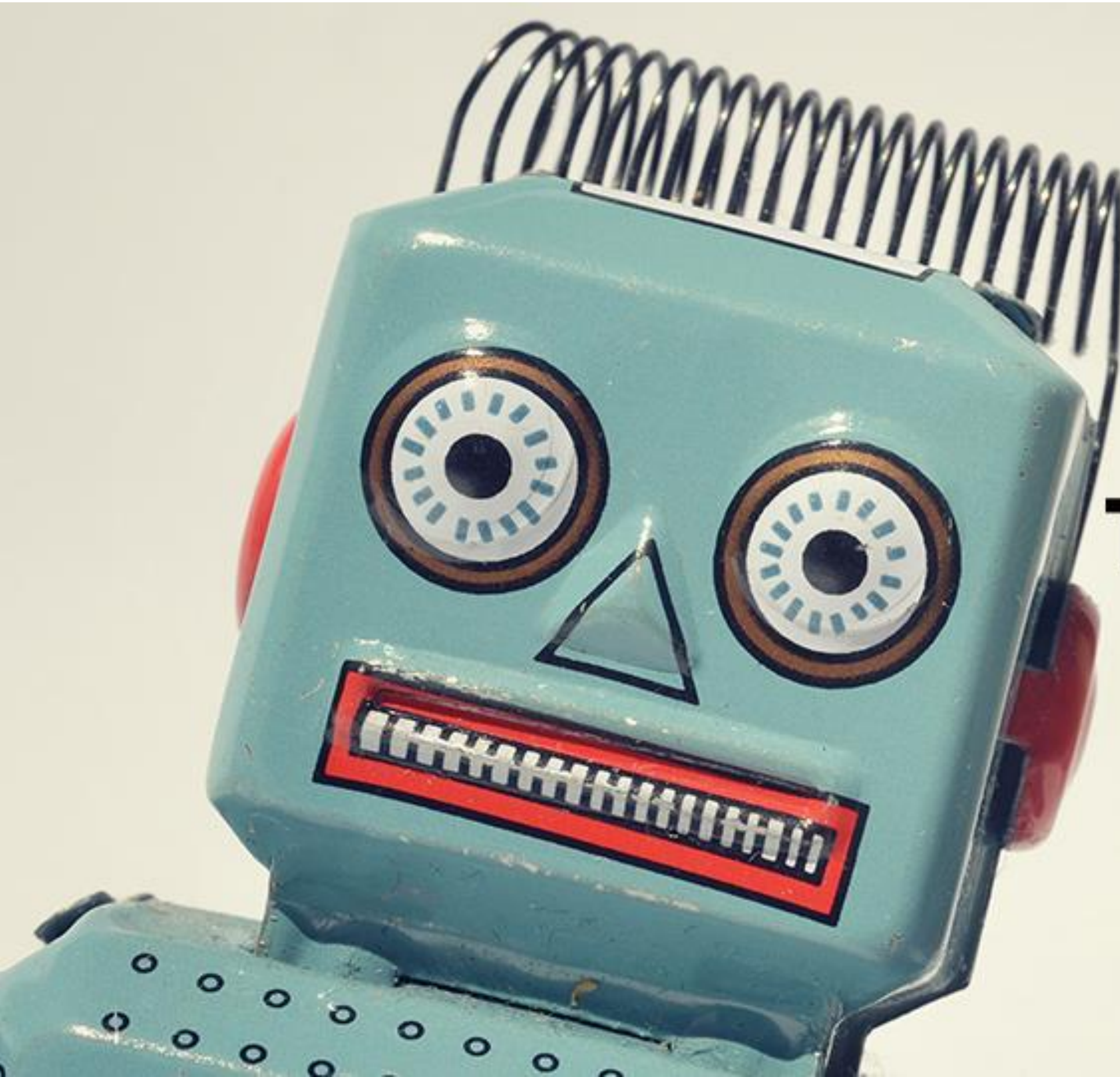
big data: how to manage the huge amount of data?



cybersecurity: how to protect “fragile” data?



connectivity and standards



THE ETHICS OF ROBOTS

**IN THE FUTURE, WILL THIS GUY MAKE
DECISIONS BEHIND THE WHEEL FOR US??**

...and regulations



OUTLINE

- GENERAL OVERVIEW ON INTEGRATED/INTELLIGENT ROBOTICS IN HEALTHCARE
- ROBOTIC TECHNOLOGIES: KEY-FACTORS TO BUILD A HEALTHCARE ROBOT, MACRO-TRENDS AND OPEN ISSUES
- **EXAMPLES OF HEALTHCARE TECHNOLOGIES AND APPLICATIONS WITH IOT FUNCTIONALITIES**



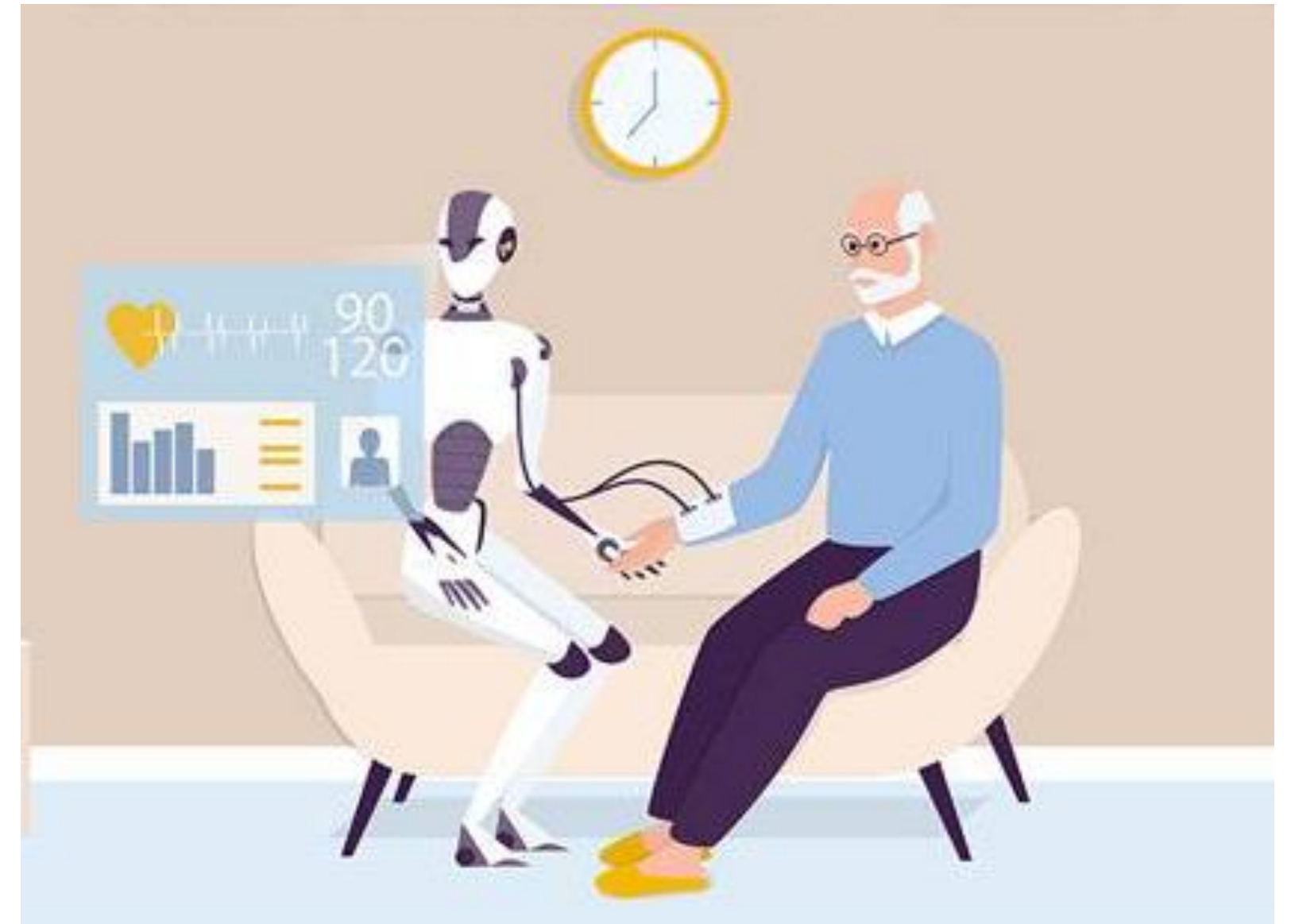
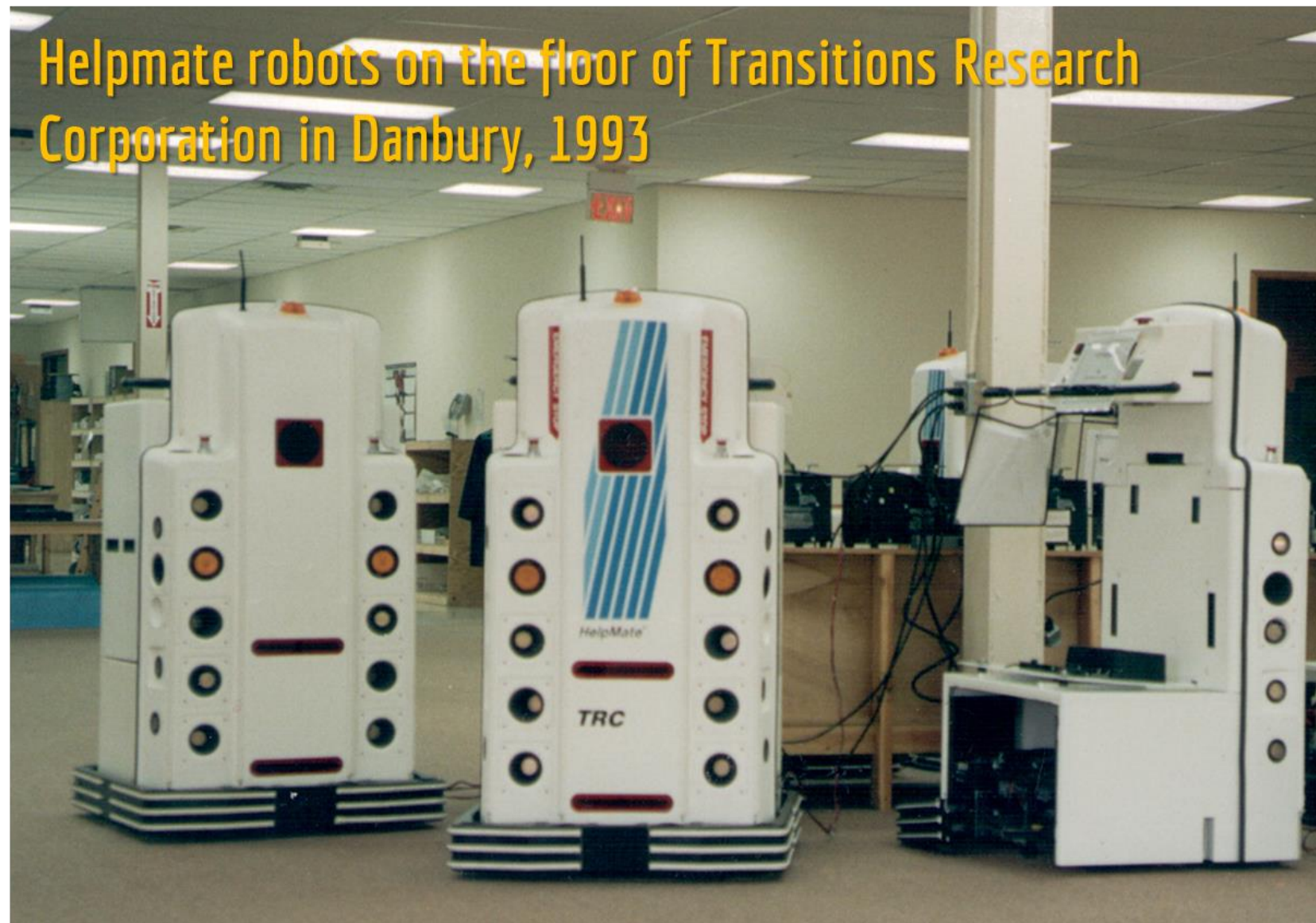
**1984: J. Engelberger
founded Transitions
Research Corporation**



Helpmate robots on the floor of Transitions Research Corporation in Danbury, 1993



Integration of robotics with the hospital's management system: how to simplify the process and solve “possible” open issues after >30 years?



**Flexible
Modular**



...not at the level of each unit

Hospital roBOT --- HOSBOT

- **Autonomous delivery** with multiple functionalities (each box)
- Host up to **3 SmartBox(es)** or boxes
- **Emergency button for safety**
- **LEDs** for showing the status and HRI
- **Sound/visual alarms** to notify that the robot is approaching
- **Carried autonomously** by RB-1 or **manually** by an operator
- **MYS RTLS** to localize each single SmartRack and SmartBox
- **Autonomous or manual recharging**

INNOVATIVE KEY-FEATURE(S): enhanced modularity, flexible connectivity at each level, *i.e.* up to robotic boxes (SmartBox)



SmartRack

Robotnik RB-1 mobile robot



SmartBox



Market Analysis (examples of autonomous robots for hospital logistics)



AETHON



telelift
SIMPLIFYING INTRALOGISTICS



PROs:

- Modularity at the cart level
- Customizability of the cart and of the anchoring with the AMR



MiR



EUREKA
SYSTEM



OMRON

CONs:

- Monolithic transport of logistic cabinet



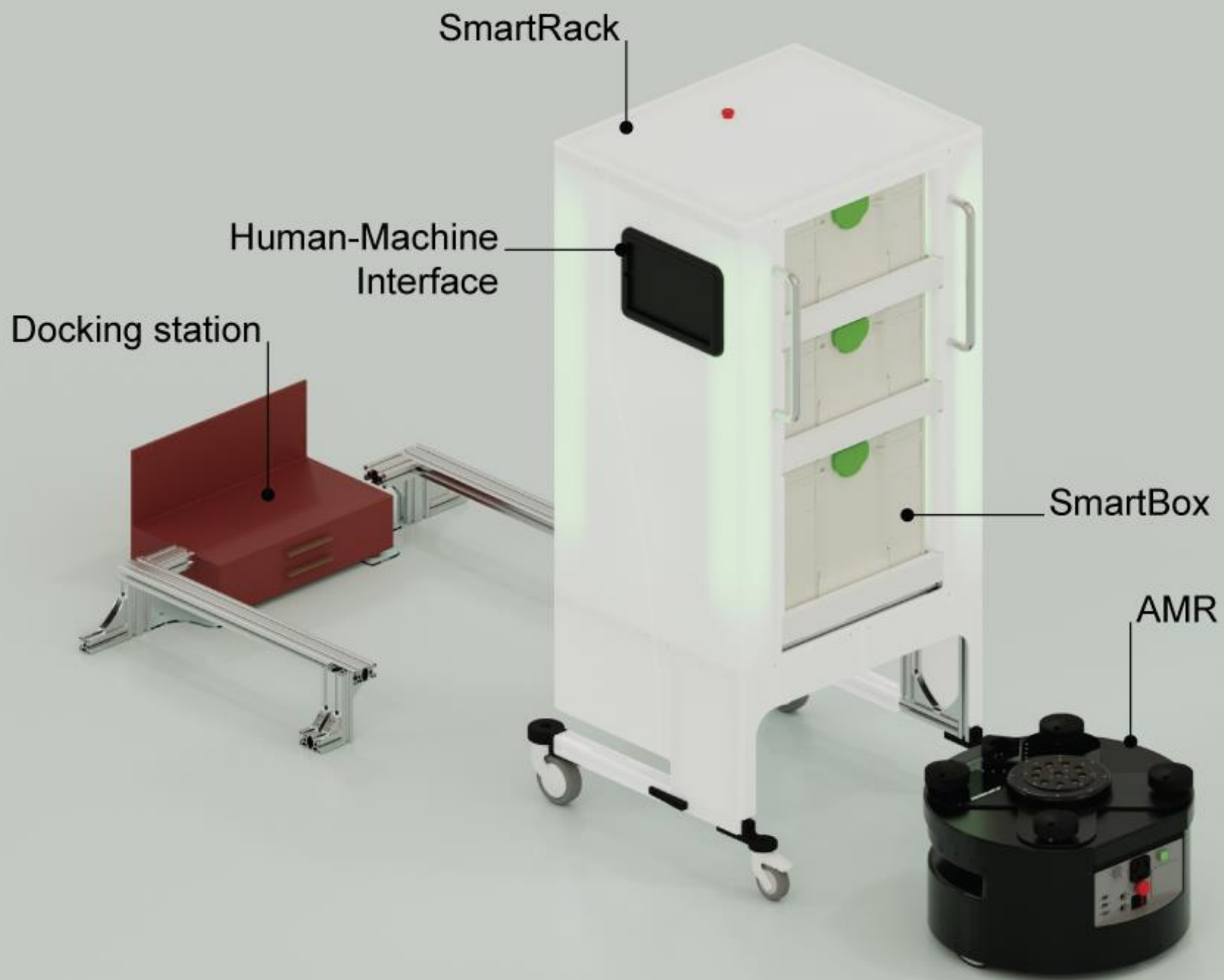
ek·robotics
INFINITE COLORS OF TRANSPORT



HOSBOT **assets** within ODIN EU project

- **Flexibility** to reach any accessible areas in the hospital avoiding infrastructural limitations (manual or robotic navigation of SmartRacks and SmartBoxes)
- **Modularity** to assemble your HOSBOT with different robotic mobile units and customised # of SmartBoxes
- **Connectivity** of each sub-unit (up to the SmartBox level) with the ODIN digital framework
- **Extended & Optimised** thanks to the customisation of robotic modules and tasks
- **Low-cost** solution as one robotic mobile unit can drive several SmartRacks and move different types of removable and standalone SmartBoxes

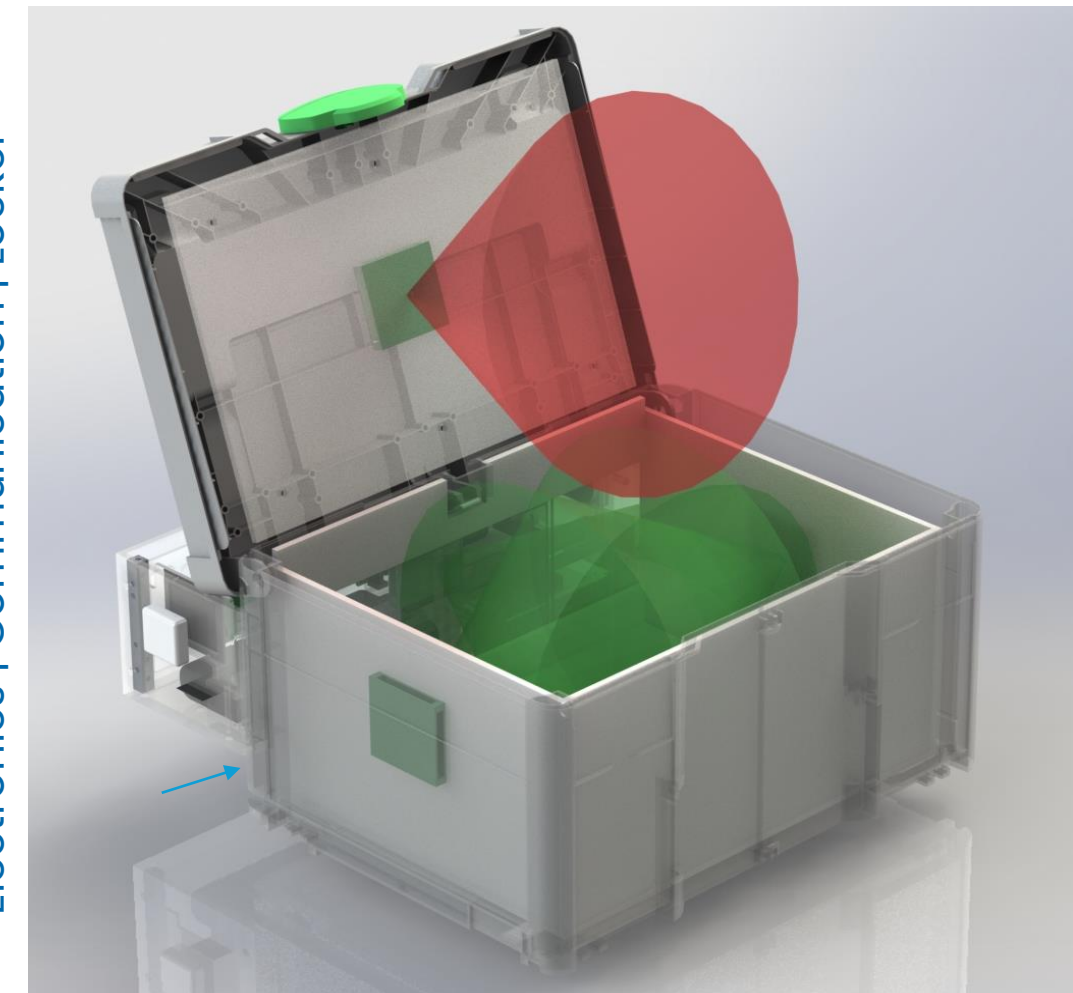




SmartBox (the robot in the robot)

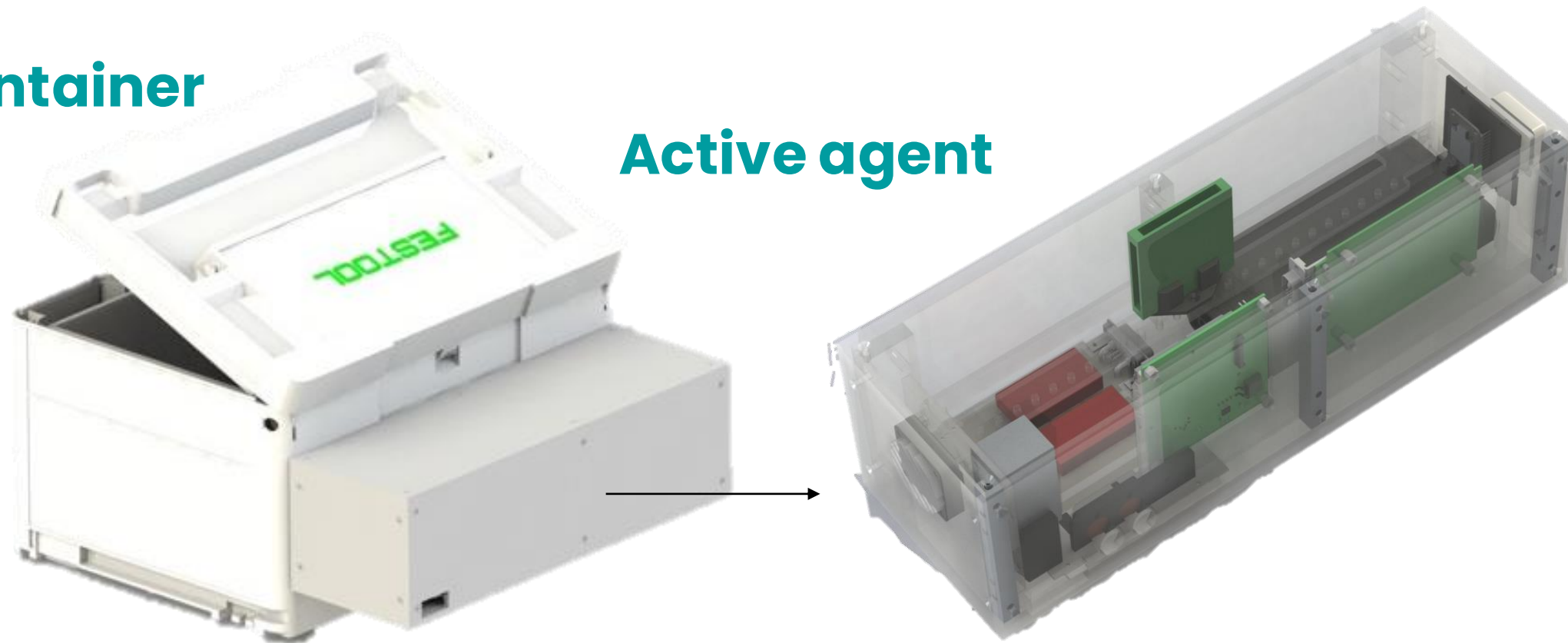
- RFID technology for **consumable detection**: up to 4 antennas
- On-board sensors for **real-time localization and tracking**
- IoT module for **linking consumable IDs with ontology**
- **Locking system** to prevent unauthorized access
- **Internal battery** with contact plugs for charging
- Can be **carried by HOSBOT or manually** by an operator

RFID antennas | battery | RTLS |
Electronics | Communication | Locker



Container

Active agent





Hosbot commands

Plan a mission

Smartbox



Back

Smartbox

Unlock

Consumables

Previous Scans

Insert

Remove

Storage



Current status and experimentation

- **TRL 5/6:** technology validated/demonstrated in a relevant environment

PILOT #1

Environment

Patient-free area of the Hospital's Emergency Department

Path length

40 meters

N° users

1

Objectives

Testing the autonomous navigation of HOSBOT & the delivery and preservation of biological samples



PILOT #2

Environment

Patient-free area of the Hospital's Smart Health Center

Path length

120 meters

N° users

25

Objectives

Testing the autonomous navigation of HOSBOT & the delivery of medical objects with the SmartBox



PILOT #3

Environment

Active area of the Logistics Department

Path length

230 meters

N° users

16

Objectives

Testing the autonomous navigation of HOSBOT & the delivery of medical objects with the SmartBox



HOSPITAL PILOTS



UMC Utrecht

3



MEDICAL
UNIVERSITY
OF LODZ



Amiens-Picardie UH (AMIS)
Amiens, France

UMC Utrecht:
University Medical Center Utrecht (UMCU)
Utrecht, Netherlands

Charité University Hospital (CUH)
Berlin, Germany

Medical University of Lodz (MUL)
Lodz, Poland

1



2



SaludMadrid
Hospital Clínico San Carlos (HCSC)
Madrid, Spain

University Hospital Campus
Bio-Medico (UCBM)
Rome, Italy

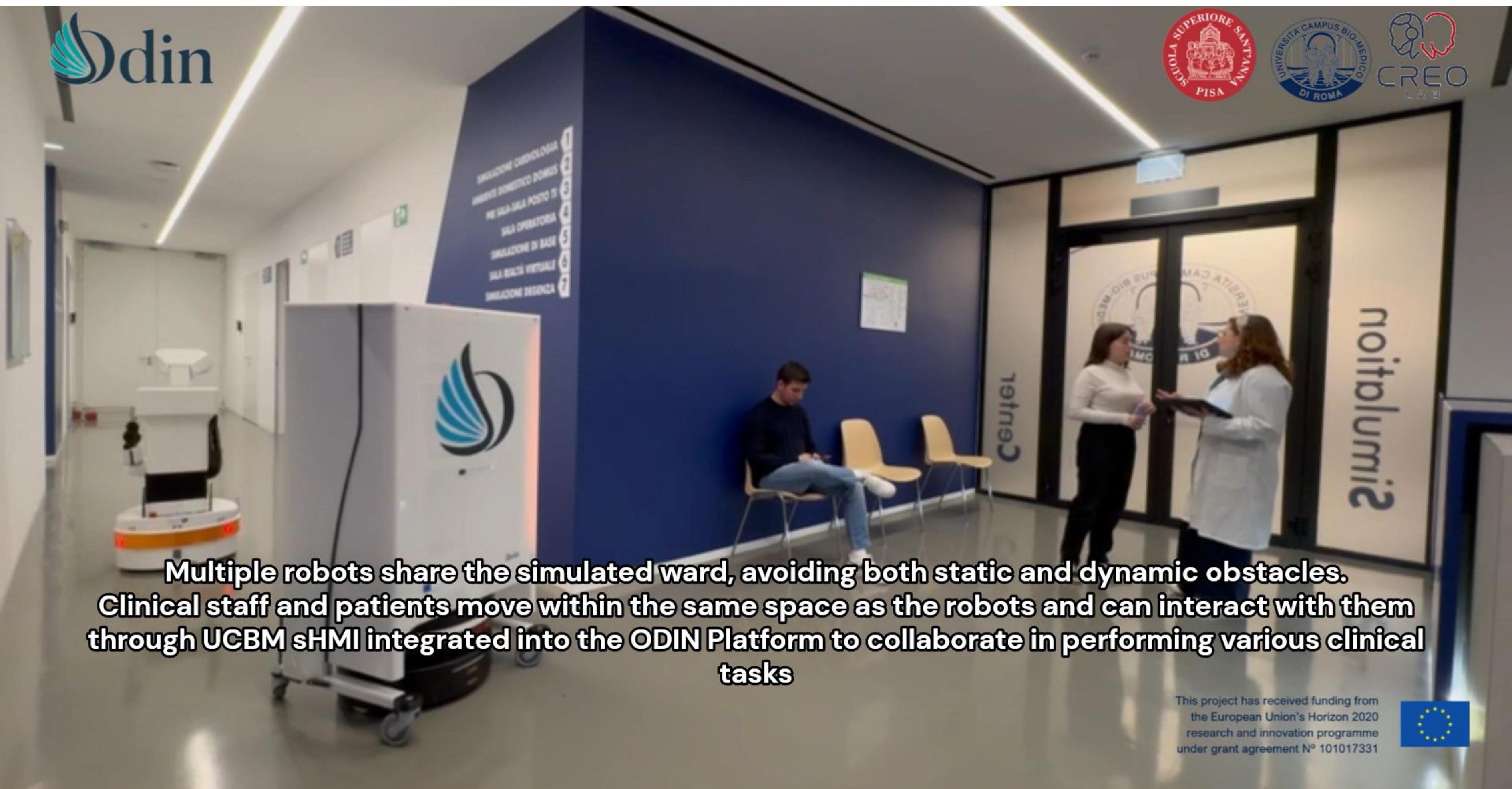


4

Integration (example) of a single robot, HOSBOT

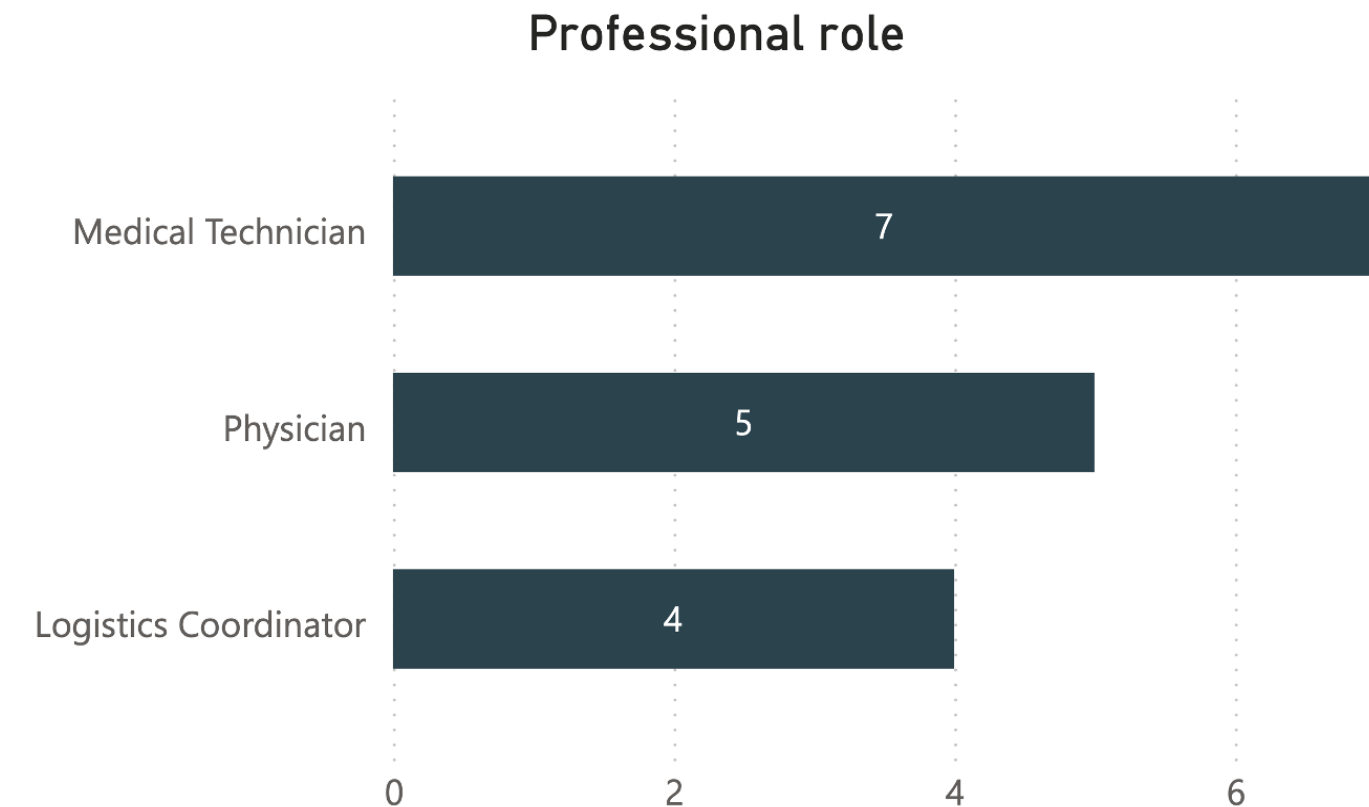
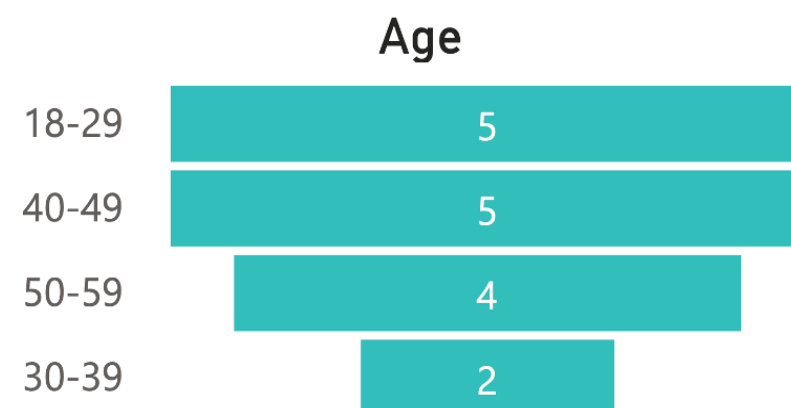
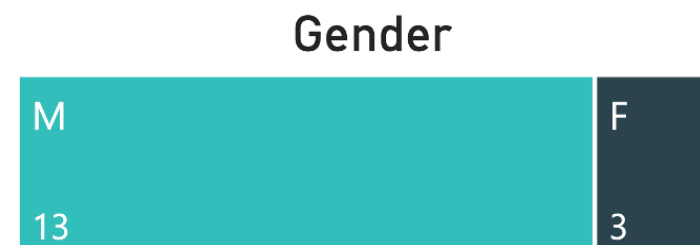


Integration (example) of a multi-agent platform



Multiple robots share the simulated ward, avoiding both static and dynamic obstacles. Clinical staff and patients move within the same space as the robots and can interact with them through UCBM sHMI integrated into the ODIN Platform to collaborate in performing various clinical tasks

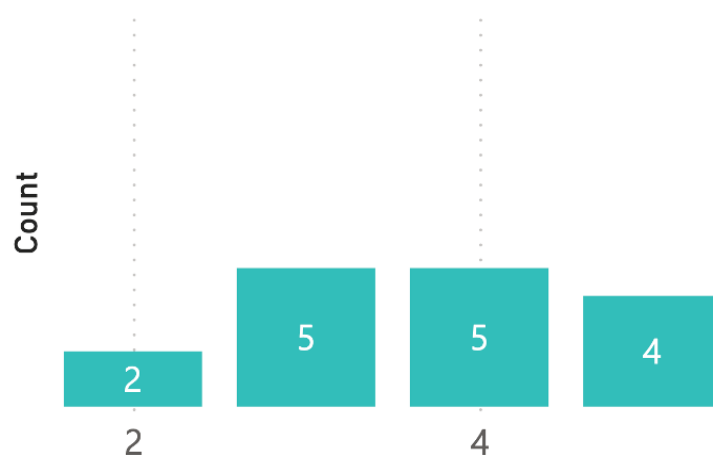
System Usability Scale (SUS)



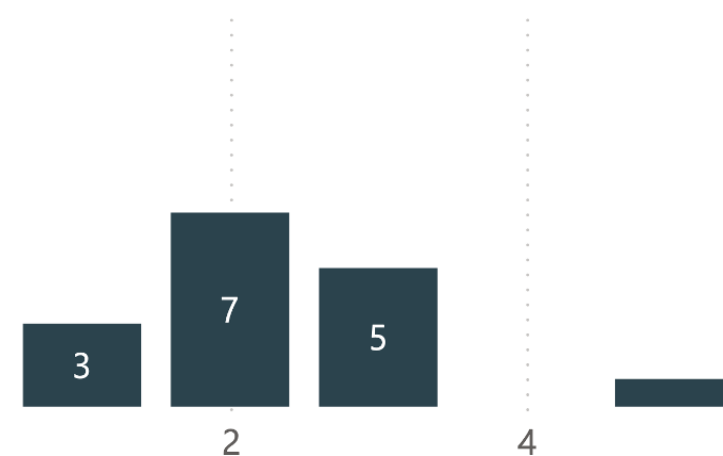
71,40
Result Average (%)

Legend:
1: Strongly disagree
...
5: Strongly agree

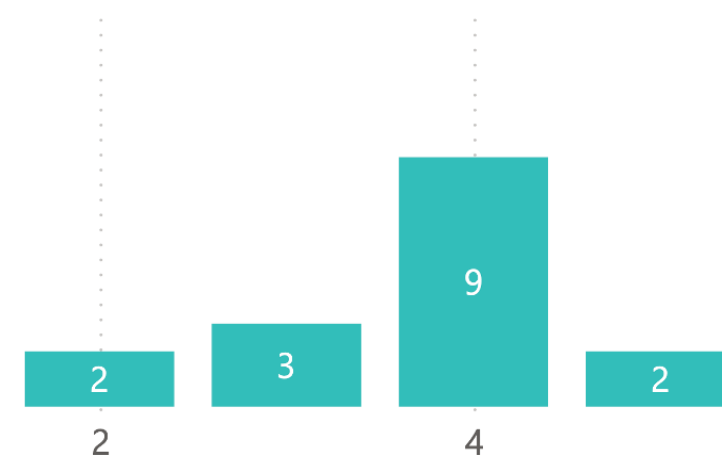
Q1: I think that I would like to use this system frequently



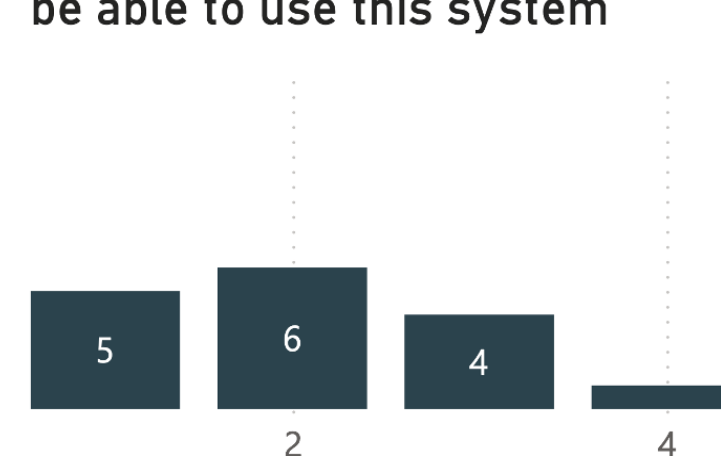
Q2: I found the system unnecessarily complex



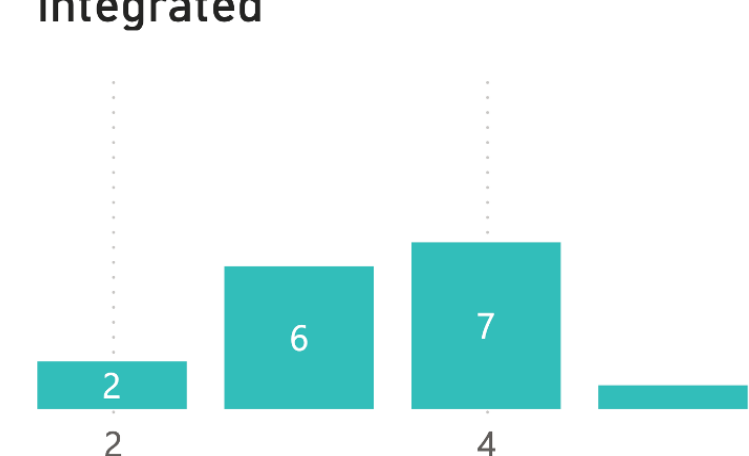
Q3: I thought the system was easy to use



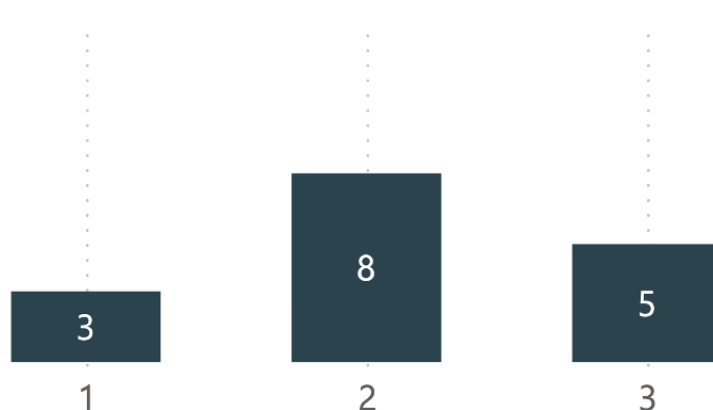
Q4: I think that I would need the support of a technical person to be able to use this system



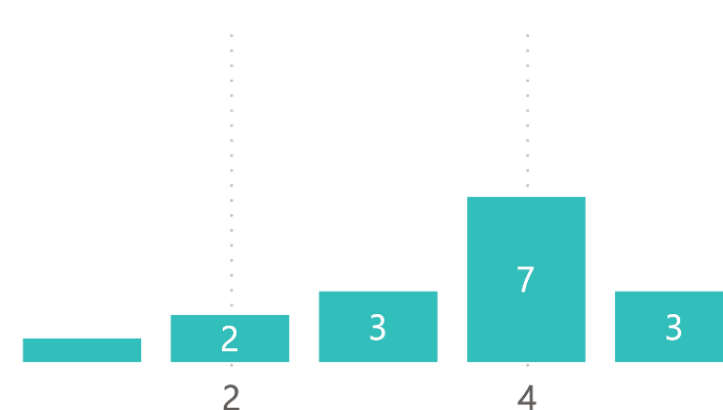
Q5: I found the various functions in this system very well integrated



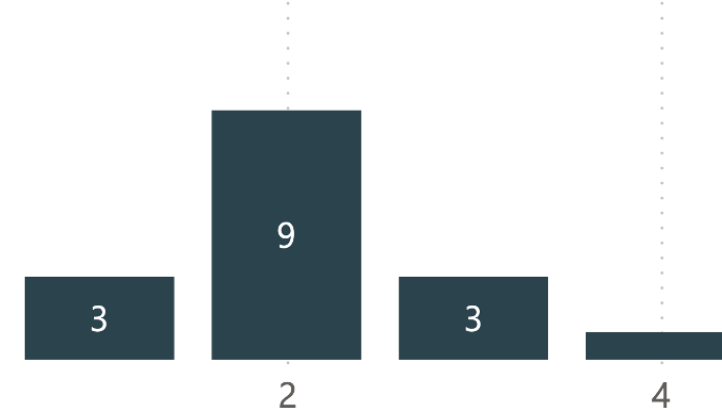
Q6: I thought there was too much inconsistency in this system



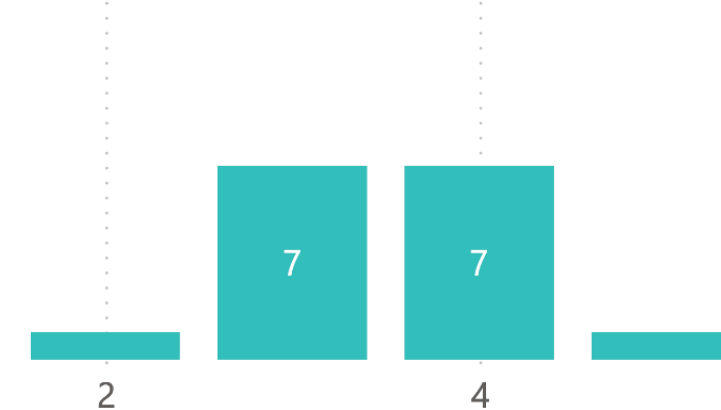
Q7: I would imagine that most people would learn to use this system very quickly



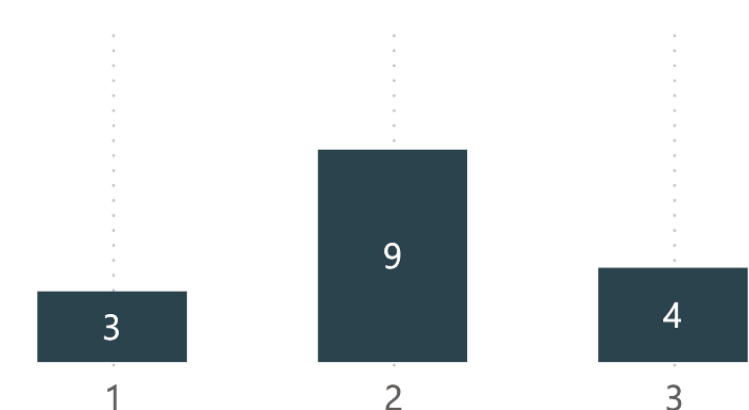
Q8: I found the system very cumbersome to use

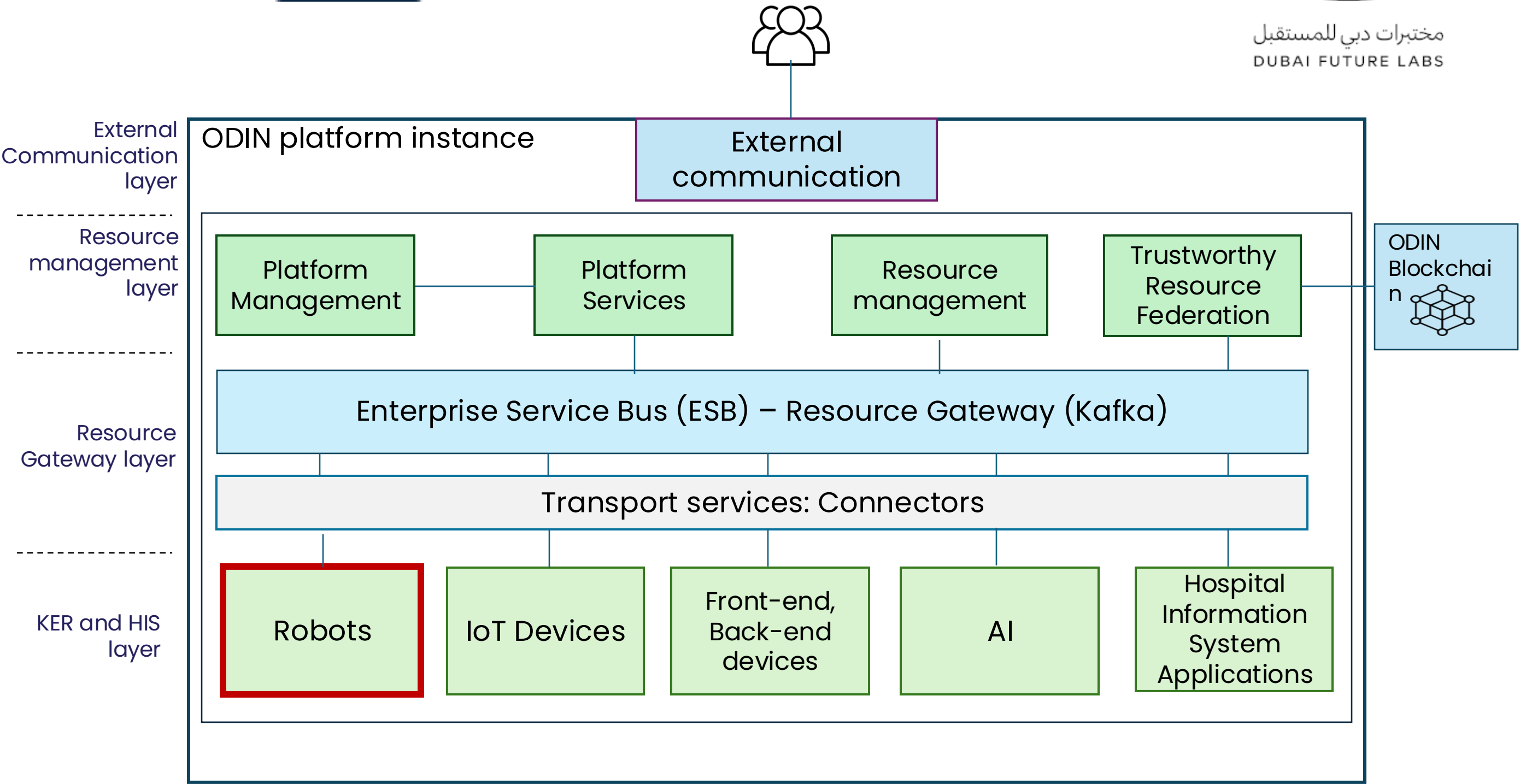
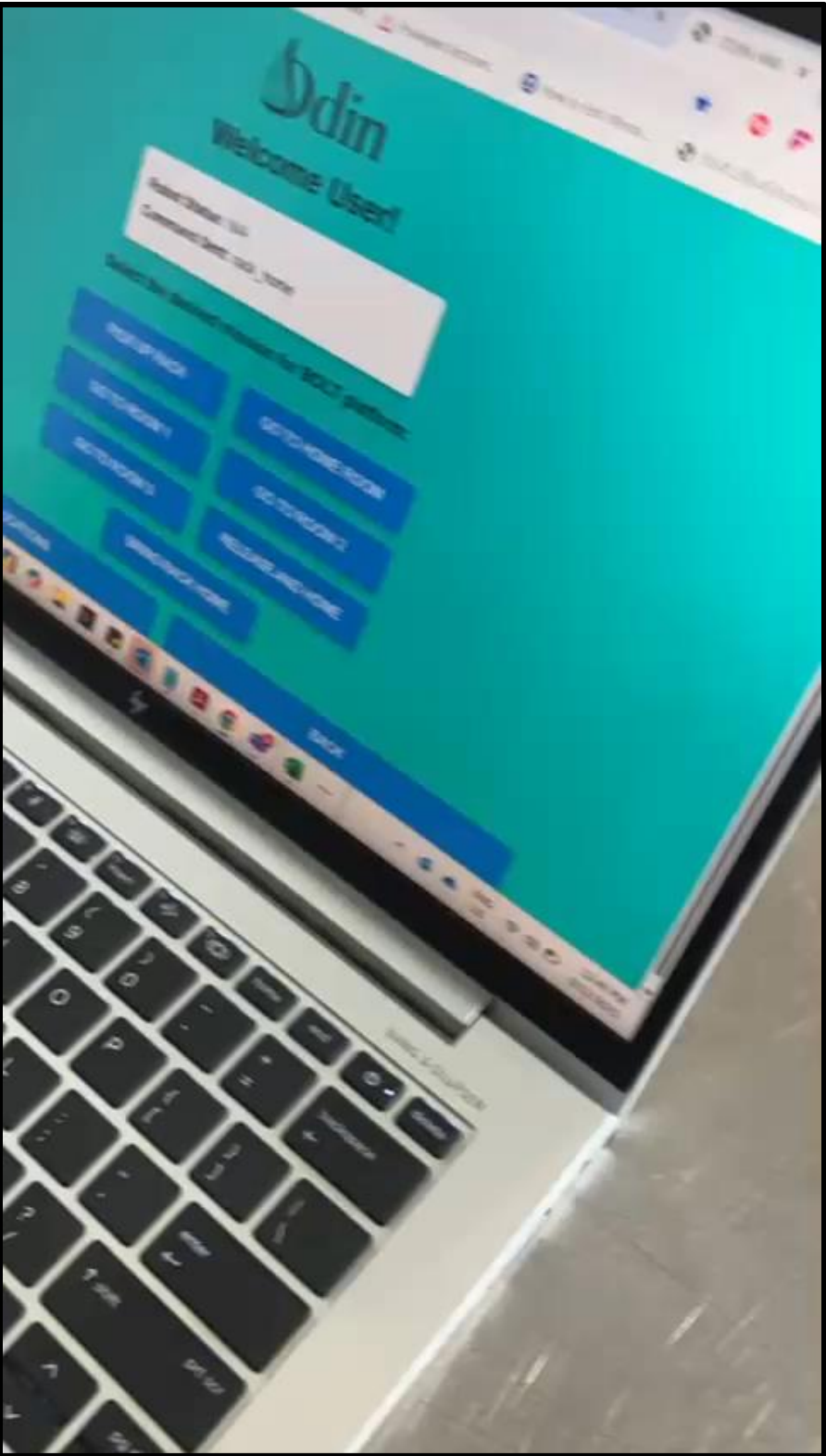


Q9: I felt very confident using the system



Q10: I needed to learn a lot of things before I could get going with this system





ODIN PLATFORM ARCHITECTURE

HOSBOT And smartbox IP



Sant'Anna
Scuola Universitaria Superiore Pisa



POLITÉCNICA

DOMANDA DI BREVETTO DI INVENZIONE PATENT APPLICATION FOR INVENTION

Titolo/Title: DISPOSITIVO DI MOVIMENTAZIONE

Titolare/Applicant: Scuola Superiore Sant'Anna
Universidad Politécnica de Madrid

Inventore/Inventor: Ciuti, Chiurazzi, Firrincieli, Dei, Fico

Data/Date - Num.: 11/06/2024 - 102024000013300

ns. rif./our ref.: 17782 A IT 2692

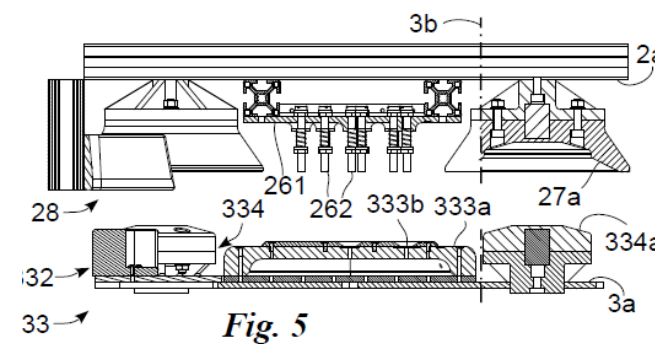
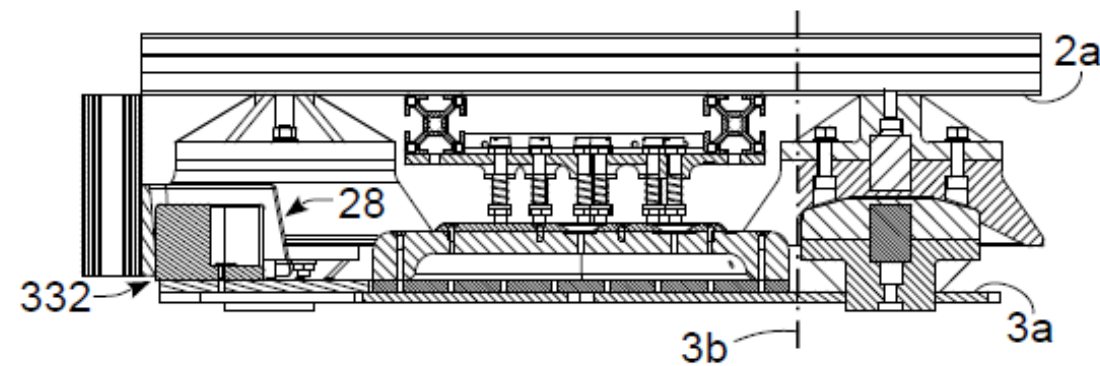


Fig. 6

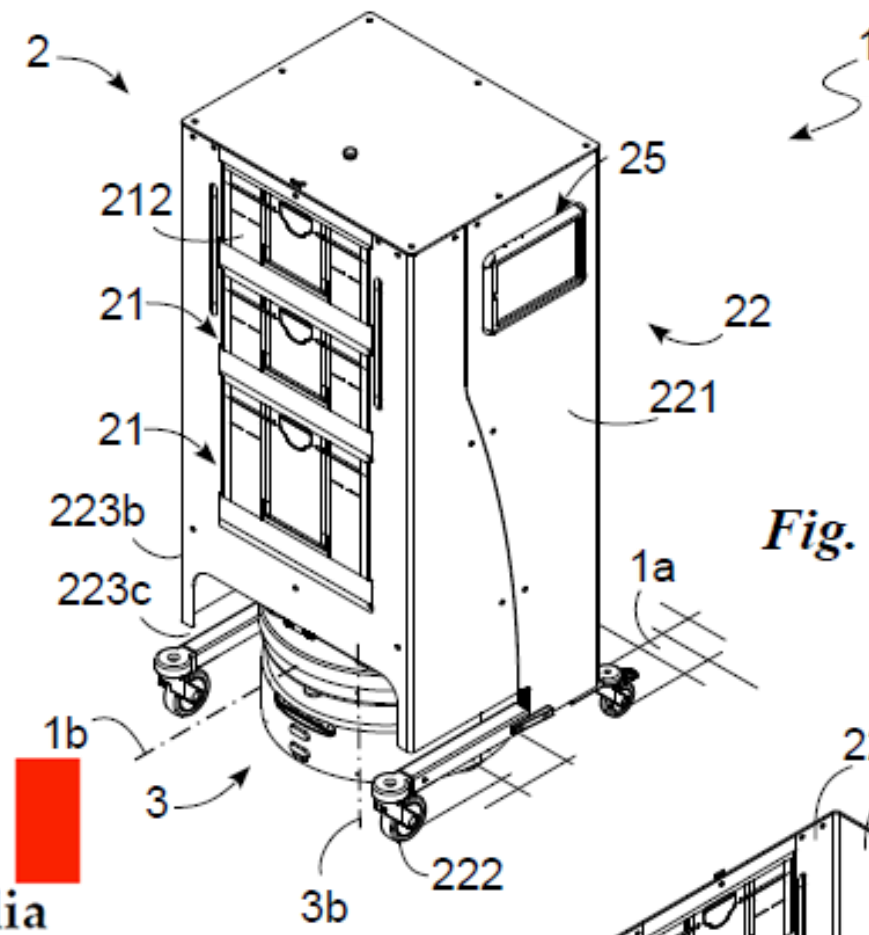
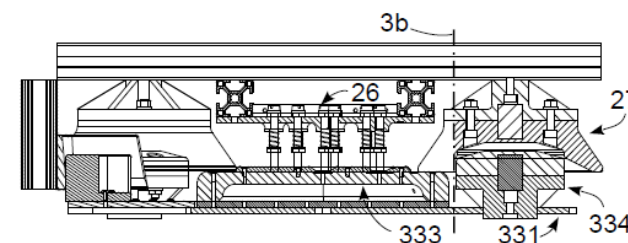


Fig. 1

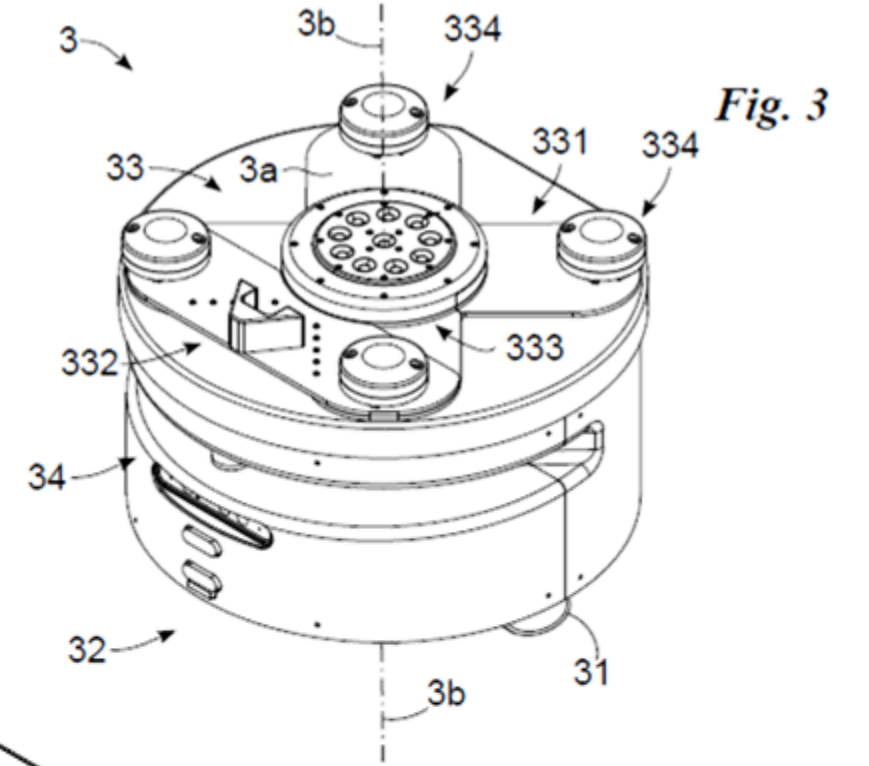


Fig. 3

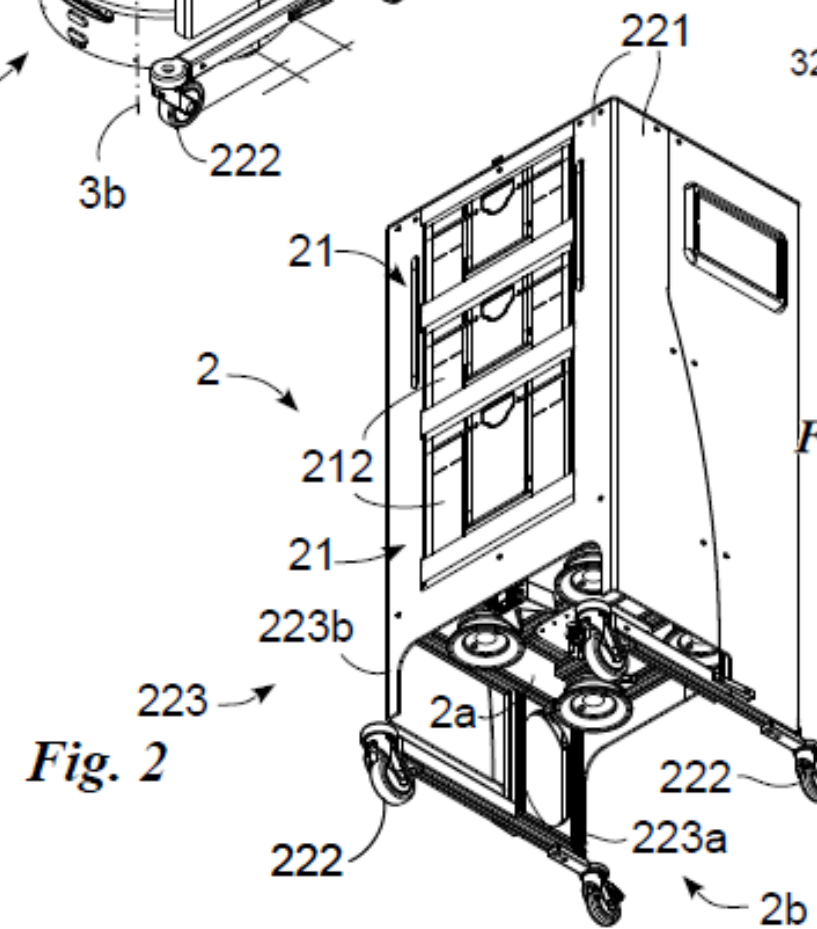
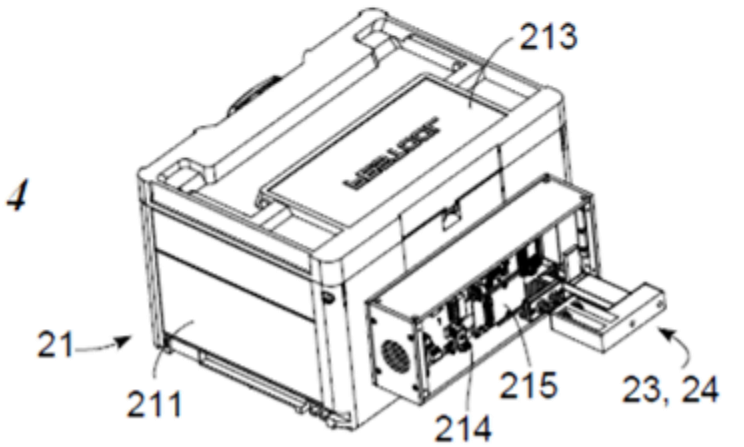


Fig. 2

Fig. 4



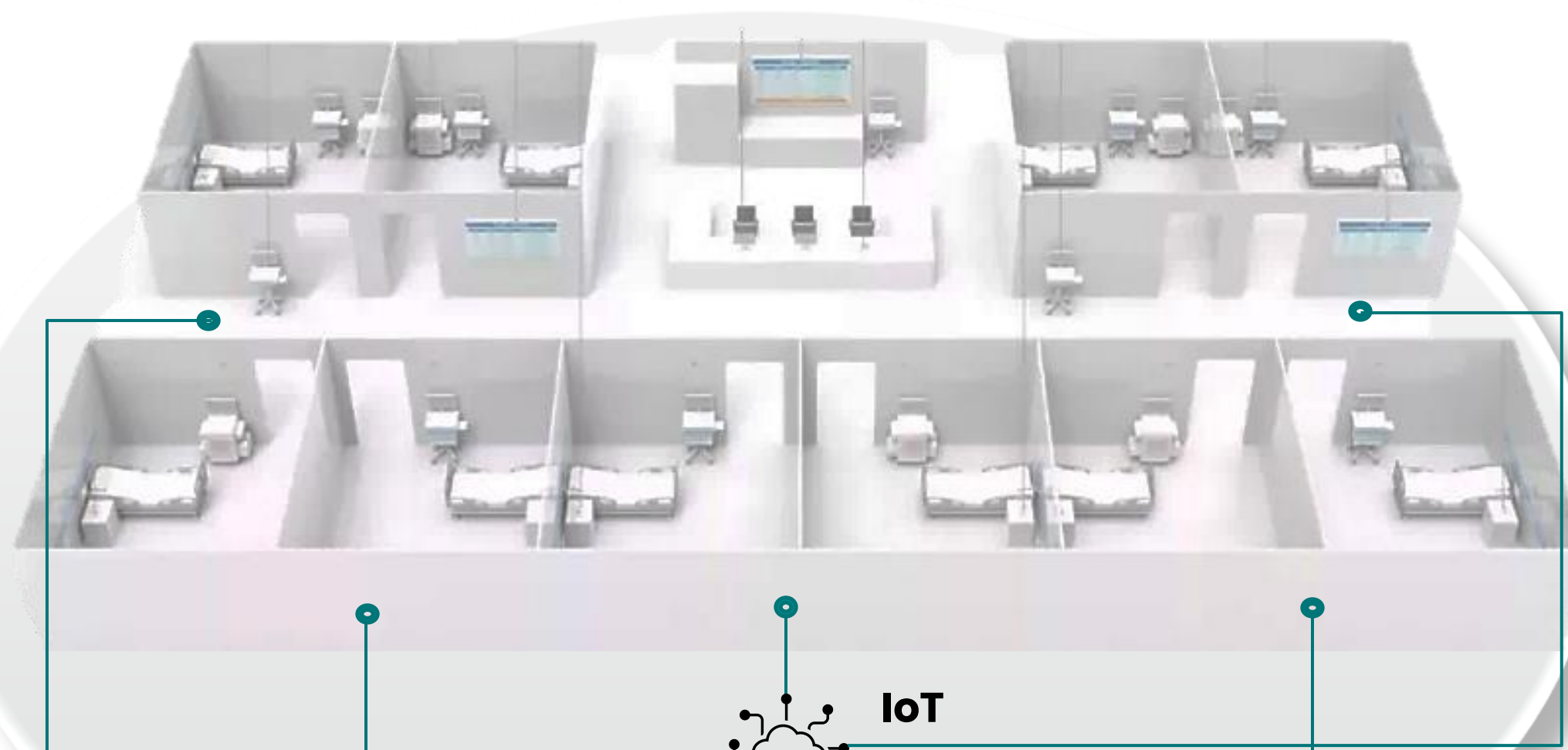
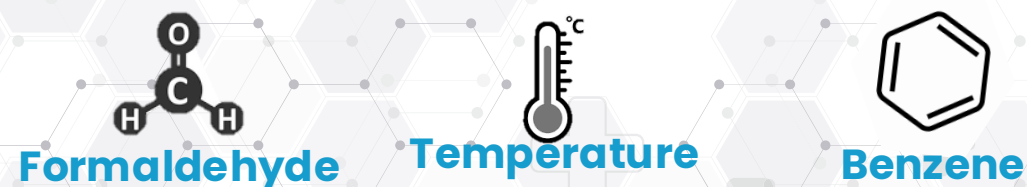
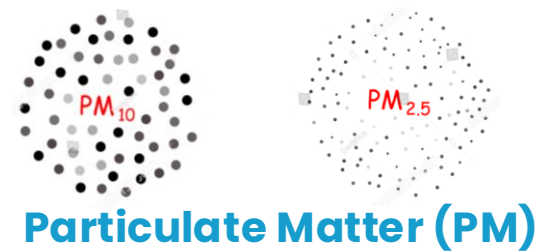
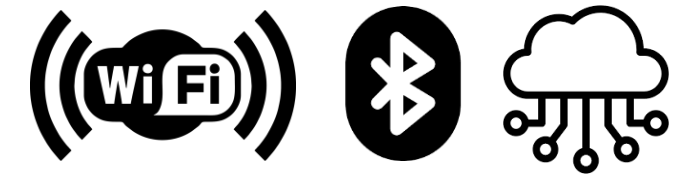
LM LUNATI & MAZZONI

European and italian patent and trademark attorneys
since 1982

Transparent Robot



Air Quality
Guidelines 2021



IoT
Multi-sensing monitoring

**SMART HOSPITAL
DIGITAL PLATFORM**



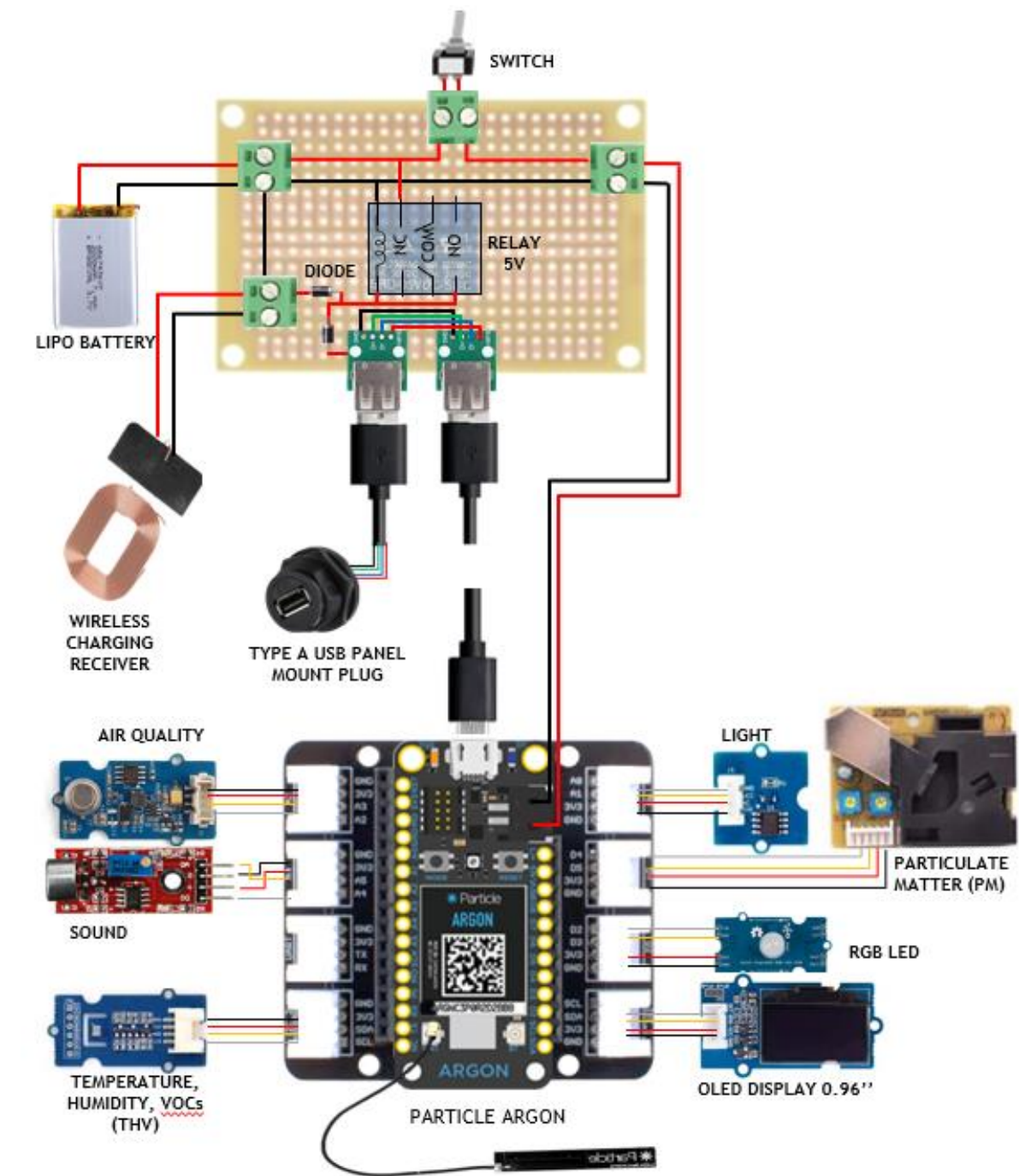
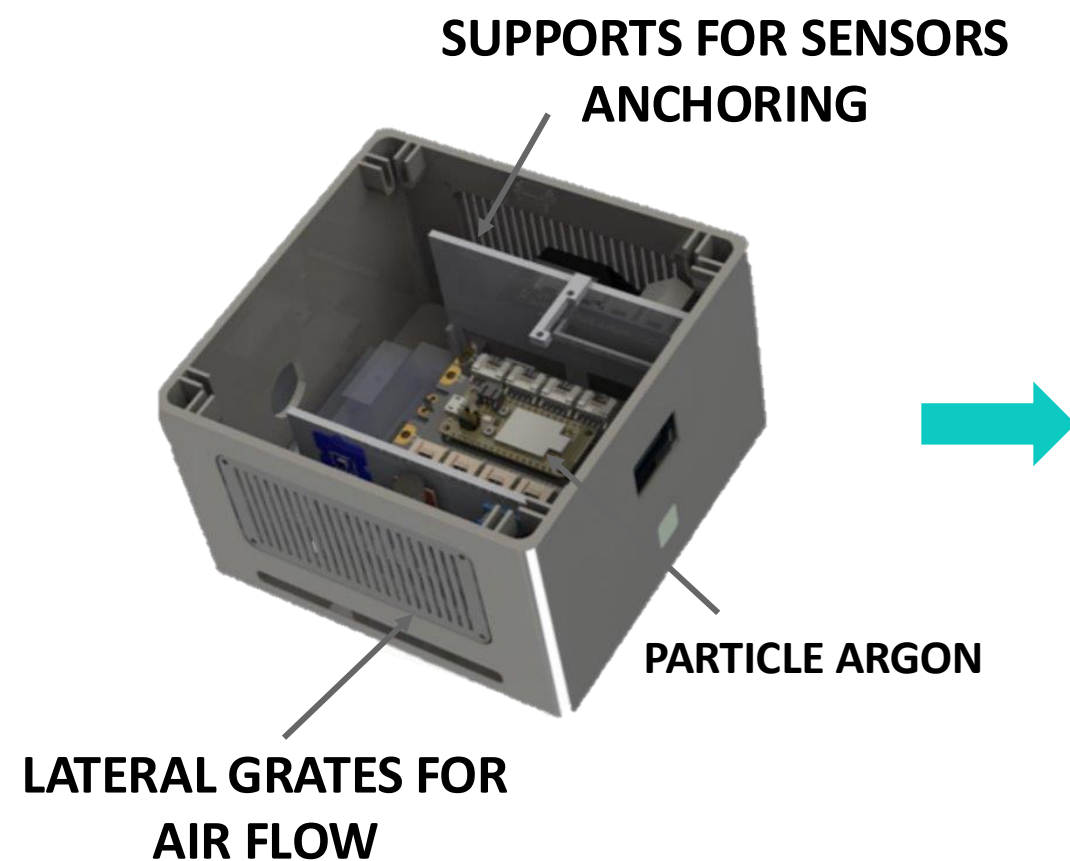
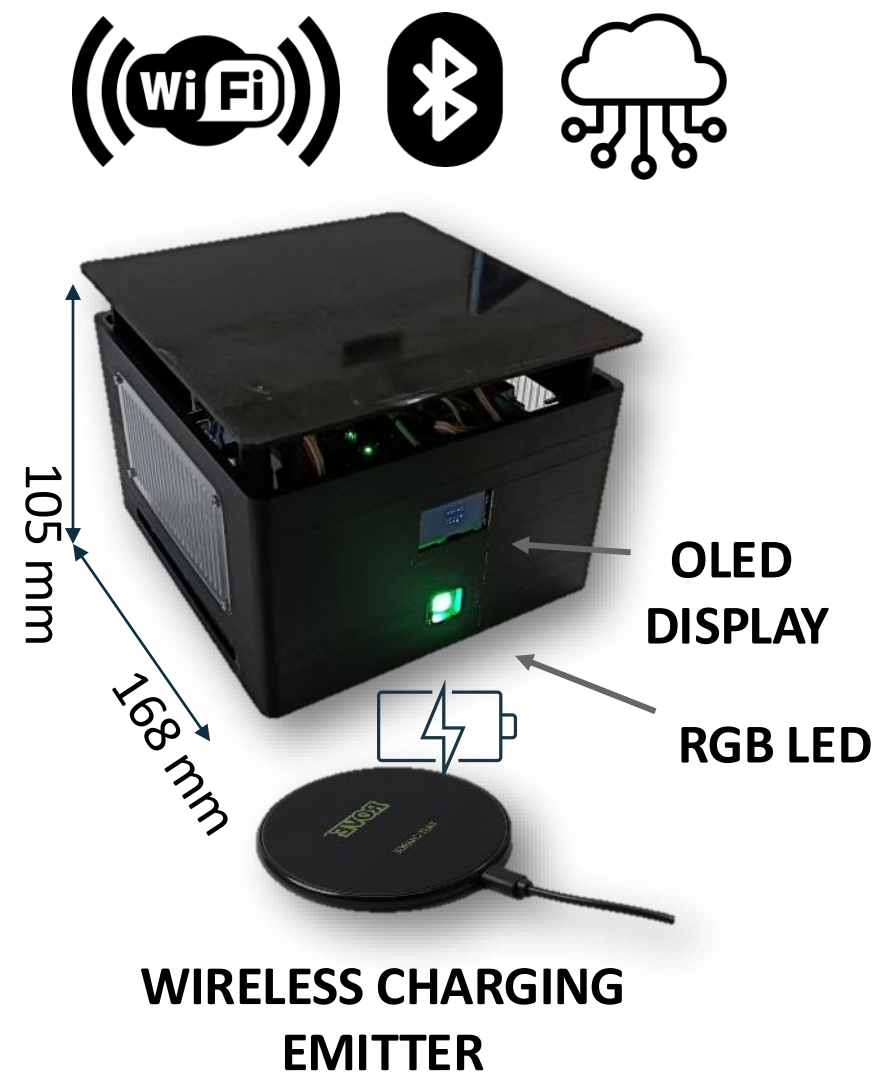
Main Features

- Modular
- IoT-based
- Real-time monitoring of IAQ
- Secured data sharing
- Integrated to a smart hospital digital platform

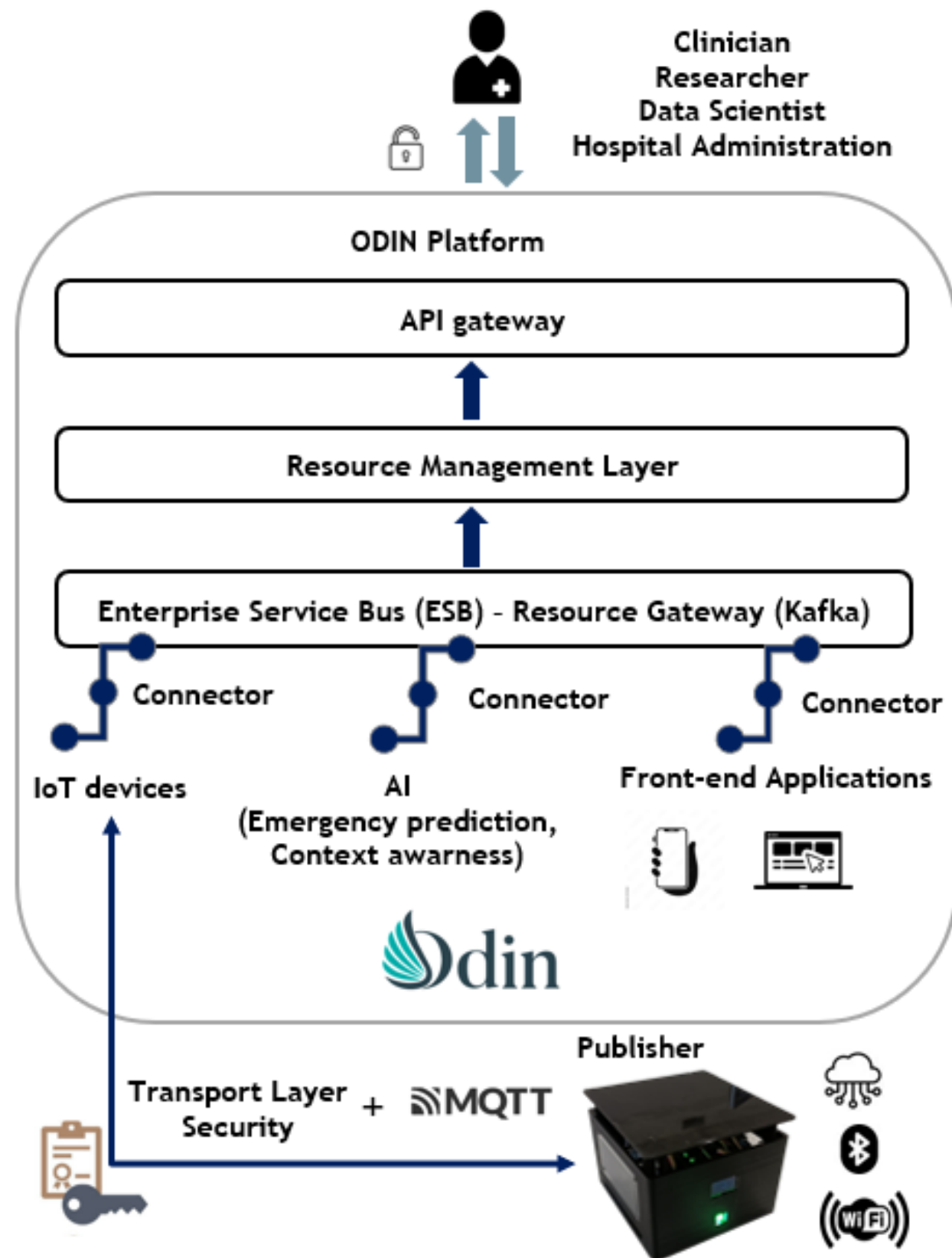
This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement N° 101017331



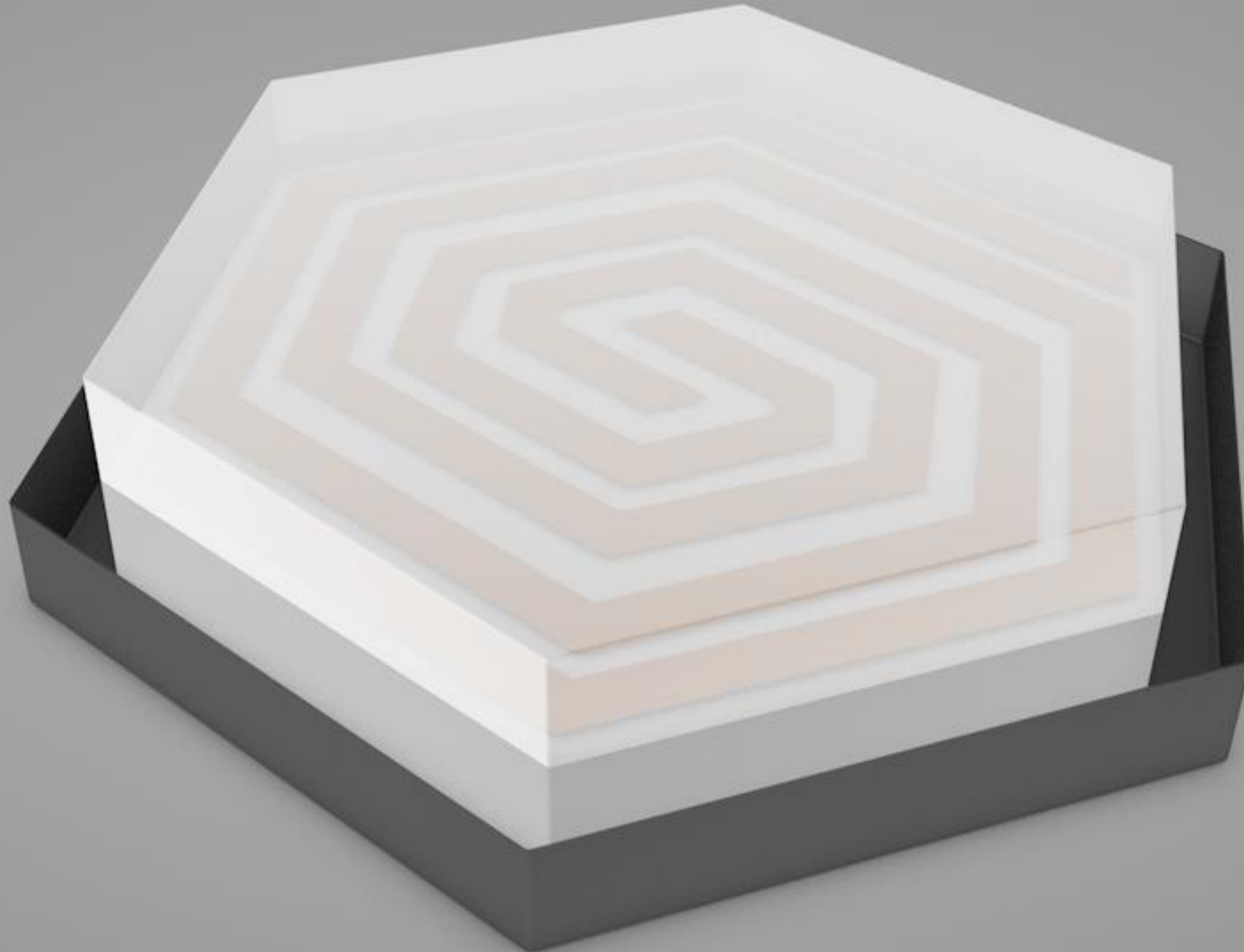
Transparent Robot

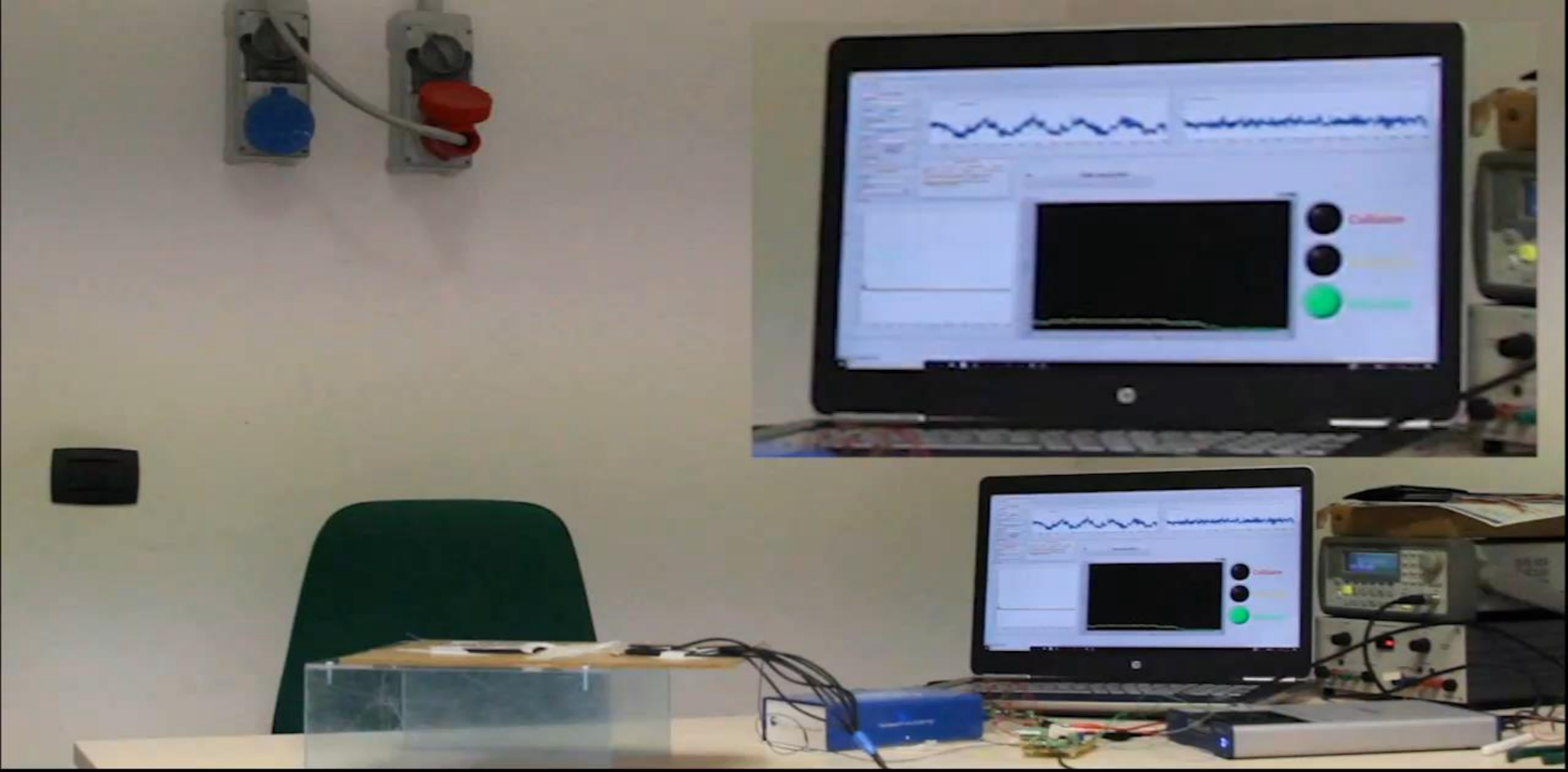


Transparent Robot

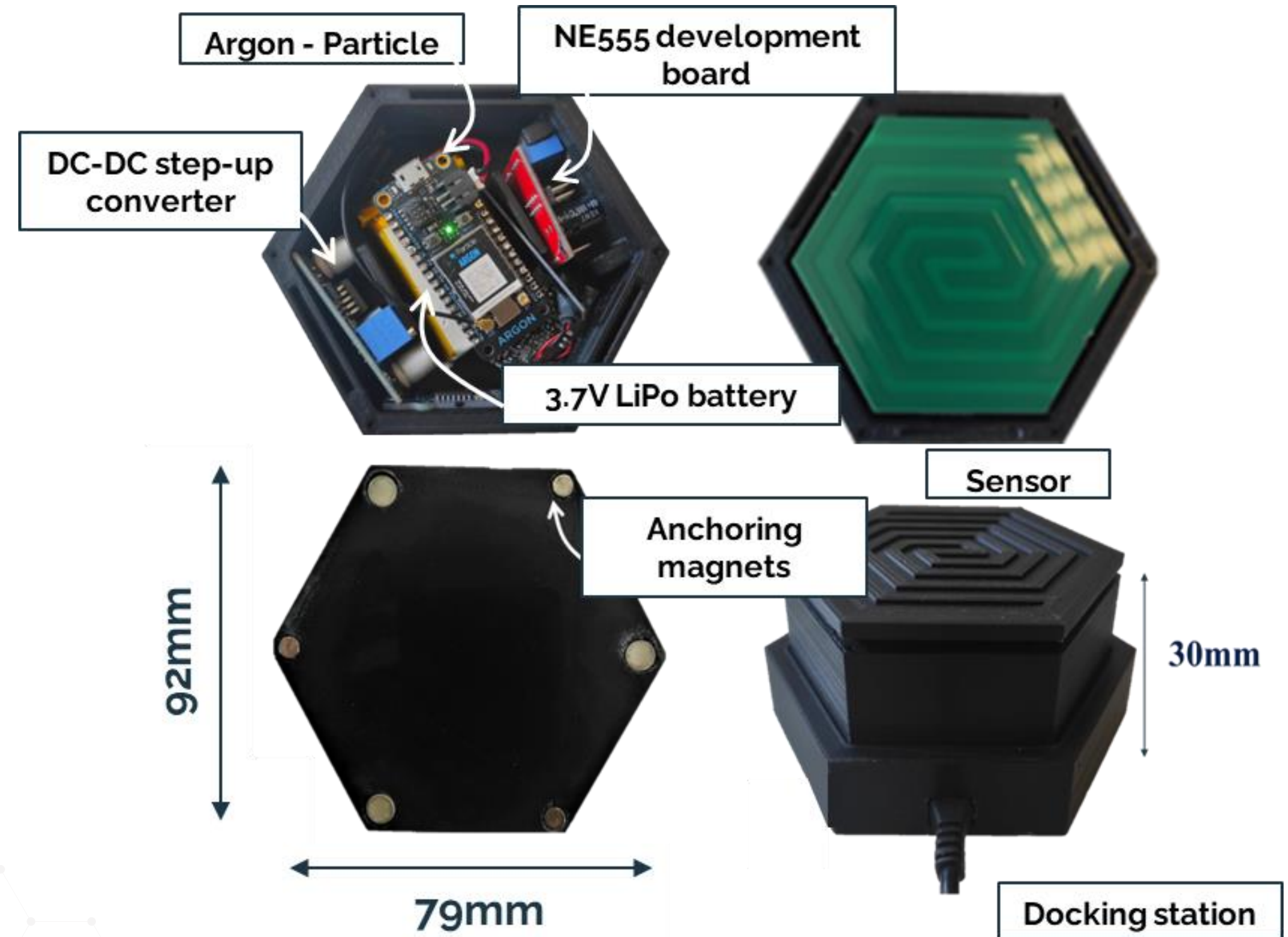
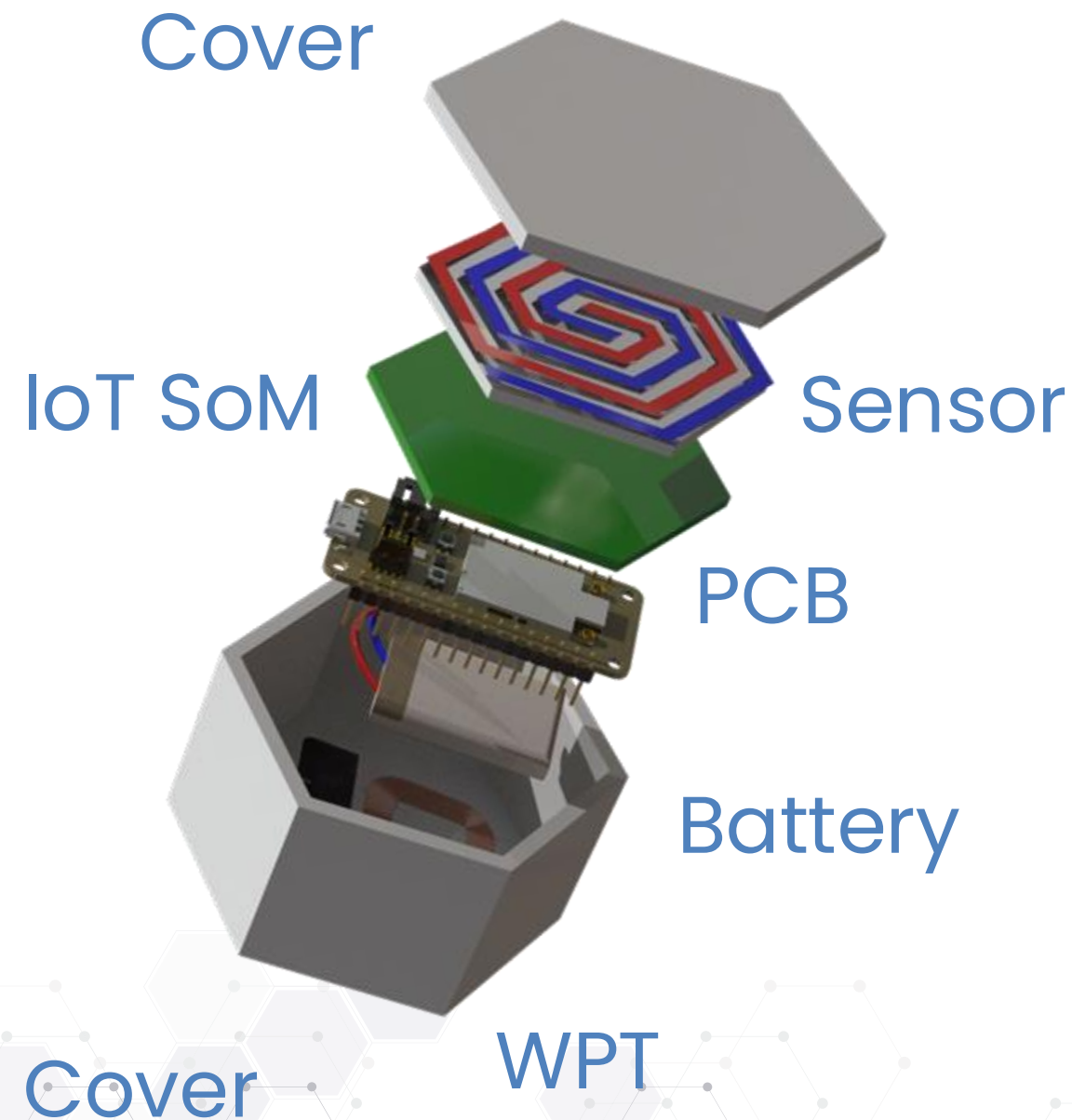
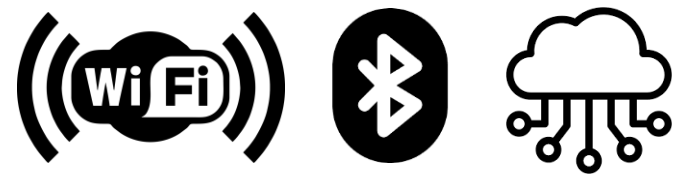


IoT-connected Proximity Sensor





IoT-connected Proximity Sensor



IoT-connected Proximity Sensor



This project has received funding from
the European Union's Horizon 2020
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under grant agreement N° 101017331





Isaac Asimov



***Joseph Engelberger
(father of robotics)***

Communication Protocols (XX.11.2025) – SLOT 1

Gastone Ciuti



Communication Protocols

- Serial Peripheral Interface (SPI)
- Inter Integrated Circuits (I²C)
- Internet of Things (IoT) Communication Protocols



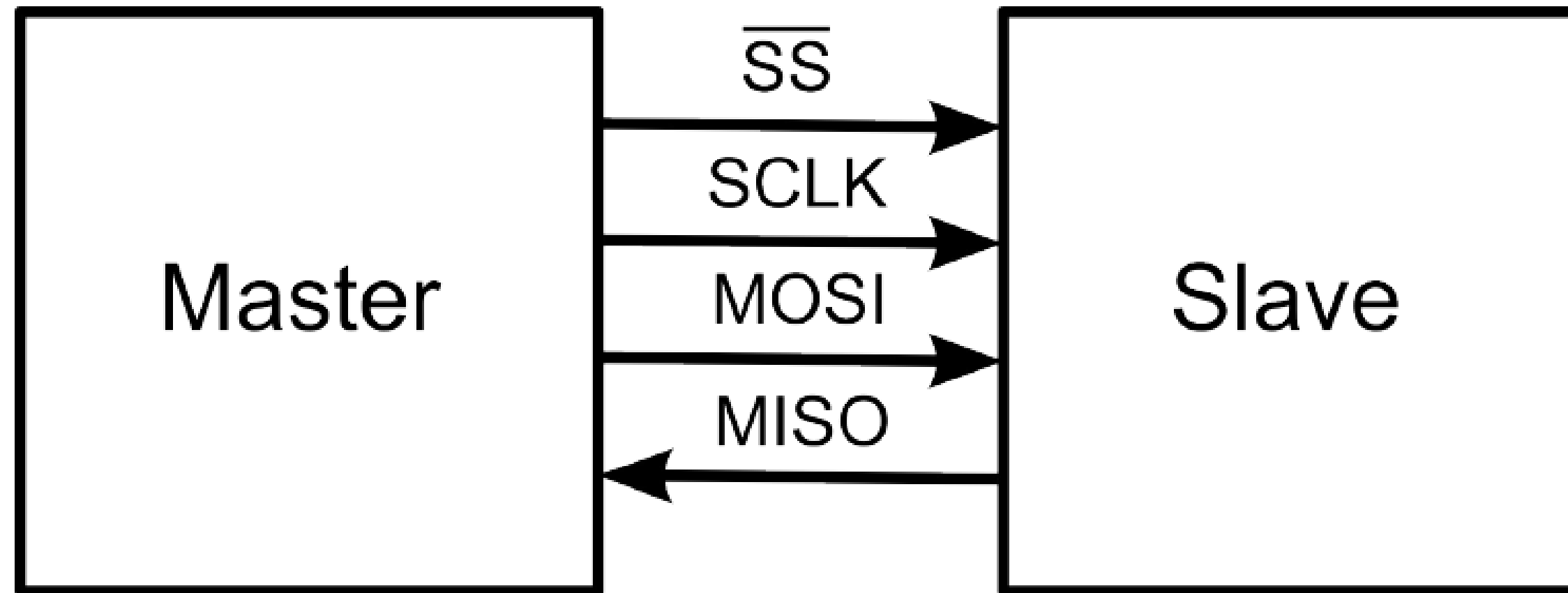
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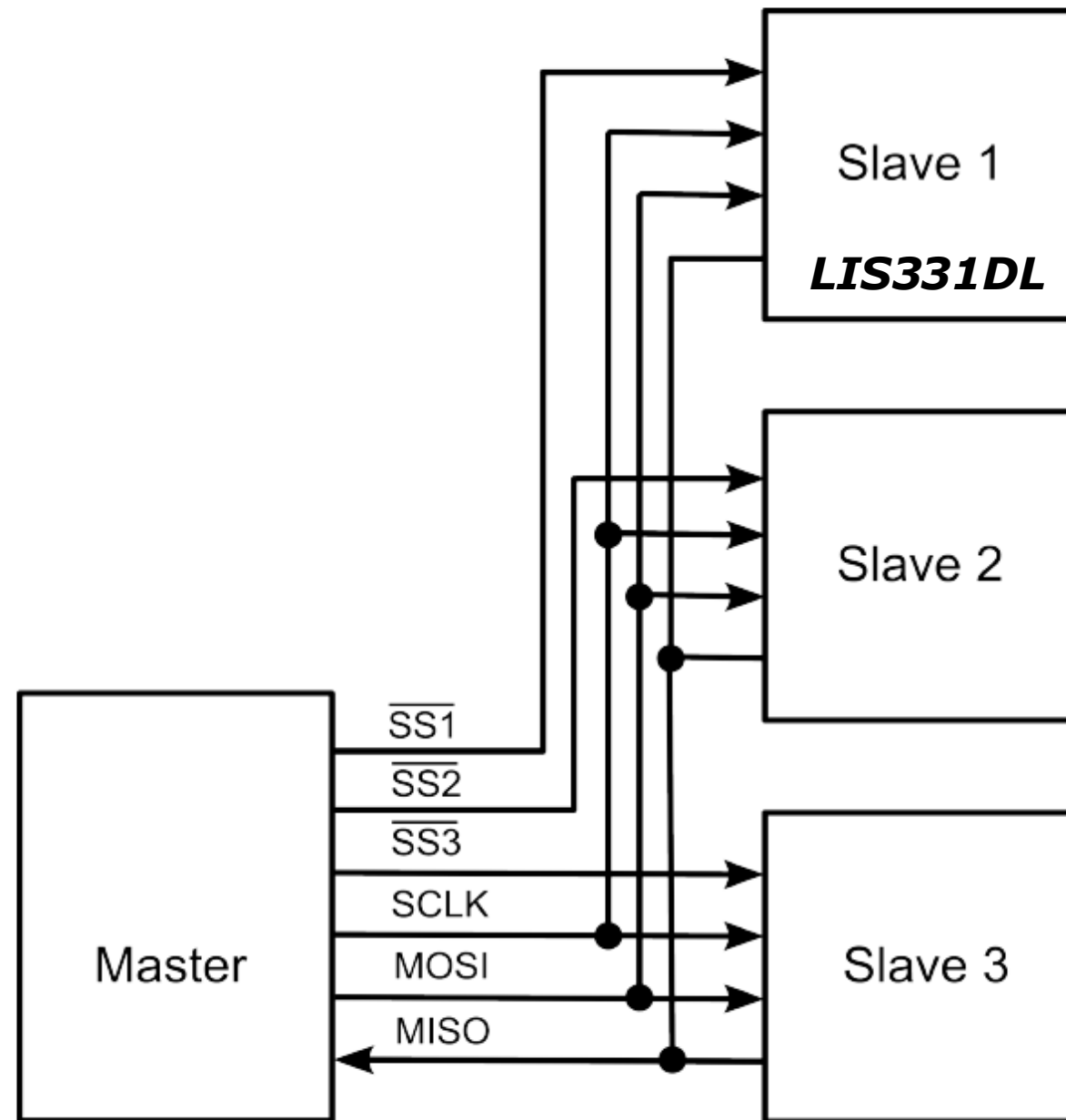


Communication protocols: SPI

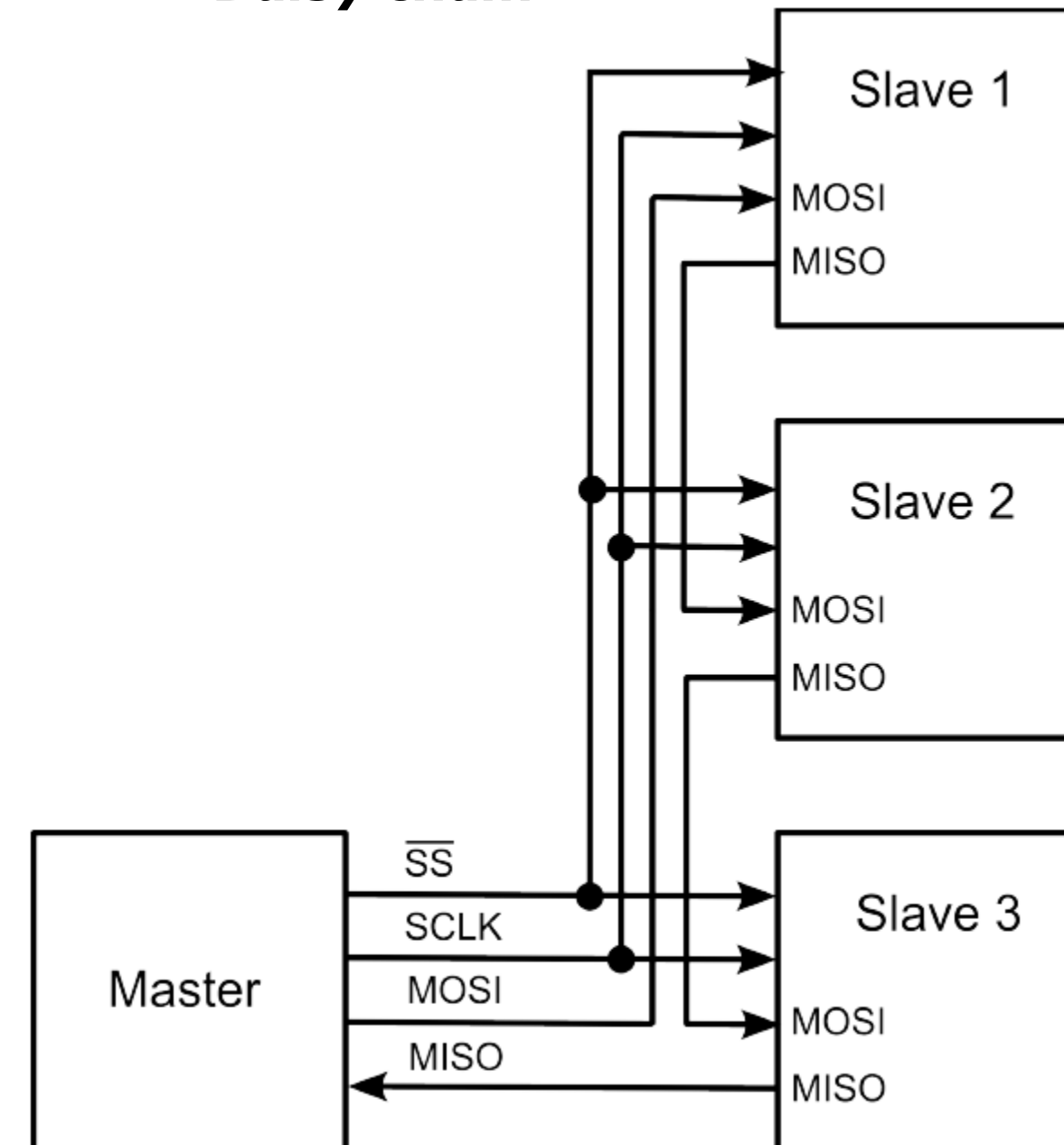
Serial Peripheral Interface



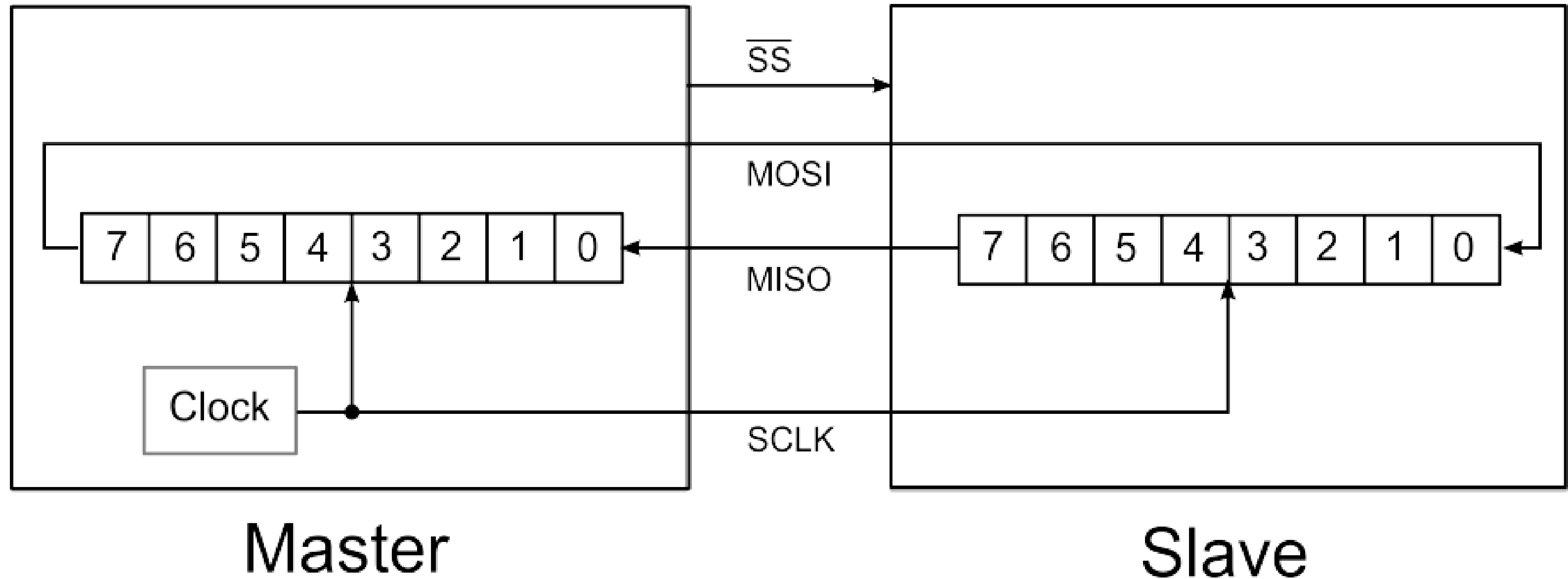
Communication protocols: SPI



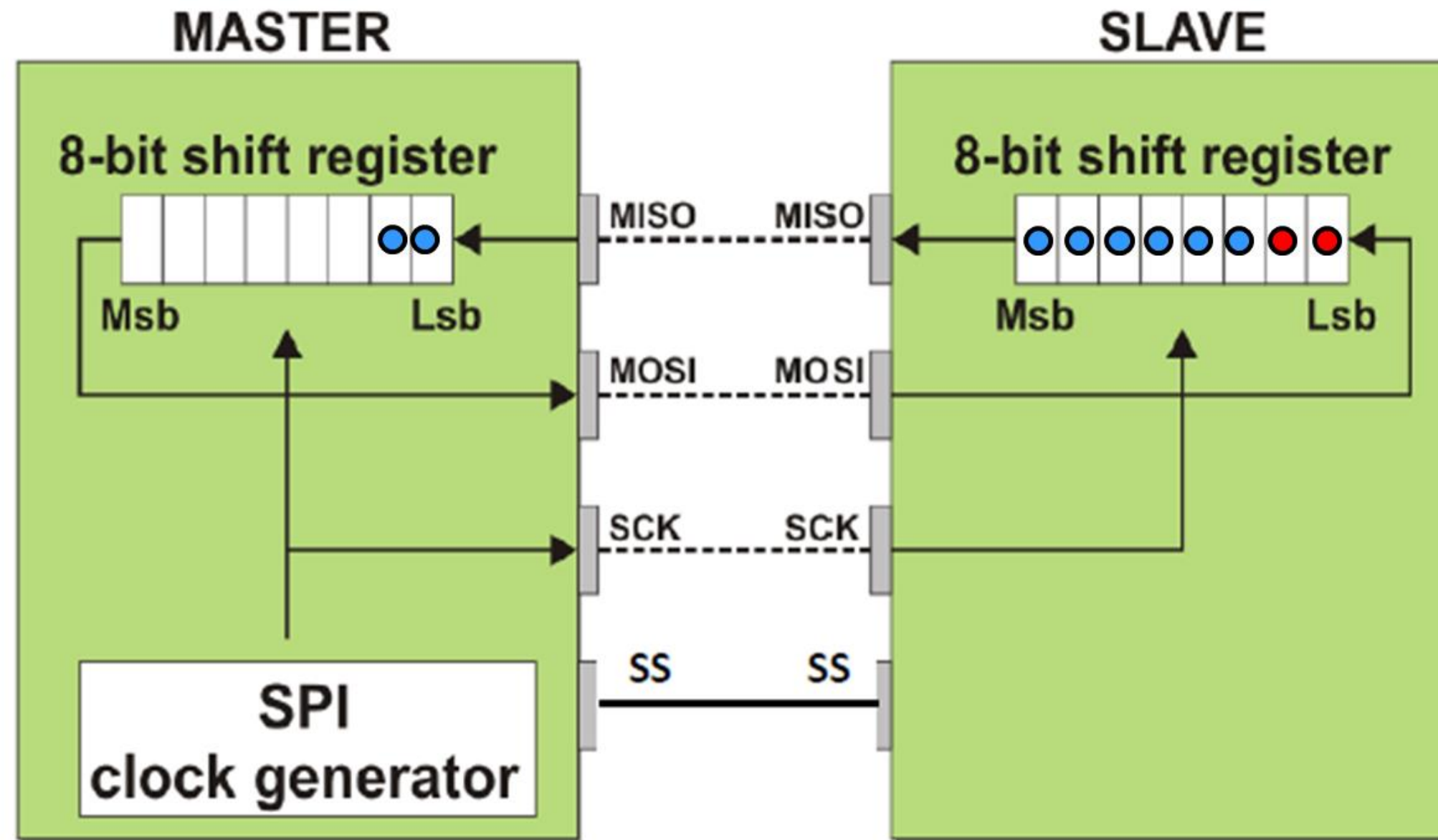
Daisy chain



Communication protocols: SPI



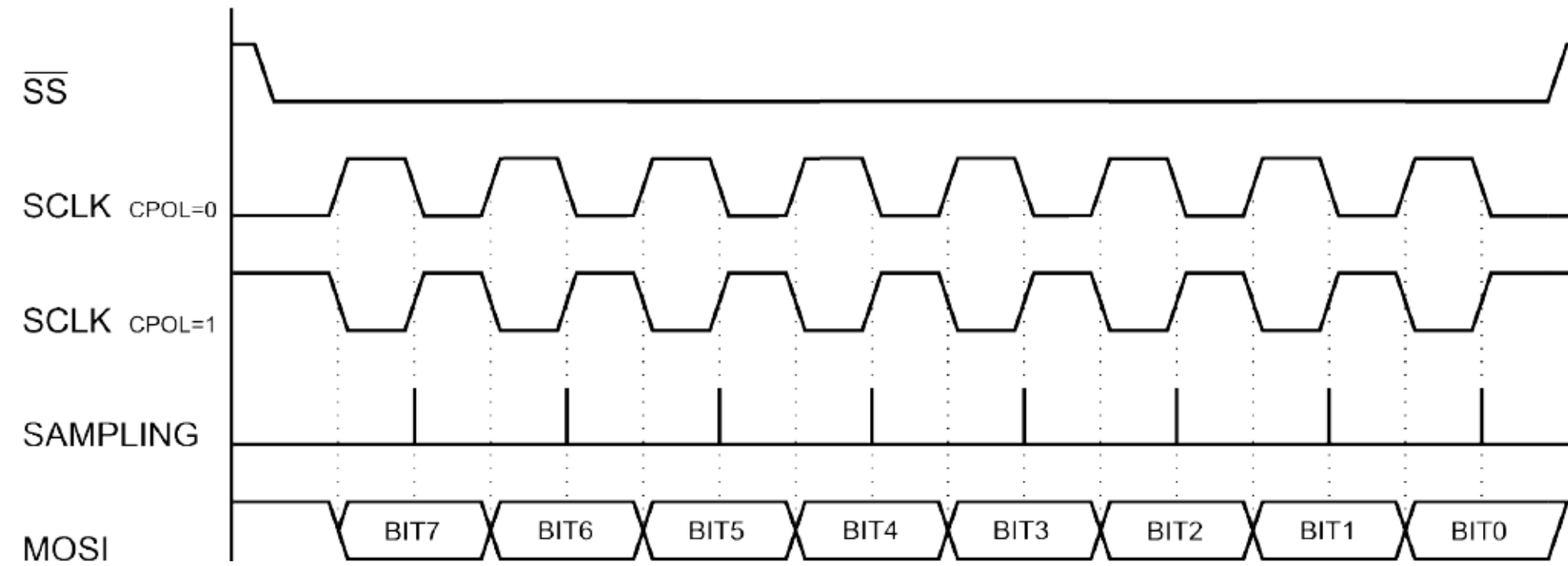
Communication protocols: SPI



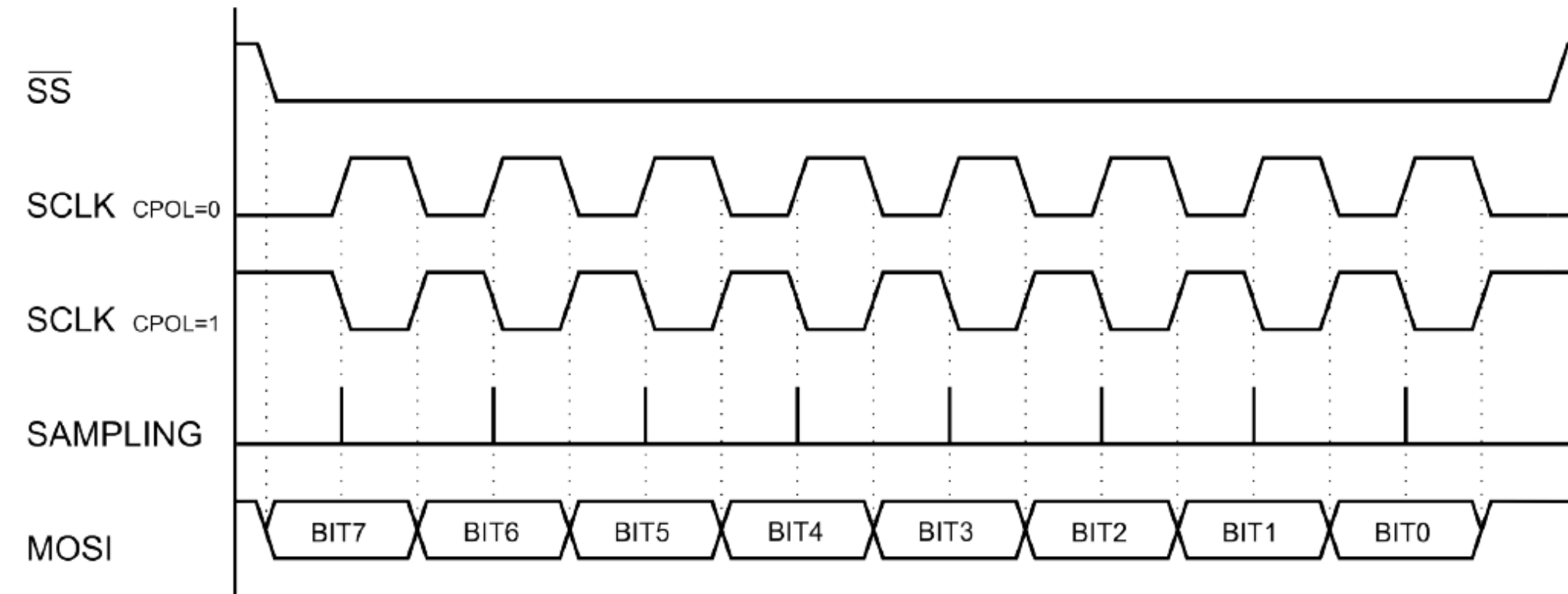
Serial – Synchronous (temporized) – Full-duplex (TX e RX at the same time)



CPHA=1



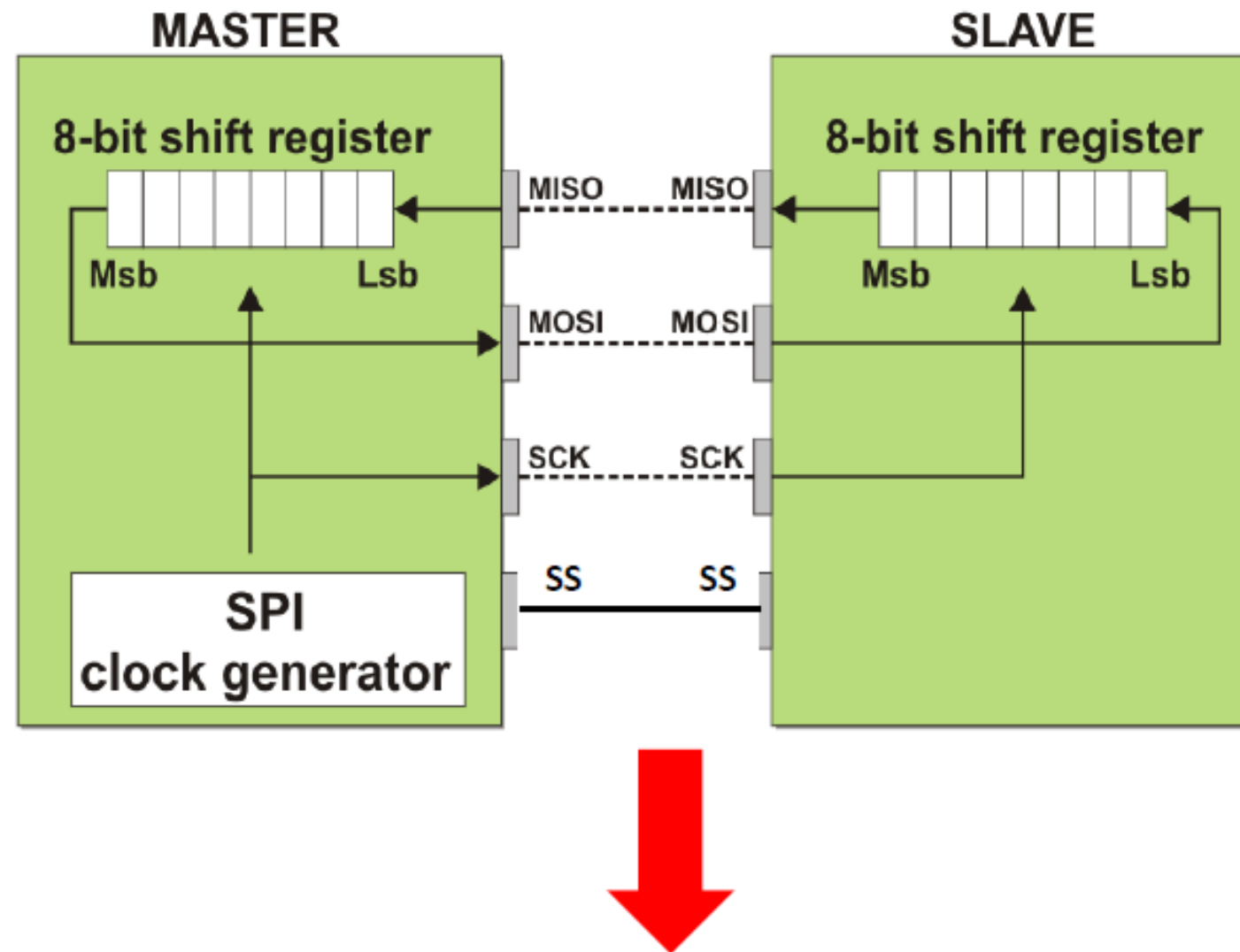
CPHA=0



Modalità	CPOL	CPHA
0	0	0
1	0	1
2	1	0
3	1	1



Communication protocols: SPI



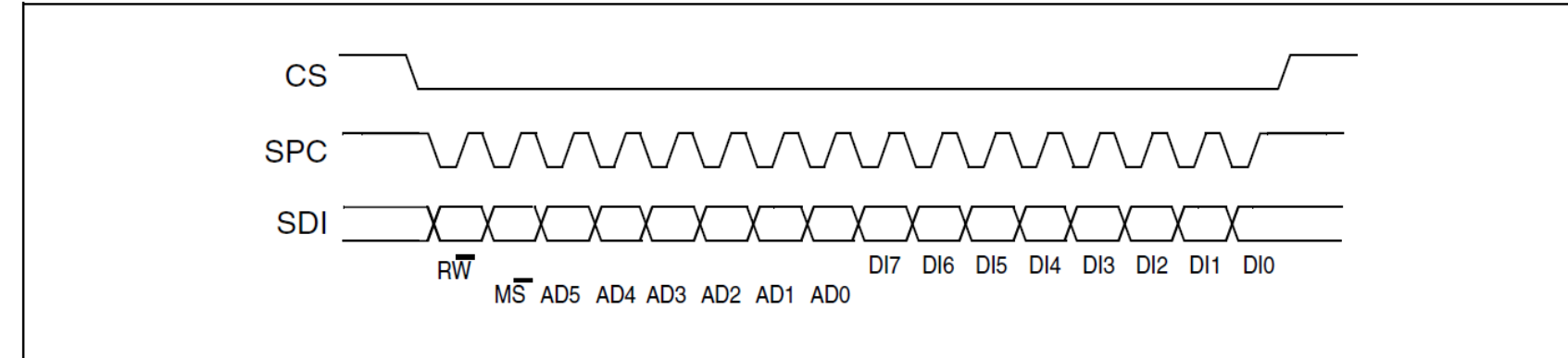
On each clock edge, both master and slave shift out a bit

1. Master configures the clock
2. Master select slave device ($\overline{CS}=0$)
3. *Full-duplex* data transmission each SPI clock cycle
4. Master deselects the slave



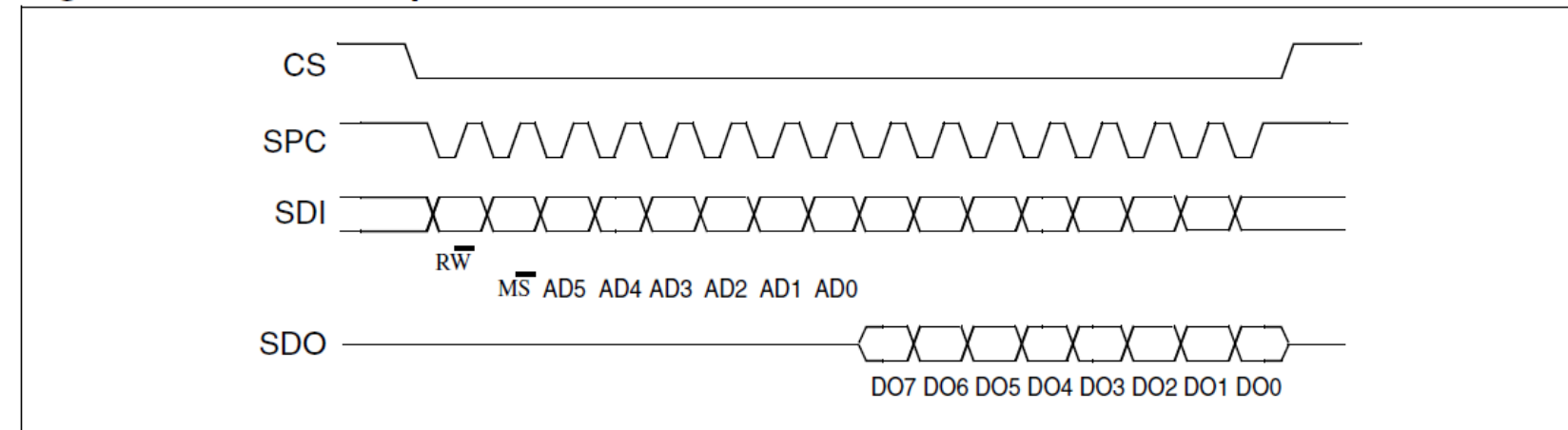
SPI write

Figure 7. SPI write protocol



The SPI Write command is performed with 16 clocks pulses. A Multiple byte write command is performed by adding blocks of 8 clocks pulses to the previous one.

Figure 5. SPI read protocol



The SPI Read command is performed with 16 clock pulses. A Multiple byte read command is performed by adding blocks of 8 clocks pulses to the previous one.

bit 0: READ bit. The value is 1.

bit 1: MS bit. When 0, do not increment the address. When 1, increment the address in multiple readings.



BINARY format [0010 1000] HEXADECIMAL format [0x28]

$$0000 = 0$$

$$0001 = 2^0 = 1$$

$$0010 = 2^1 = 2$$

$$0100 = 2^2 = 4$$

$$1000 = 2^3 = 8$$

$$1001 = 2^3 + 2^0 = 8 + 1 = 9$$

$$1111 = 2^3 + 2^2 + 2^1 + 2^0 = 8 + 4 + 2 + 1 = 15$$

$$00101111 = 0010 \mid 1111 = 2F$$

$$0111 \ 0110 = 0111 \ 0110 = 76$$

$$1100 \ 1110 = 1100 \ 1110 = CE$$

$$0111 \ 0101 = 0111 \ 0101 = 0x75 = 2^0 + 2^2 + 2^4 + 2^5 + 2^6 = 117$$

$$1111 \ 1111 = FF = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 = 1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 = 255$$

10

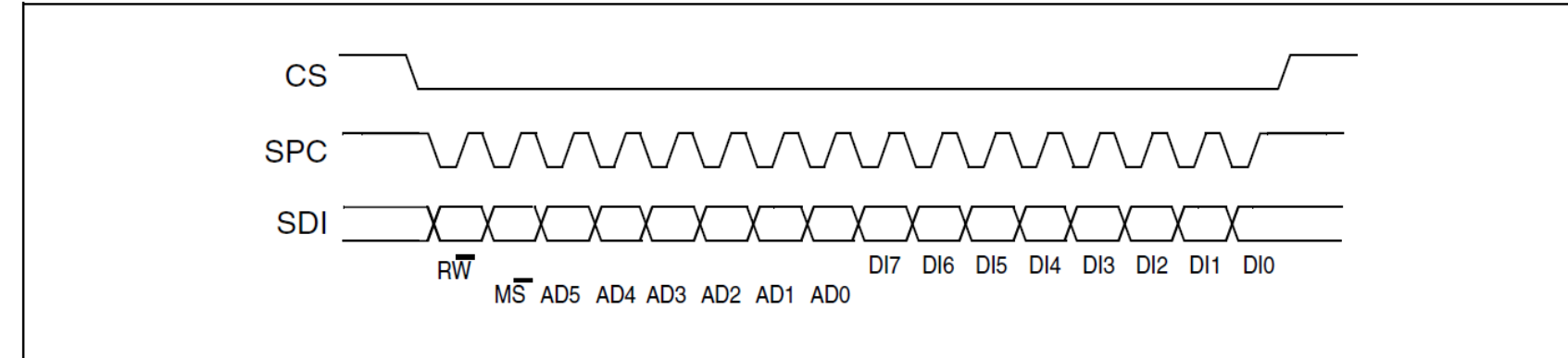
15

esadecimale	binario
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111



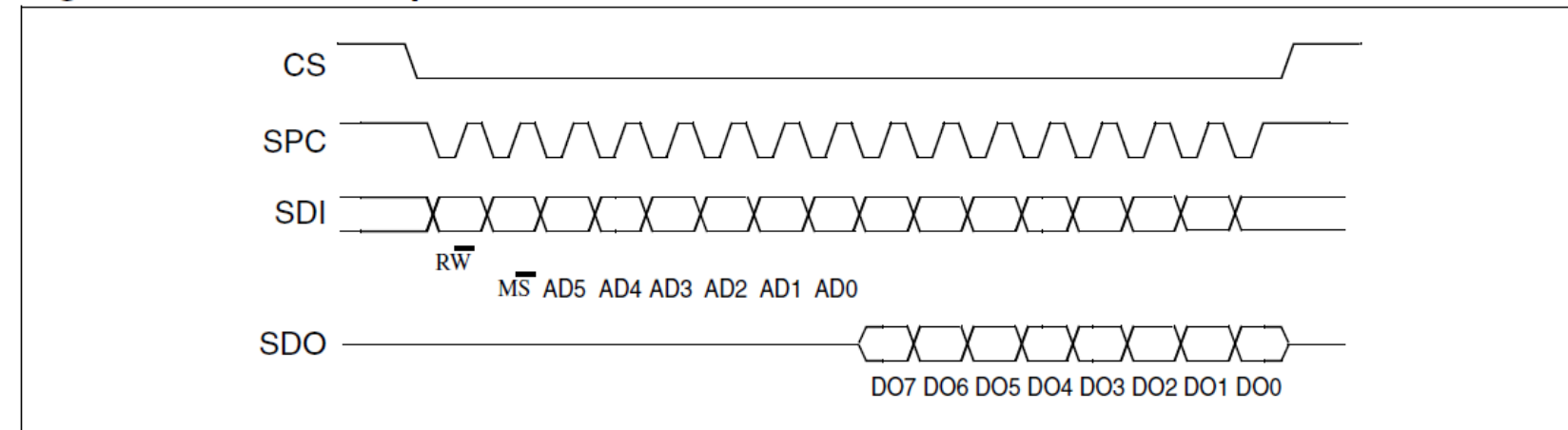
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Table 4. Registers address map

Name	Type	Register address		Default	Comment
		Hex	Binary		
Reserved (do not modify)		00-0E			Reserved
Who_Am_I	r	0F	00 1111	00111011	Dummy register
Reserved (do not modify)		10-1F			Reserved
CTRL_REG1	rw	20	10 0000	00000111	
CTRL_REG2	rw	21	10 0001	00000000	
CTRL_REG3	rw	22	10 0010	00000000	
HP_filter_reset	r	23	10 0011	dummy	Dummy register
Reserved (do not modify)		24-26			Reserved
Status_Reg	r	27	10 0111	00000000	
--	r	28	10 1000		Not used
Out_X	r	29	10 1001	output	
--	r	2A	10 1010		Not used
Out_Y	r	2B	10 1011	output	
--	r	2C	10 1100		Not used
Out_Z	r	2D	10 1101	output	
Reserved (do not modify)		2E-2F			Reserved
FF_WU_CFG_1	rw	30	11 0000	00000000	
FF_WU_SRC_1(ack1)	r	31	11 0001	00000000	
FF_WU_THS_1	rw	32	11 0010	00000000	
FF_WU_DURATION_1	rw	33	11 0011	00000000	
FF_WU_CFG_2	rw	34	11 0100	00000000	
FF_WU_SRC_2 (ack2)	r	35	11 0101	00000000	
FF_WU_THS_2	rw	36	11 0110	00000000	
FF_WU_DURATION_2	rw	37	11 0111	00000000	
Reserved (do not modify)		38-3F			Reserved

CTRL_REG1 (20h)

Control register #1.

DR	PD	FS	ST P	ST M	Zen	Yen	Xen
DR	Data rate selection. Default value: 0 (0: 100 Hz output data rate; 1: 400 Hz output data rate)						
PD	Power-down control. Default value: 0 (0: power-down mode; 1: active mode)						
FS	Full scale selection. Default value: 0 (0: +/- 2 g; 1: +/- 8 g)						
STP, STM	Self test enable. Default value: 00 (00: normal mode; 10: self test P; 01 self test M; 11: forbidden)						
Zen	Z-axis enable. Default value: 1 (0: Z-axis disabled; 1: Z-axis enabled)						
Yen	Y-axis enable. Default value: 1 (0: Y-axis disabled; 1: Y-axis enabled)						
Xen	X-axis enable. Default value: 1 (0: X-axis disabled; 1: X-axis enabled)						

The **DR** bit allows the selection of the data rate at which acceleration samples are produced. The default value is 0, which corresponds to a data rate of 100 Hz. By changing the content of DR to 1, the selected data rate is set equal to 400 Hz.

The **PD** bit allows the device to exit power-down mode. The device is in power-down mode when PD= “0” (default value after boot). The device is in normal mode when PD is set to 1.



Communication Protocols (25.11.2024) – SLOT 2

Gastone Ciuti

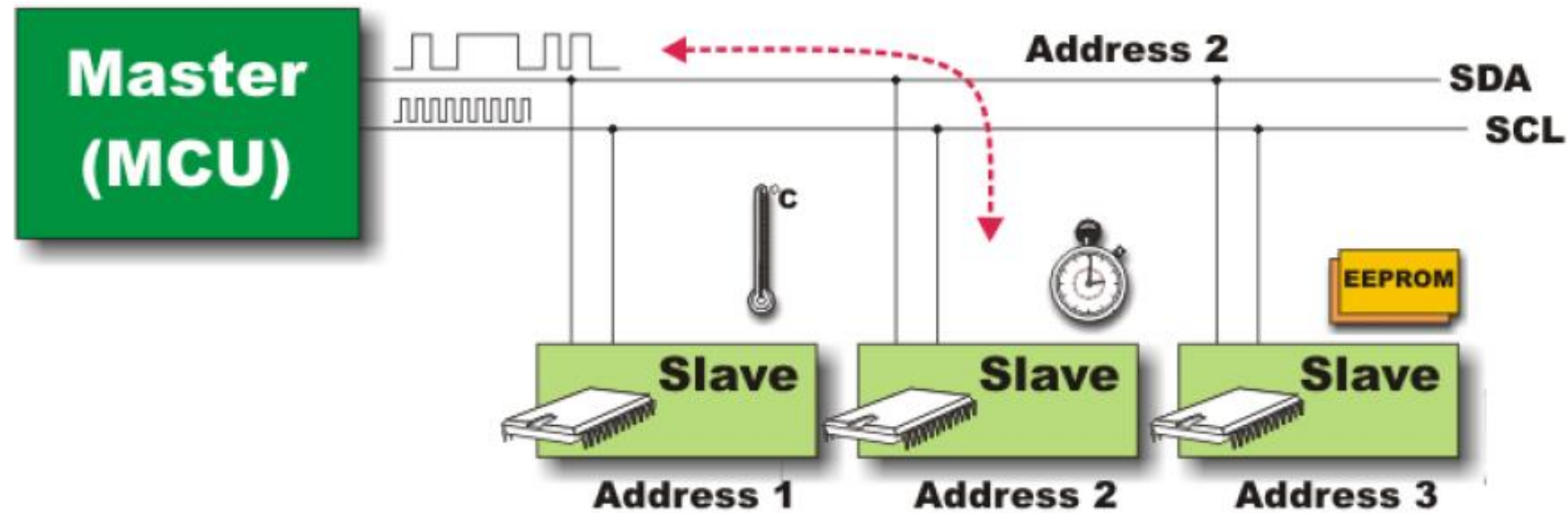


Communication Protocols

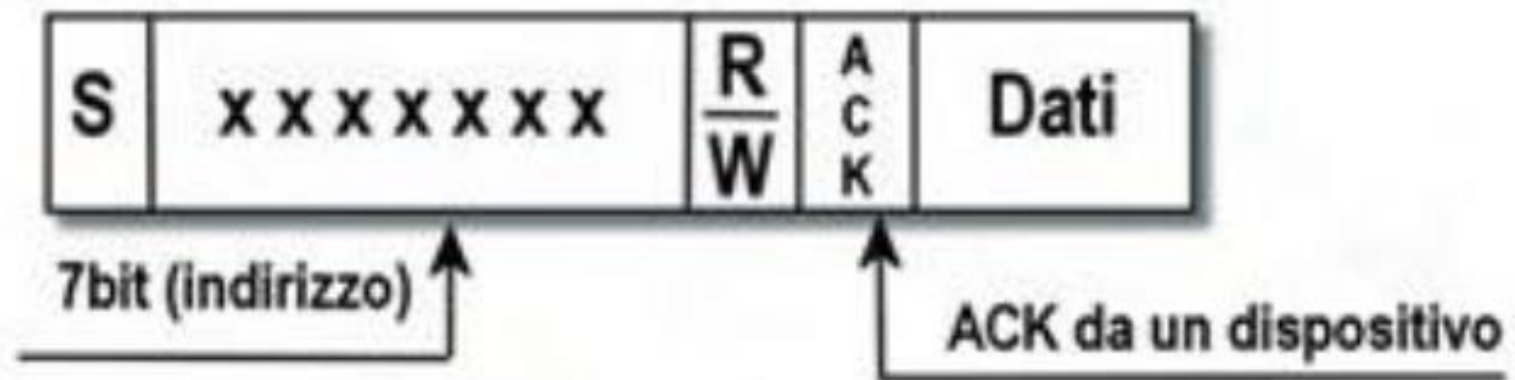
- Serial Peripheral Interface (SPI)
- Inter Integrated Circuits (I²C)
- Internet of Things (IoT) Communication Protocols



Communication protocols: I2C



Inter Integrated Circuits

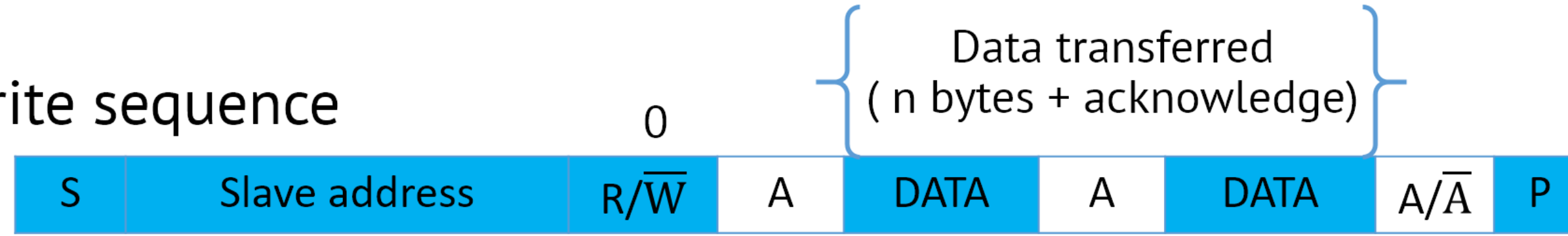


- Bus with 2 communication lines
- Half-duplex
- Multi-master architecture
- *Clock stretching*

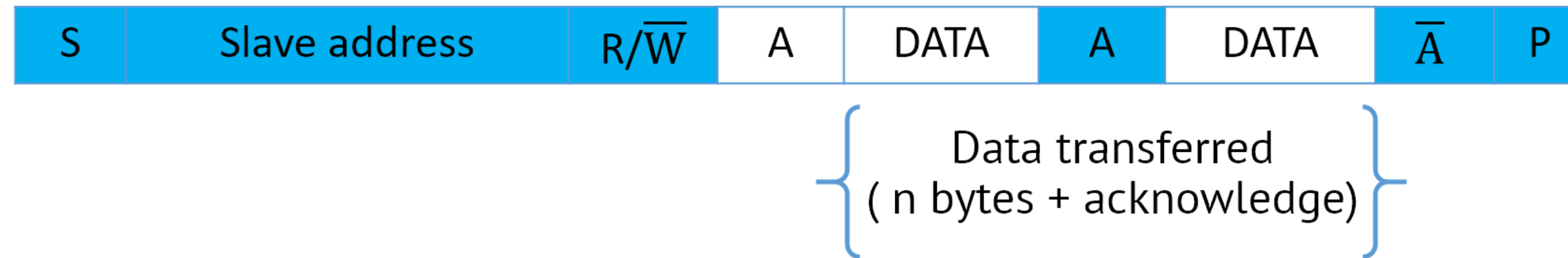
Serial – Synchronous (temporized) – Half-duplex (TX e RX NOT at the same time)

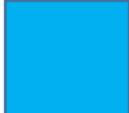


Write sequence



Read sequence



 Master to slave

 Slave to master

A = Acknowledge (SDA low)

$\overline{A}$ = Not acknowledge (SDA High)

S = Start condition

P = Stop condition



Communication protocols: SPI vs. I2C

- The SPI interface is full-duplex (I2C half-duplex).
- Speed: SPI (~ 80-100MHz) vs. I2C (3.4Mbit /s in high-speed mode).
- SPI: interface – not a real protocol – and simple to use (lack of a protocol).
- SPI: arbitrary length of the data to be transmitted (I2C: 8 bits; address format can be either 7-bit or 10-bit).
- SPI: use of a larger number of lines and lack of a protocol and transmission confirmation (*i.e.*, acknowledge, sometimes it is an advantage).
- SPI: lack of peripheral addressing (excluding SS signal).



Communication Protocols

- Serial Peripheral Interface (SPI)
- Inter Integrated Circuits (I²C)
- Internet of Things (IoT) Communication Protocols



IOT Communication Protocols





LoRa (Long Range) is a low-power, wide-area network (LPWAN) protocol designed for **long-range communication**. It operates in the sub-GHz frequency bands, such as 433 MHz, 868 MHz, and 915 MHz, depending on regional regulations.

Key Features:

Long Range: can cover several kilometres to tens of kilometres in open environments.

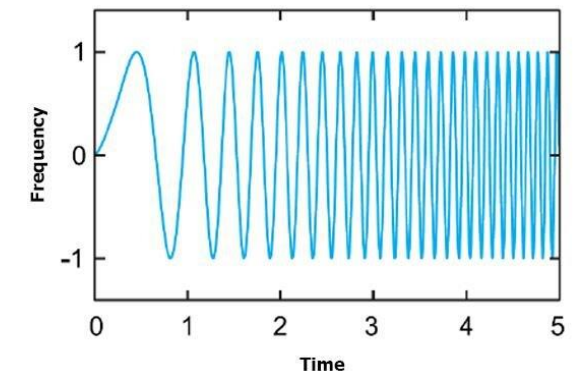
Low Power Consumption: ideal for battery-operated devices in remote sensing and monitoring applications.

Robustness: high resistance to interference and multipath effects due to Chirp Spread Spectrum modulation*.

Flexible Data Rates: allows adjustment of data rates to balance range and throughput.

Applications: smart agriculture, environmental monitoring, smart cities, Industrial IoT

*Chirp Spread Spectrum (CSS) modulation: in digital communications, chirp spread spectrum (CSS) is a spread spectrum technique that uses wideband linear frequency modulated chirp pulses to encode information. A **chirp** is a sinusoidal signal whose frequency increases or decreases over time (often with a polynomial expression for the relationship between time and frequency).





Zigbee is a specification for a suite of high-level communication protocols using **low-power digital radios**. It is based on the IEEE 802.15.4 standard and operates in the 2.4 GHz, 900 MHz, and 868 MHz frequency bands*

*ISM, 784 MHz in China, 868 MHz in Europe, 915 MHz in the US and 2.4 GHz in most of the rest of the world

Key Features:

Low Power Consumption: suitable for devices that need long battery life.

Mesh Networking: supports mesh networks, enhancing network reliability and coverage.

Interoperability: promotes interoperability among different manufacturers.

Security: incorporates robust security features, such as Advanced Encryption Standard AES-128/64/32.

Applications: home automation, smart lighting, security systems, healthcare monitoring

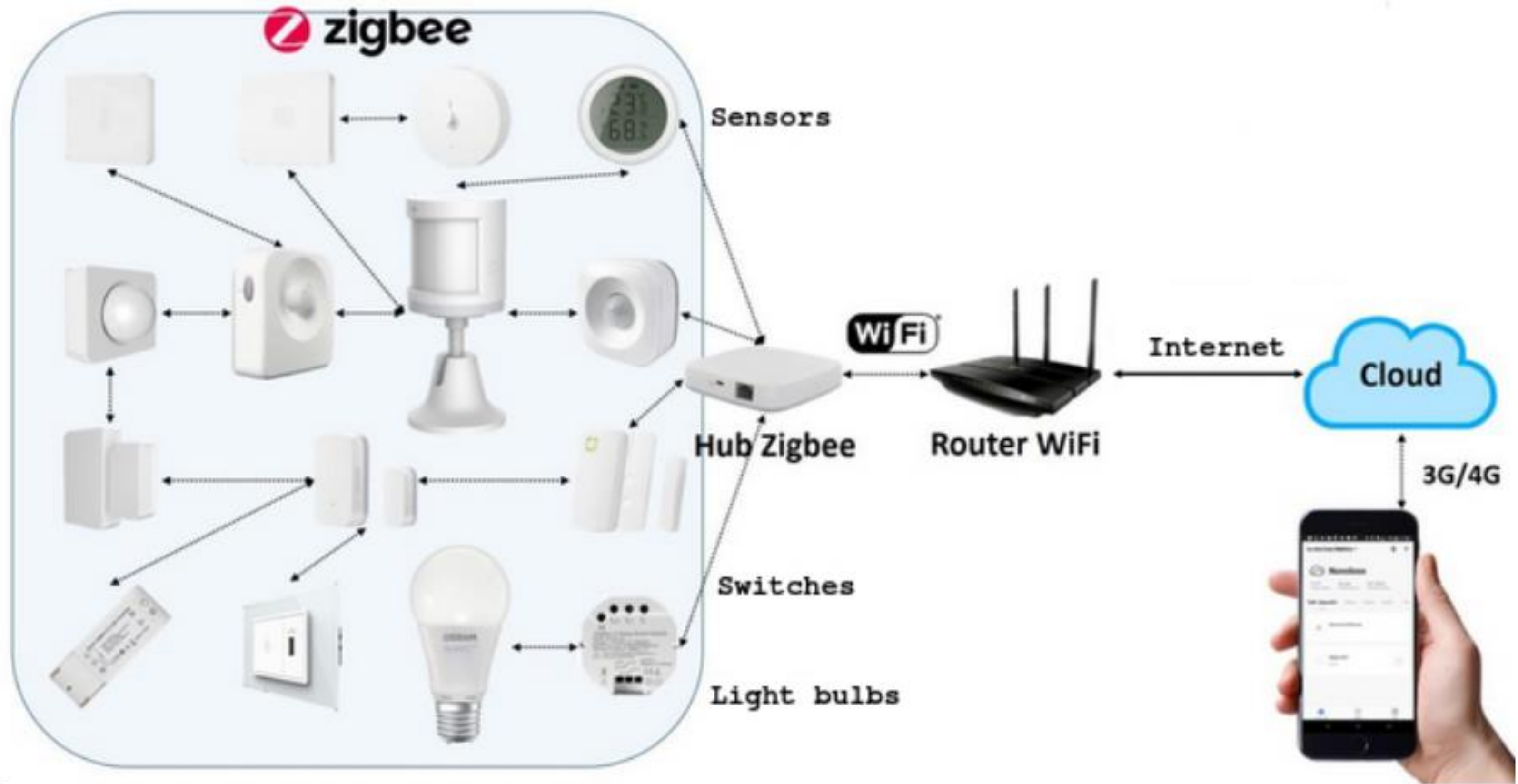


Connectivity Standards Alliance (CSA), formerly the Zigbee Alliance, is a group of companies that maintain and publish the Zigbee and Matter standard, along with several others (Amazon, Apple, Comcast, Google, Yandex, Ikea, and Samsung SmartThings)

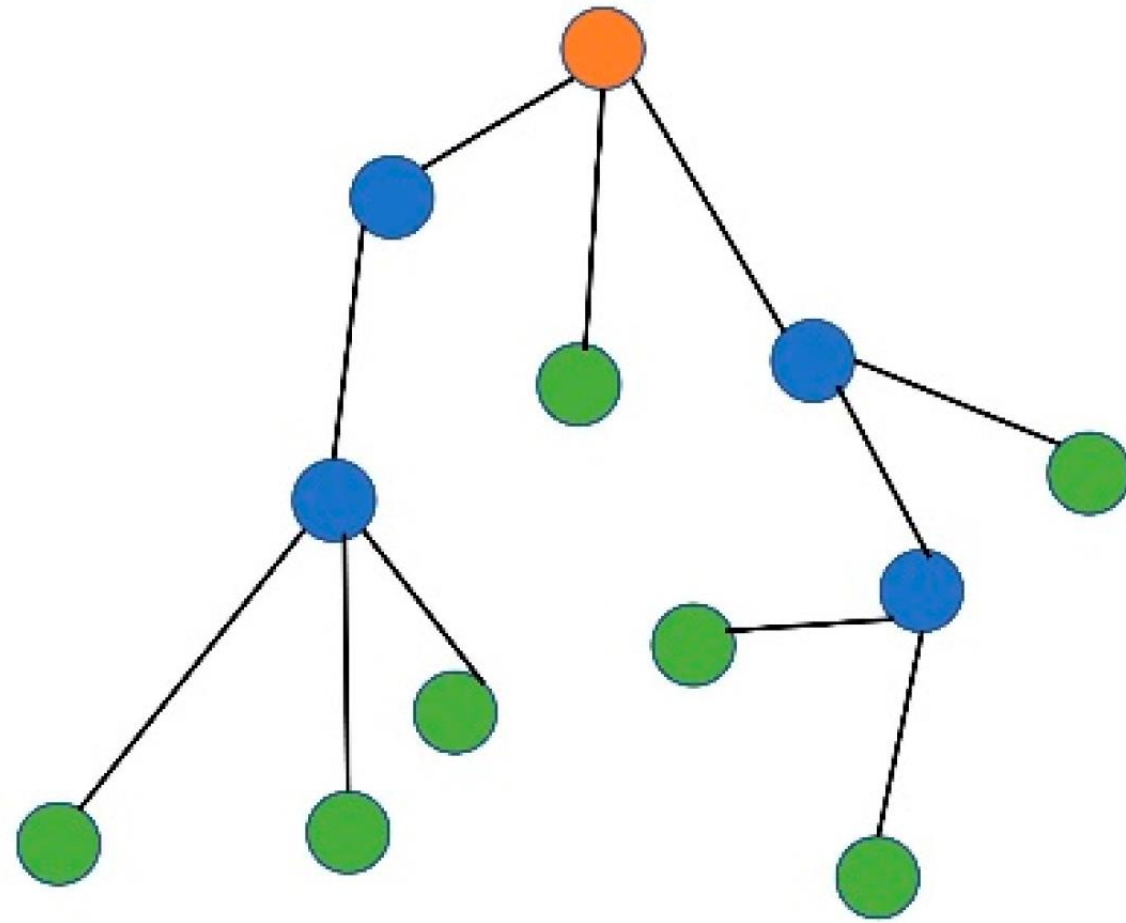
[LINK](#)

Furthermore, ZigBee devices must comply with the standards of the IEEE 802.15.4-2003 Low-Rate Wireless Personal Area Network (WPAN) standard to operate on the 2.4 GHz, 915 MHz and 868 MHz frequencies of the ISM (Industrial, Scientific and Medical) band, "not licensed", which does not require authorisation from a government authority.

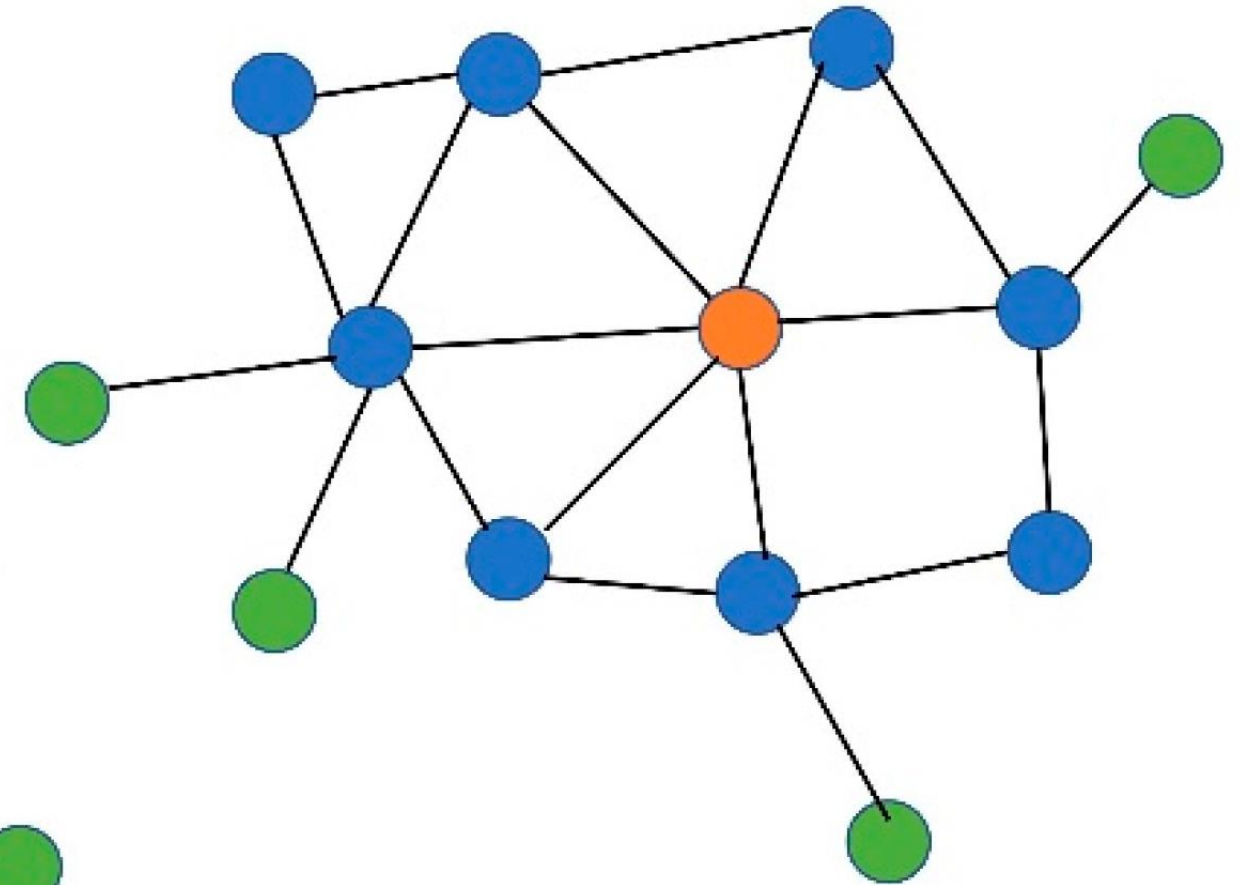




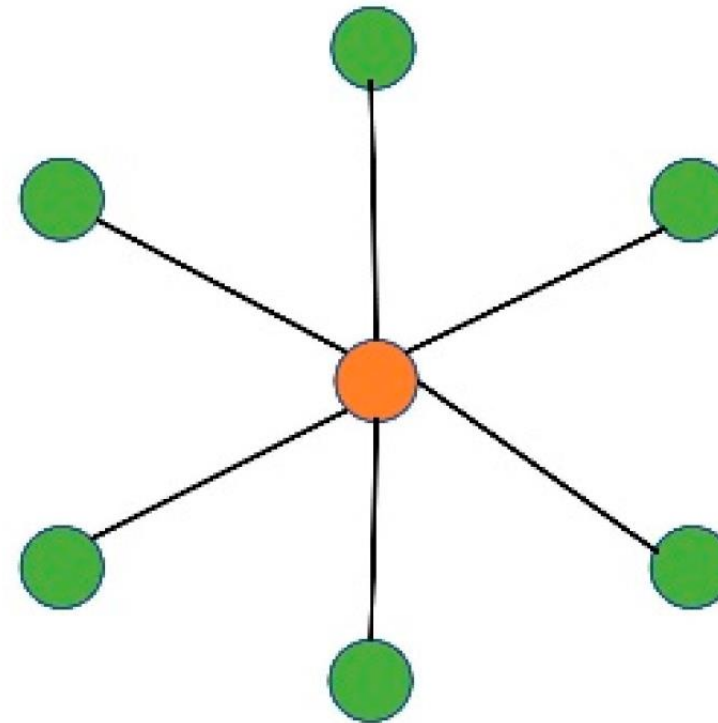
ZigBee Tree Topology



ZigBee Mesh Topology



ZigBee Star Topology



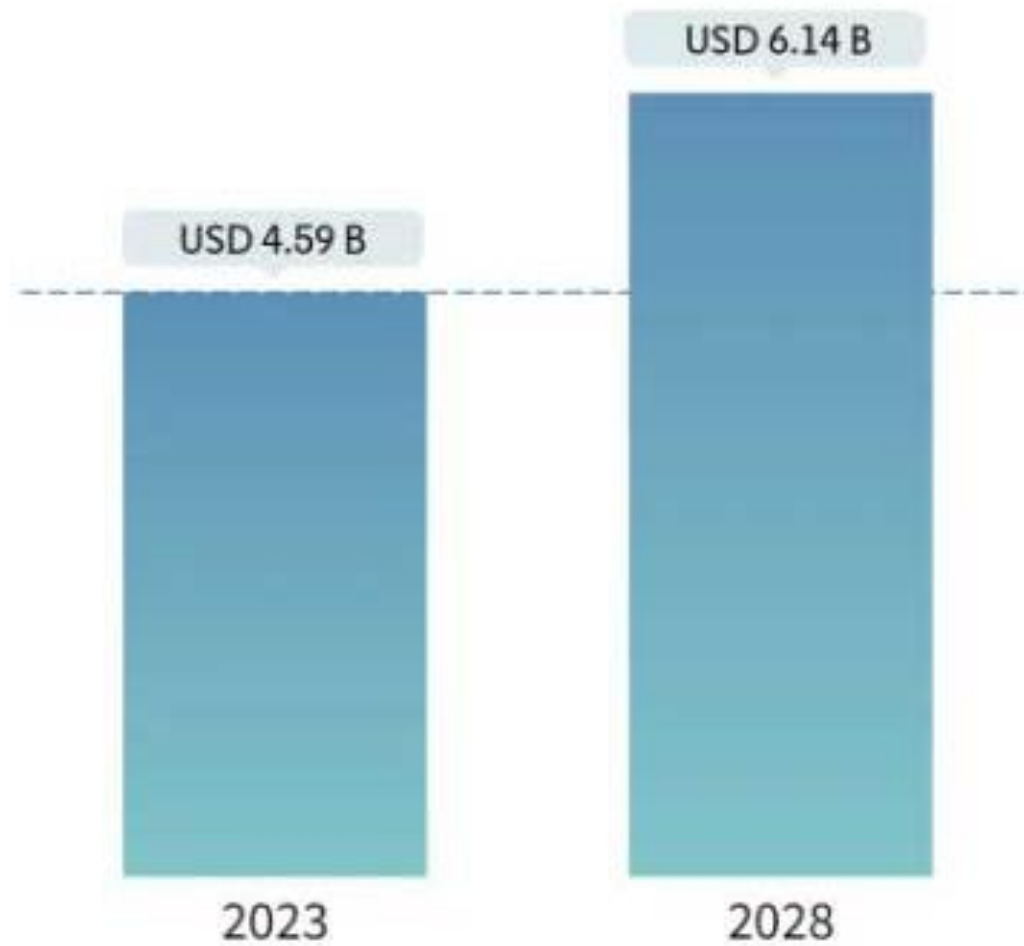
-  Coordinator Nodes
-  Router Nodes
-  End Nodes



ZigBee Market

Market Size in USD Billion

CAGR 6.01%



Source : Mordor Intelligence



Study Period

2018 - 2028

Market Size (2023)

USD 4.59 Billion

Market Size (2028)

USD 6.14 Billion

CAGR (2023 - 2028)

6.01 %

Fastest Growing Market

Asia Pacific

Largest Market

North America

Major Players



*Disclaimer: Major Players sorted in no particular order





Z-Wave is a proprietary wireless communication protocol that operates at 900 MHz; it is designed for home automation, providing **reliable and low-latency communication** between devices.

Key Features:

Low Power Consumption: designed for battery-operated devices with extended battery life.

Mesh Networking: enhances coverage and reliability through mesh networking.

Interoperability: strong emphasis on interoperability among devices from different manufacturers.

Frequency Bands: operates in the sub-GHz frequency bands, minimising interference.

Applications: home automation, smart thermostats, security systems, lighting control

Comparison with Zigbee: frequency (900 MHz vs. 2.4 GHz); # devices (232 vs. 65k); interoperability (ZigBee+); power consumption (ZigBee+ with respect to Z-Wave and Wi-Fi)





433 MHz RF is a popular choice for short-range, low-power wireless communication. It operates in the unlicensed 433 MHz ISM band

Key Features:

Simplicity: easy to implement and use.

Low Power Consumption: suitable for battery-operated devices.

Cost-Effective: inexpensive components and implementation.

Short Range: typically covers distances up to a few hundred meters.

Applications: remote controls, wireless sensors, home automation, simple data transmission





Wi-Fi is a widely used wireless networking technology based on the IEEE 802.11 standards. It operates in the **2.4 GHz and 5 GHz frequency** bands.

Key Features:

High Data Rates: supports high data rates, suitable for bandwidth-intensive applications.

Wide Range: covers significant distances within homes and buildings.

Ubiquity: extensive adoption and integration into numerous devices.

Internet Connectivity: provides direct access to the internet.

Applications: smart home devices, video streaming, security cameras, IoT hubs





Bluetooth Low Energy (BLE) is a wireless personal area network technology designed for **short-range communication with low power consumption.**

Key Features:

Low Power Consumption: optimised for devices requiring extended battery life.

Short Range: typically covers distances up to 100 meters.

Interoperability: strong support for interoperability between devices.

Data Rates: supports moderate data rates suitable for sensor data transmission.

Applications: wearable devices, health monitors, beacons, smart home devices





NB-IoT (Narrowband IoT) is a low-power, wide-area network (LPWAN) technology developed by 3GPP for IoT applications, **operating within existing cellular networks.**



Key Features:

Low Power Consumption: designed for battery-powered devices with long lifespans.

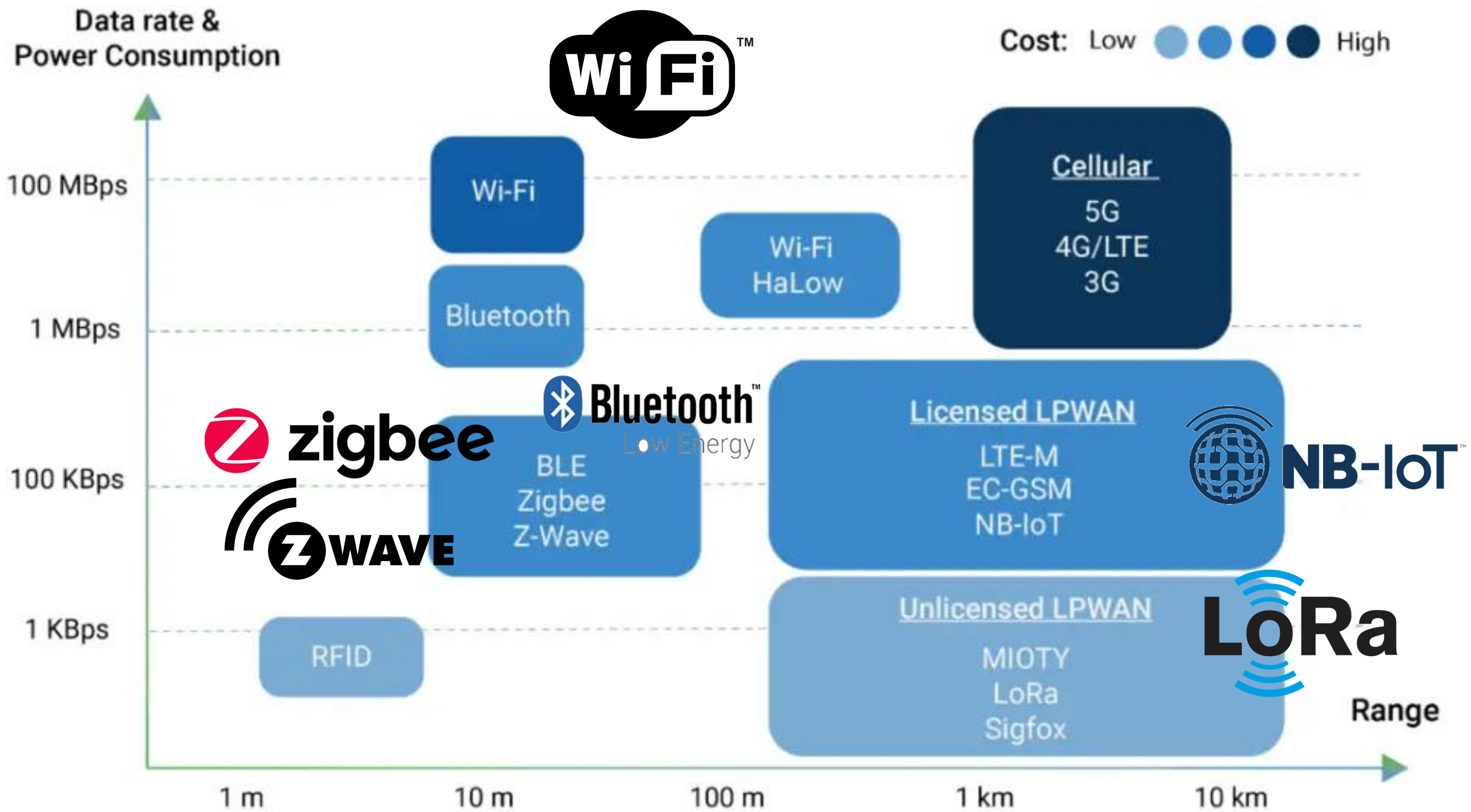
Wide Area Coverage: provides extensive coverage, including deep indoor penetration.

Scalability: supports a large number of devices per cell.

Security: leverages existing cellular security features.

Applications: smart metering, asset tracking, environmental monitoring, eIndustrial IoT





...a few conclusions

The diversity of IoT communication protocols reflects the wide range of applications and requirements in the IoT landscape. Each protocol offers unique features and advantages, making them suitable for specific use cases. **LoRa** excels in long-range, low-power applications, while **Zigbee** and **Z-Wave** are ideal for home automation with their mesh networking capabilities. The **433 MHz RF** protocol is a cost-effective solution for simple, short-range communication, whereas **Wi-Fi** and **BLE** are ubiquitous and versatile for various IoT devices.

Variable	Wi-Fi	Z-Wave	ZigBee
Year first launched in Market	1997	2003	2003
PHY/MAC Standard	IEEE 802.11.1	ITU-T G.9959	IEEE 802.15.4
Frequency Band	2.4 GHz	900 MHz*	2.4 GHz
Nominal Range (0 dBm)	100 m	30 – 100 m	10 – 100 m
Maximum Data Rate	54 Mbit/s	40-100 kbit/s	250 kbit/s
Topology	Star	Mesh	Mesh
Power Usage	High	Low	Low
Alliance	Wi-Fi Alliance	Z-Wave Alliance	ZigBee Alliance





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